OECD

OECD ECONOMIC SURVEYS

1998

SPECIAL FEATURES
Reforming the Federal Fiscal System
Policies to Promote Growth and Employment

GERMANY

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1997-1998

GERMANY

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF GERMANY

THE LAND

Area, 1993 (thousand sq. km)	357.0
Agricultural area, 1993 (thousand sq. km)	195.4
Forests, 1993 (thousand sq. km)	104.3

Major cities, June 1995 (thousand inhabitants)	
Berlin	3 470
Hamburg	1 707
Munich	1 241
Cologne	964
Frankfurt	651
Essen	616
Dortmund	600
Stuttgart	587
Düsseldorf	572
Bremen	549

THE PEOPLE

Population (thousands), end 1996	82 012
Number of inhabitants per sq. km	230
Net natural increase in population (thousands), 1996	–87
Net migration (thousands) 1996	282

Labour force (thousands), 1997	38 260
Employment (thousands), 1997	33 876
of which:	
Agriculture, forestry, fishing	928
Industry (including construction)	11 518
Private services	14 366

PRODUCTION

GDP, 1997 (billions of DM)	3 642
GDP per head, 1997 (US$)	25 594
Gross fixed investment 1997:	
Per cent of GDP	20.1
Per head (US$)	5 141

Origin of GDP, 1997 (per cent)	
Agriculture, forestry, fishing	1.1
Industry (including construction)	31.9
Services	67.0

THE GOVERNMENT

Public consumption, 1997, (per cent of GDP)	19.4
General government current revenue, 1997 (per cent of GDP)	44.9
Public debt, end 1997 (ratio to general government current revenue)	144.8

Composition of Parliament:	Seats
Social Democratic Party (SPD)	252
Christian Democratic Party (CDU)	244
Christian Social Union (CSU)	50
Greens	49
Free Democratic Party (FDP)	47
PSD	30

Last general election: October 1994
Next general election: 1998

FOREIGN TRADE

Exports of goods and services, 1997 (per cent of GDP)	26.6
Main exports, 1997 (per cent of total merchandise exports):	
Products of agriculture, forestry and fishing	0.9
Basic materials and semi-finished goods	22.4
Manufactured foods and tobacco	4.1
Other consumer goods	10.5
Investment goods	56.6
Other exports	5.5
Total	100

Imports of goods and services, 1997 (per cent of GDP)	24.9
Main imports, 1997 (per cent of total merchandise imports):	
Food	9.5
Raw materials and semi-finished goods	14.5
Finished goods	68.4
of which:	
Primary products	11.2
End products	57.2
Other exports	7.6
Total	100

THE CURRENCY

Monetary unit: Deutschemark

Currency units per US$, average of daily figures:	
Year 1997	1.734
May 1998	1.775

Note: An international comparison of certain basic statistics is given in an Annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Germany by the Economic and Development Review Committee on 8 June 1998.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 10 July 1998.

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The previous Survey of Germany was issued in August 1997.

Assessment and recommendations

Overview of current policy issues

When the German economy was last reviewed in June 1997, there were signs that export growth was leading to a pick up in output but domestic demand and employment were still depressed. Developments since this time have been broadly compatible with this picture. International competitiveness has improved and exports have been buoyant, but employment and investment have remained weak in comparison with previous recoveries and domestic demand growth has been slow. These latter features are the expression of an economy which has been going through a difficult fiscal adjustment and major structural change: investment and employment have been weak, partly because capital and labour are being used more efficiently. The first two chapters of this year's *Survey* examine recent trends and the prospects for a quickening of the expansion, both from the medium-term perspective that structural adjustment has strengthened prospects for more balanced longer-run growth and against the short-term background of a relatively favourable fiscal and monetary climate, in which domestic demand should reinforce the growth impulse given by exports. Membership of the European Economic and Monetary Union offers opportunities and challenges. Monetary policy is being increasingly shaped by the requirements of the Euro area as a whole, which should be consistent with continued monetary support for the expansion, while fiscal policy will be conditioned by the requirements of the Stability and Growth Pact. Among

the most important policy requirements for successfully adapting to this new environment, the *Survey* discusses the changes which might be required in the federal fiscal arrangements in order to ensure both the proper implementation of the Pact and longer-run public-sector efficiency. Most fundamentally, for Germany, as for other participating countries in the Euro area, the potential economic benefits of the EMU will depend upon creating a structurally sound and economically adaptable economy. In this context the *Survey* discusses the progress made towards more flexible labour and product markets, the final chapter containing a follow-up along the lines of the *OECD Jobs Strategy*.

Growth is picking up and should become more balanced

GDP grew by some 2¼ per cent in 1997, leading to a marked recovery in industrial production and to a rise in capacity utilisation which prompted a gradual pick up in investment in plant and machinery. Driving the expansion has been a sharp rise in exports, based on an improvement in competitiveness due to the marked reduction in unit labour costs and a fall in the effective exchange rate. These favourable conditions remain in place in 1998. The Asian crisis has not led to any significant appreciation of the deutschemark in effective terms; business confidence remains high in both parts of the country, and strong incoming orders reflect rising demand in Europe, which has offset a declining tendency from the Asian region. Of even greater significance has been the rise in domestic orders, reflecting not only an increase in investment activity, but also a tentative recovery in consumption (especially with respect to autos). In the course of the year, real disposable incomes should begin to expand following the period of fiscal retrenchment and this should allow a gradual pick-up in personal consumption. At the same time, with the outlook for inflation favourable in the Euro area, interest rates should remain at low levels, so that monetary conditions

should be expansionary. GDP could grow by around 2¾ per cent in 1998 with some acceleration in 1999 to around 3 per cent, since many of the favourable factors will impart a strong momentum to the economy. With employment already showing signs of recovering in western Germany, an improvement in labour market conditions should lead to a decline in the aggregate unemployment rate. But the projected output growth would not be strong enough for this to be very marked, nor to achieve a satisfactory rate of convergence in the eastern Länder.

Weakness over recent years reflects substantial structural adjustment...

A feature of the economy in recent years has been the weak performance of investment and employment, with increases of labour productivity in manufacturing of around 8 per cent in each of the past two years being associated with a major shakeout of employment in that sector. Both employment and investment have been much weaker than model forecasts based on past behaviour would have suggested. At the same time, patent activity remains high, and anecdotal evidence points to significant restructuring across industry together with a greater emphasis on product innovation and a recovery of R&D activity. Taken together, the situation suggests that during the current business cycle the story is not simply one of factor substitution, but rather of a substantial increase of total factor productivity. This has been driven by the need to preserve competitiveness and to raise profitability and has been facilitated by important structural reforms – notably with respect to greater work-time flexibility and tighter sick leave provisions which have reduced labour costs and allowed capital and labour to be used more flexibly. Employment in telecommunications, energy, mining, transport and construction have all registered declines, in part due to changes in regulatory regimes designed to promote competition or to reduce subsidies and support levels. Such structural reform is necessary to enhance growth and employment prospects in the longer run.

... which, if maintained, would lead to higher growth and employment in the medium term

In assessing medium term prospects, the important questions are whether this rapid rise in productivity represents a one-off or a trend increase in the potential growth rate due to structural reform and how this will affect the structural rate of unemployment. It is too early at this stage to arrive at definite conclusions, but it seems likely that total factor productivity growth is set to remain strong for some time. OECD estimates based on filtering productivity data to derive underlying trends place potential output growth at around 2½ per cent currently after 2¼ per cent throughout most of the 1990s. Higher productivity growth could serve to reduce the structural rate of unemployment – as defined by the non-accelerating wage rate of unemployment (NAWRU) – if the productivity increase is not bid away in the form of higher wages. Hence the OECD's existing estimate of a structural rate of unemployment of around 9½ per cent might now be on the high side. Again, it is too early to say whether labour market flexibility has changed wage behaviour to the extent that it will remain moderate once employment starts to recover, but the evidence presented in this *Survey* permits a certain optimism that structural reforms may have begun to reduce the natural rate of unemployment. Given the excessive level of wages relative to productivity in the new Länder, growth is still too biased towards capital/labour substitution, so that the prospects for rapid convergence are less bright. Labour-market institutions in the east are adapting, but the pace of adjustment is insufficient relative to the scale of the problems faced.

Monetary conditions have been expansionary and long-term interest rates have fallen

Since the previous review, monetary conditions have constituted a favourable background both for stabilising inflation expectations and for supporting growth. Policy rates were edged upwards by 30 basis points in October 1997, the Bundesbank believing at the time that inflation prospects were worsening with the weakness of the

deutschemark against the dollar. The money supply was also growing rapidly, prompting the Bundesbank to seek some slowing. Since the intervention, monetary conditions have been more stable. Against the background of financial turbulence in Asia, long-term interest rates have fallen to under 5 per cent and the yield curve has flattened. While the sharp decline represents to some extent a ''flight to quality'' in the wake of the Asian turbulence, the favourable inflation climate has also been a factor: once allowance is made for measurement problems, inflation has fallen to rates equivalent to price stability. Although part of the decline is due to a pronounced fall in raw material prices, which might be reversed at some stage, a great deal is also due to the fact that wage rises have been moderate and unit labour costs have been falling. Bond market confidence has risen to the extent that the real rate of interest has declined to levels which have previously only been seen temporarily in conditions of accelerating inflation.

Only marginal changes in the policy stance should prove necessary in the run up to EMU

In the course of 1998, German monetary policy will be increasingly driven by European considerations and by the concern on the part of the Bundesbank to hand over to the new European Central Bank at the start of 1999 a monetary policy stance which is adapted to the EMU area at large. From the European viewpoint, the process of converging money-market and policy interest rates could result in a monetary stimulus in the former high inflation countries which would push up rates in general. Indeed, forward rates had originally priced in such a judgement. However, the expected level of rates has fallen since the beginning of 1998 to levels closer to the core countries, indicating that inflation expectations might be converging in Europe to levels which would allow interest rates in the Euro area to settle near to German rates. Bundesbank actions have helped to condition such expectations and to send a signal in favour of continuity, the target growth band for M3

being lowered by ½ percentage point to 3 to 6 per cent for 1998. With activity projected to pick up in the Euro area, some further increases in policy rates by the Bundesbank could be required in the course of the year, but they should not need to be large. Any such increases should still leave the monetary policy stance broadly appropriate for German needs.

Having met the Maastricht budget criterion, fiscal policy is now entering a more difficult phase of consolidation...

In spite of weaker-than-expected tax revenues, fiscal policy was successful in lowering the general government budget deficit to 2.7 per cent of GDP in 1997, below the 3 per cent Maastricht ceiling. Severe curbs on expenditure proved necessary, especially with respect to investment, while the overall budget position benefited from a substantial reversal in the finances of the social security funds, which started to rebuild their legally-required reserves. The challenge is now to place these gains on a sustainable basis and to continue with consolidation in order to achieve the government's objective of a leaner and more effective public sector – the government aims to lower expenditures in relation to GDP to pre-reunification levels – and reduce the burden of taxation. Fiscal plans are currently being framed against this background and the requirements of the Stability and Growth Pact, although fiscal consolidation is set to slow in 1998, with both the authorities and the OECD projecting a decline in the deficit of only around a quarter point to 2½ per cent of GDP. Thereafter the convergence plan presented under the Stability and Growth Pact projects a reduction in the deficit to 1½ per cent of GDP in 2000. Improvements in the primary surplus would be sufficient to reduce gross debt to just under 60 per cent of GDP in 2001 under the assumptions adopted. However, the pace of adjustment is rather slow during 1998 and 1999, when the deficit is not scheduled to fall below 2 per cent in spite of favourable growth prospects. This could still be too high to avoid breaching the 3 per cent deficit ceiling specified

under the Pact if the economy were to experience an unexpected cyclical slowdown of the average magnitude experienced in the past – or to avoid the pro-cyclical fiscal restraint which would be needed to prevent such a breach.

... and progress towards medium-term goals requires further action in tax reform and in improving public sector efficiency

For the current year, it will be crucial for fiscal policy to achieve the planned moderate reduction in the deficit and this will not be easy given the rising public pressure for more expenditures and the fact that, after three years, the effectiveness of spending caps might have been eroded. Furthermore, in 1999 and beyond additional action is clearly required to achieve the necessary sustainable consolidation. This will require action in a number of areas.

- First, there is an urgent need for a comprehensive tax reform which would, *inter alia*, significantly reduce tax expenditures and other preferential tax treatment. Although there has been welcome progress in some areas of tax policy, and the tax/GDP ratio has fallen, the reduction has resulted from a further erosion of the tax base, rather than a lowering of rates so that fiscal pressure remains very high. Marginal rates need to be reduced – including for higher incomes where tax avoidance is particularly profitable – and the tax base needs to be widened significantly for both corporate and income tax.
- Second, social spending needs to be kept under close control. A great deal has already been achieved during the past year in the area of health care, but to translate short term cost savings into longer term gains will require giving the statutory health funds – as well as the private funds – greater freedom to deal directly with health suppliers.
- Third, further steps will be necessary in the pension reform process and to this end greater attention will need to be given to expanding the role of supplementary private coverage. Important advances have been achieved in the area of pensions with the introduction of an

adjustment for life expectancy in the annual pension adjustment formula. This will have the effect of lowering the average replacement rate and result in a significant easing of the fiscal burden. However, the reform does not suffice on its own to cope with the fiscal burden of population ageing and burdens have been raised by a more generous treatment of time spent outside the work force caring for children.

- Fourth, continuing efforts are necessary at all levels of government – and especially in the new states – to improve public sector efficiency. This is not simply a matter of improving budget procedures and cutting employment, in which areas some success has already been achieved, but also of examining the efficiency of the regulatory structure and identifying those organisational features of government which distort allocational incentives at all levels of the German federal system.

Greater public sector efficiency requires a restructuring of federal fiscal relations

The current federal system appears to provide significant disincentives for lower levels of government to follow policies which would be efficient from a country-wide viewpoint. The constitutional objective that living conditions should be broadly equalised throughout the federation is currently implemented through a complex fiscal equalisation system. In order to equalise revenues per capita in all parts of the country, sums are transferred from the wealthier to the poorer states and federal transfers are also directed towards states with below average per capita revenues. These transfers are made on an annual basis. The immediate consequence of the system is that the marginal effective tax retention facing states which generate extra tax revenues is confiscatory: an additional DM 1 million in tax revenues generates only between DM 80 000 and DM 290 000 in extra revenues after fiscal equalisation. Such high marginal rates may be expected to lead to inefficient decisions with respect to attempting to increase the

state's own tax base. Crucial to any reform is the political requirement to continue transitional support to the new states. One system which would meet present ambitious social and political objectives, while providing better incentives to improve economic performance, would be to allocate lump-sum payments over a multi-annual period to poorer states with only a small redistribution of tax revenues, at low rates, on the basis of financial capacity. Exploring the broader options in this direction remains a priority.

The Stability and Growth Pact requires a domestic equivalent, which distributes binding budget caps across all levels of government

The momentum for reforming federal/state fiscal relations has increased with the adoption of the Stability and Growth Pact, which requires the setting up of a domestic equivalent. Negotiations are moving in the direction of allocating binding budget caps across all levels of government if Germany were to be subject to the excessive budget deficit procedure under the Maastricht Treaty, with deficit violations to be sanctioned by fines. Such an agreement will change the current federal system importantly. For sanctions to be effective, the implied fiscal burden should not trigger burden sharing or equalising transfers from other governments. At present, two states receive federal supplementary transfers for the purpose of debt service, and legal conditions would need to be created which prevent any bailing out. If bailing out in one form or another is excluded, the Länder's capability to observe the pact will depend crucially on their scope to vary their revenues and spending. These are at present rather limited.

Deeper reforms covering the allocation of taxing powers and expenditure responsibilities will also have to be considered

Although the present federal system was originally based on the principle of subsidiarity, allowing some fiscal independence to the individual states, in reality the constitutional objective of broadly equalising living conditions has meant that the system has evolved into a highly co-operative union. States have little individual tax autonomy, though they do wield a powerful collective fiscal influence. Taxes are mainly shared and responsibilities for expenditures are diffuse. Many projects are jointly financed, reducing the link between expenditure decisions and their budgetary consequences in a way which is usually considered inefficient for effective decision making. Pressures to reduce the size of government and make it more efficient suggest the need to give states greater control of their finances. Experience in Germany and in other countries would indicate some priority areas in which a start could be made. Jointly-financed programmes could be limited to those which address potential externalities between states. In addition, consideration needs to be given to introducing an element of tax autonomy while preserving administrative simplicity. This could be achieved with respect to income tax by allowing for a Länder specific surcharge within the context of a tax reform which reduced overall rates while maintaining a uniform tax code throughout the country. Tax competition would imply downward pressure on rates, in contrast to the inefficient competition which characterises the present system, in which non-transparent subsidies and other support are often used to attract enterprises.

Labour-market reform is increasingly effective, but needs to be further pursued

With the public sector shrinking slowly and manufacturing growth reliant on productivity gains, the present pattern of growth places a heavy emphasis on reallocating resources toward the private service sector. This may require further structural measures in both the labour and product markets, building on the significant progress which has been

achieved over the last two years. In the labour market, reforms have facilitated and encouraged changed labour market behaviour and, as noted above, there are grounds for supposing that economic potential might have risen in recent years. But for this to be translated into a permanent decline in the rate of unemployment will require continued flexible labour market behaviour. To ensure that flexibility is preserved and enhanced there is a need to underpin labour market reform in several areas. The most difficult and sensitive area concerns the wage determination framework. Here the authorities need to support the ongoing negotiations to reform the branch wage system. At the same time, the authorities could strengthen the role of work's councils as contracting parties in enterprises not subject to collective agreements, while also defining more realistically the principle that collective agreements can only be altered in the worker's favour. This has been taken by the courts to mean higher wages or reduced work time and is ill-suited to current conditions in which wage moderation has often been agreed in return for employment security. Minor jobs (jobs entailing a limited number of hours per week), which are free of social security contributions, need to be maintained and should not be made subject to arbitrary ceilings on a company or sectoral basis. Finally, to encourage those at low wage levels or on social assistance, consideration needs to be given to introducing in-work benefits or greater earnings disregards. However, such a reform would only be sustainable if wage negotiations and labour-market developments resulted in a wider dispersion of wages at the lower end of the income scale.

Active labour-market measures need to be reappraised

While labour-market policies have succeeded in improving flexibility, developments have been less positive in active labour market policy (ALM) and in training. In response to assessments that active labour market measures (job creation and retraining) were not providing an effective bridge

back into normal employment (especially in the east), policy was changed at the start of 1997 to emphasise this bridging function better. Initial difficulties were experienced in implementing the policy, and at the start of 1998 regulations were changed, once again giving communes wide powers to set up job-creation projects. Funds were increased with the objective of raising participation in the short term by some 200 000 persons. This policy shift, while leading to a decline in the headline unemployment rate, risks making the programmes a permanent feature in eastern Germany, which would be detrimental to the development of the primary labour market. Similarly, a new initiative to encourage communes to place recipients of social assistance in work is welcome in principle, but runs the danger of interacting with the changed ALM measures to crowd-out the primary labour market, while requalifying people for unemployment benefits: simply shifting recipients from one category to another via a short spell of work.

Much has been achieved in improving skills and dynamism in product markets but barriers remain in some areas

To solve fully the underlying issues leading to unemployment requires that attention be devoted to developing skills and technology and to ensuring the development of an entrepreneurial climate. To this end, the government has achieved a great deal within the framework of its 50 point programme for growth and employment announced in early 1996, but there is a need to extend policy action in several key areas:

- With respect to skills, the government needs to continue with the modernisation of the apprenticeship system and ensure that it is not unreasonably disadvantaged in comparison with the publicly-financed full-time education system. At present, the publicly-financed education system is free and offers a less-costly route for firms to recruit. Preserving the incentives for taking on apprentices might call for the introduction of fees as part of a wider reform of the higher education system. The

introduction of a tax on enterprises which do not offer training, which would be accompanied by subsidies for those who do, risks leading to labour market distortions and should not be pursued. To ensure that all school leavers obtain apprenticeships, the federal government has started to link the awarding of procurement contracts to the training record of the company and has increased its own training. As there is no guarantee that such new places – and in particular the skills which are to be created – are in fact required, this policy threatens simply to put off the labour market problem to the future when apprentices might be looking for posts with skills not in demand. Moreover, the policy raises objections from the point of view of competition policy.

- To encourage an entrepreneurial climate, the regulatory system needs to be simplified and entry barriers reduced, particularly in the service sector where the skilled-trades law *(Handwerksordnung)* is important. Resort to start-up subsidies needs to be kept under close control, since this will only extend dependence on the state and is no effective substitute for well-functioning financial markets.
- Significant progress has been made in opening up financial markets and in reforming corporate governance to improve transparency, but barriers to their development, particularly with respect to the development of pension funds, need to be lifted.
- While some success has been registered by the federal government in reducing explicit subsidies, low tax revenues indicate that tax expenditures in fact remain high, and the Länder have been active in increasing support programmes. More determined action to reduce subsidies and tax expenditures needs to be undertaken perhaps as part of a much needed tax reform. Action will also be required, however, at state level.

Summing up

In sum, Germany has been passing through a period of extraordinary change, arising from reunification, the transformation of eastern and central Europe, the rapidly evolving world economy, the completion of the European single market and now the advent of the common currency, the Euro. These external changes have placed – and will continue to place – great pressure on German institutions, both governmental and private, requiring increased flexibility and adaptation to the newly-evolving circumstances. Within this context, important advances have been made in improving the framework conditions for the labour market, the markets for goods and services and financial markets, including the system of corporate governance. In response, companies have been undergoing major restructuring, and the social partners are in the process of creating new and more flexible labour market institutions. The role for policy continues to be to underpin these advances, both through institutional change and through appropriate macroeconomic policy. However, the tax system remains in need of reform, legislative and regulatory processes remain cumbersome and inefficient inter-governmental relations within the federal system act to impede economic development. The catch-up process in eastern Germany is proving painful and slow. While productive potential in Germany as a whole should have been raised by the ongoing structural change, which is leading to greater competitiveness, sustained employment-creating growth will require further action to fill in the gaps in the reform programme, giving scope for a reallocation of resources to a modern service sector. If this is carried through, the German economy should experience more balanced, employment-creating growth, which would be translated into lower unemployment and higher incomes.

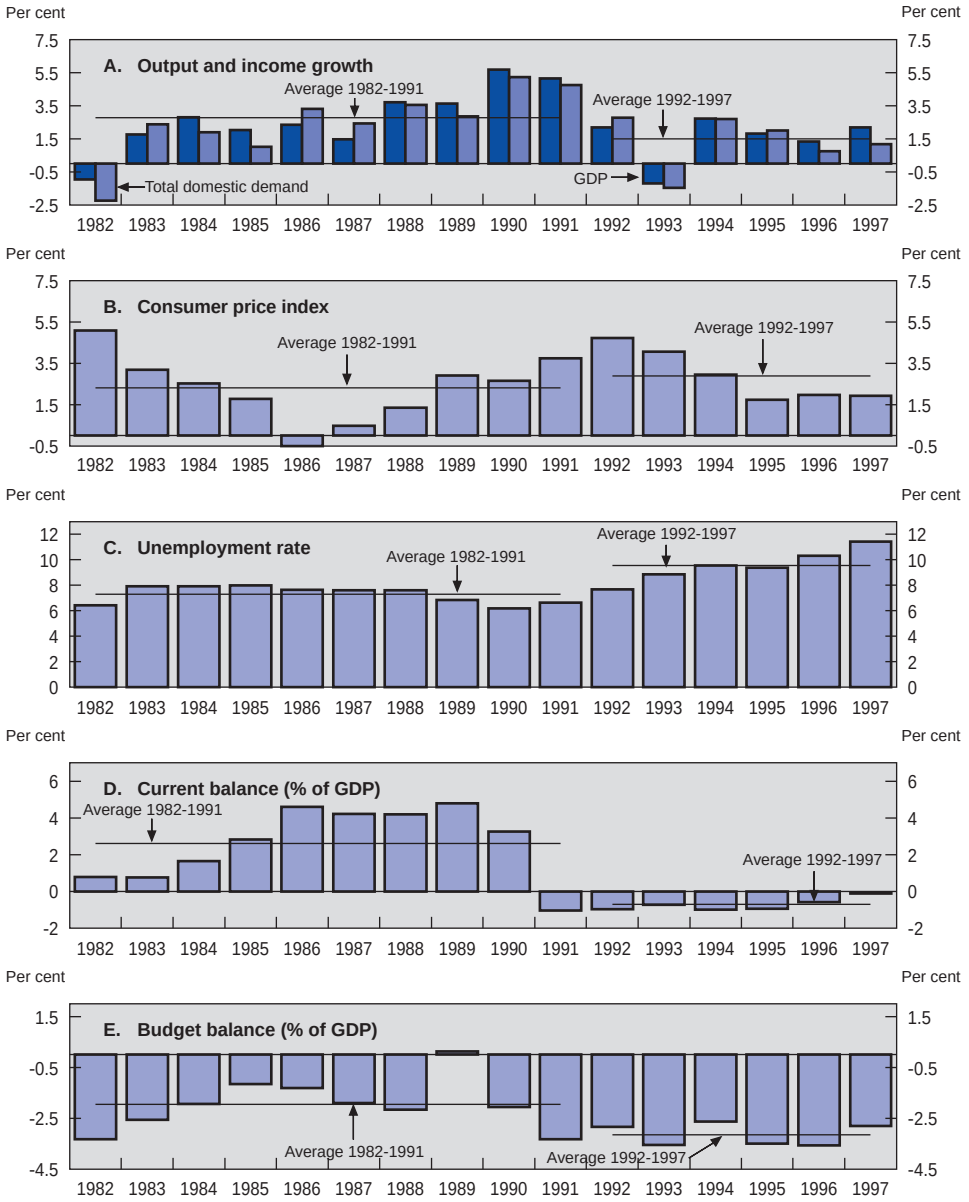
I. Establishing the basis for longer-run growth

Overview

Against the background of easing monetary conditions and a commitment to fiscal consolidation, business confidence began to recover in 1996 and into 1997, when GDP expanded by some 2¼ per cent (Figure 1). The pick-up was based largely on export growth, consumption and investment remaining depressed (Table 1). Employment continued to decline, with unemployment (not seasonally-adjusted) reaching the record level of 12.6 per cent at the start of 1998. The process of convergence in the eastern Länder stalled in 1997, as GDP grew by only 1½ per cent: productivity in industry (excluding construction) rose by 12½ per cent, while employment continued to shrink. The success of the fiscal consolidation process, continued price stability and an expansionary monetary policy stance now constitute a favourable macroeconomic framework for pan-German growth in the coming two years. Activity is projected to accelerate somewhat in 1998 and again in 1999, broadening gradually as improving real incomes and rising investment in machinery and equipment compensate for slower, but still substantial, export growth; the improved international competitiveness of Germany has not been significantly affected by the new constellation of exchange rates in Asia. The labour market is projected to recover slowly, with unemployment falling back, though it will remain high throughout this year and next. However, growth in the new Länder is not expected to be greater than in the old states so that there will be only limited progress in the convergence of economic performance.

While macroeconomic policy settings have been playing a crucial role in these developments, structural factors are important in explaining why employment and investment have both been surprisingly weak compared with previous upturns. The economy has been going through a period of marked structural

Figure 1. **MACROECONOMIC PERFORMANCE**[1]



1. Western Germany up to 1991.
Source: OECD.

Table 1. **Demand and output**

Percentage change from previous period, 1991 prices

	1994	1995	1996	1997	1998	1999
Private consumption	1.2	1.8	1.4	0.2	1.4	2.5
Government consumption	2.1	2.0	2.6	–0.4	0.9	1.2
Total fixed investment	3.5	0.8	–1.2	0.2	2.6	4.4
Construction	6.5	0.3	–3.1	–2.2	–0.6	2.2
Business	2.3	–0.3	–7.0	–2.2	. .	. .
Residential	11.6	2.6	0.1	–0.3	. .	. .
Government	0.2	–5.9	–6.1	–8.9	. .	. .
Machinery and equipment	–1.0	1.6	1.9	3.9	7.2	7.3
Stockbuilding [1]	0.8	0.4	–0.3	1.1	0.5	0.1
Total domestic demand	**2.7**	**2.0**	**0.8**	**1.2**	**2.1**	**2.7**
Net exports [1]	–0.0	–0.2	0.6	1.0	0.6	0.2
Exports of goods and services	7.9	6.6	5.1	10.7	9.3	7.1
Imports of goods and services	7.8	7.3	2.8	7.0	7.5	6.8
GDP at market prices	**2.7**	**1.8**	**1.4**	**2.2**	**2.7**	**2.9**
Memorandum items:						
Total employment	–0.7	–0.4	–1.3	–1.4	0.1	0.8
Household savings rate	11.6	11.3	11.4	10.9	11.1	11.0
Relative ULC in manufacturing	1.0	7.4	–2.8	–7.9	. .	. .
Export market growth in manufacturing	10.6	9.9	7.5	9.7	8.1	7.3
GDP deflator	2.4	2.1	1.0	0.6	0.9	1.3
Private consumption deflator	3.0	1.7	2.0	1.9	1.5	1.6
Unemployment (in millions)	3.70	3.61	3.97	4.38	4.41	4.25
Unemployment (per cent of labour force)	9.6	9.4	10.3	11.4	11.5	11.1
Current account (per cent of GDP)	–1.0	–1.0	–0.6	0.0	0.4	0.7

1. Contributions to changes in GDP (as a percentage of real GDP in the previous period).

Source: Statistisches Bundesamt; OECD.

adjustment, which appears to reflect a complex set of responses both to domestic and international competition and to structural policy initiatives, which have changed the framework of goods and factor markets in important respects (Chapter IV). Coming at a time when they are making efforts to raise productivity and lower costs, firms have been benefiting from structural reforms designed to increase flexibility, but whose transitional impact has been to magnify or accelerate the adjustment. The question now at issue is whether the reforms and the changed labour market behaviour are sufficient to result in permanently higher growth and a marked decline in the level of unemployment. After reviewing recent macroeconomic developments and short-term prospects, the chapter focuses on some of principal structural and microeconomic forces which may be at work and their prospective impact in the medium term.

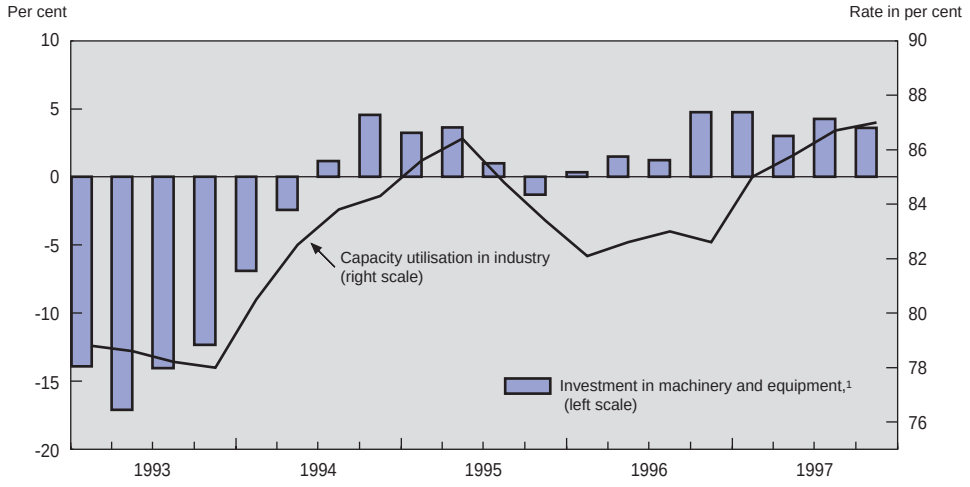
Improving macroeconomic fundamentals

External strength and domestic demand weakness

GDP grew by some 2¼ per cent in 1997, the pick-up in growth being driven by exports, leading to a marked recovery in industrial production and to a rise in capacity utilisation, with a gradual firming of investment in machinery and equipment (Figure 2). More recently, export growth, while remaining strong, has slowed to more sustainable rates, and investment in machinery and equipment has strengthened. However, a survey conducted by the Chamber of Commerce in February 1998 confirms that, at the beginning of the year, business conditions differed markedly across sectors. Companies which were more dependent on the domestic market assessed their current conditions less positively or weakly, while export-oriented enterprises reported improving business conditions and profits.

The strong growth in export volumes has been underpinned both by improved international competitiveness and buoyant market growth. The growth of export volumes in 1997, at 10½ per cent, was around 1 percentage point higher than the rate at which world trade has expanded. Virtually all sectors of

Figure 2. **CAPACITY UTILISATION AND INVESTMENT**



1. In 1991 prices. Percentage change over one year.
Source: IFO Institut; OECD.

German industry have gained from the strong foreign demand, although manufacturers of producer-related goods, which increased their exports by around 12½ per cent, benefited most. The German motor industry was the main contributor but there was also strong demand for electrical engineering goods and office and data-processing equipment. German exporters have been successful in almost all foreign markets. While the increase in exports to other member countries of the European Union has been particularly significant, there has also been strong growth in exports to North America. In addition, there has again been a marked increase in German merchandise exports to both Central and Eastern Europe, which is now more important a market than the United States, and to the expanding economies of Latin America. Even before the onset of the turbulence in Asia, exports to that region were modest in comparison with other export destinations (Table 2).

Table 2. **Trade by region**

Percentage change from previous year

	1992	1993	1994	1995	1996	1997	Share 1997
Exports to[1]							
EU	0.7	–13.2	9.3	8.2	3.1	8.4	55.3
EFTA	–4.7	–6.1	10.6	6.8	–3.1	6.4	5.3
Central and Eastern Europe	–5.7	9.9	12.8	14.4	17.1	27.8	10.4
North America[2]	0.1	7.6	16.0	0.3	9.7	29.8	9.3
Japan	–11.3	4.9	14.0	6.3	12.5	–2.8	2.3
Asia[3]	4.3	15.4	23.2	13.0	5.3	8.3	5.4
Central and South America	5.8	6.7	10.6	8.3	4.7	28.1	2.7
Other	8.7	10.5	0.7	3.1	4.2	14.6	9.3
Total	**0.5**	**–5.8**	**10.1**	**7.7**	**4.9**	**12.5**	**100.0**
Imports from[1]							
EU	–1.0	–16.3	8.3	9.1	3.5	5.5	53.8
EFTA	1.9	–4.7	11.0	5.5	1.8	11.9	5.6
Central and Eastern Europe	1.7	–1.1	22.0	17.1	6.8	19.8	9.6
North America[2]	0.0	–5.2	9.4	4.3	7.5	17.1	8.2
Japan	–5.7	–9.1	–0.5	3.6	–5.0	8.6	4.9
Asia[3]	–5.9	4.3	8.6	7.9	–4.9	11.9	5.2
Central and South America	–5.6	–15.6	14.7	1.2	1.0	15.3	2.1
Other	–2.3	13.3	4.8	–7.7	8.4	33.4	10.6
Total	**–1.4**	**–10.1**	**8.7**	**7.0**	**3.4**	**11.2**	**100.0**

1. F.o.b., including adjustment.
2. United States and Canada.
3. Dynamic Asian economies.

Source: Deutsche Bundesbank, *Zahlungsbilanzstatistik.*

Overall, the contribution of net exports to real GDP growth amounted to 1 per cent in 1997, even though imports of goods and services increased by 7 per cent in volume. International industrial integration has resulted in a growing demand for intermediate and semi-finished products from other countries. The value of such imports rose by some 14 per cent in 1997. The pick-up in imports may have been also related to some re-stocking of raw material inventories. Notwithstanding the improvement in the trade balance, the current account still showed a slight deficit, the primary reason for which was the larger deficit on services (Table 3).

Table 3. **The current account of the balance of payments**

DM billion

	1991	1992	1993	1994	1995	1996	1997
Goods	**31.6**	**43.6**	**68.1**	**82.5**	**93.2**	**107.3**	**130.0**
Exports (f.o.b.)[1]	668.1	671.4	632.7	696.4	749.9	786.5	885.2
Imports (f.o.b.)[1]	636.6	627.9	564.5	613.9	656.7	679.2	755.2
Services	**–35.3**	**–47.4**	**–54.7**	**–64.0**	**–65.1**	**–65.5**	**–71.2**
of which: Tourism	–34.4	–39.5	–43.0	–49.3	–49.1	–50.3	–51.5
Factor income	**35.5**	**28.1**	**21.8**	**11.4**	**–1.8**	**–7.7**	**–4.2**
of which: Investment income	32.2	26.3	21.1	11.0	–0.4	–5.9	–2.4
Transfers	**–61.4**	**–54.1**	**–58.4**	**–62.9**	**–58.7**	**–54.8**	**–56.4**
of which:							
Net contribution to the EU	–21.9	–25.3	–27.3	–31.7	–30.0	–27.5	–28.4
Other public transfers	–26.8	–13.9	–15.5	–14.6	–12.1	–9.9	–10.5
Current account	**–29.6**	**–29.8**	**–23.2**	**–32.9**	**–32.4**	**–20.7**	**–1.7**
(As a per cent of GDP)	–1.0	–1.0	–0.7	–1.0	–0.9	–0.6	–0.0

1. Special trade, according to the foreign trade statistics, including supplementary trade items; imports exclude freight and maritime transport insurance costs, which are included under Services.

Source: Deutsche Bundesbank, *Zahlungsbilanzstatistik.*

Domestic demand has been significantly retarded by depressed construction activity and weak consumption. The combination of fiscal consolidation and structural reform – especially in the domain of health expenditures (Chapter II) – restricted the growth of general government spending to below 1 per cent in 1997 after adjustment for inflation and also led to a marked fall in public fixed

investment (Table 1). Private consumption stagnated in real terms (Table 1), personal disposable income being held back by the continuing decline in employment, which has restricted wage growth and eroded household confidence, as well as by the fiscal squeeze on real disposable income.

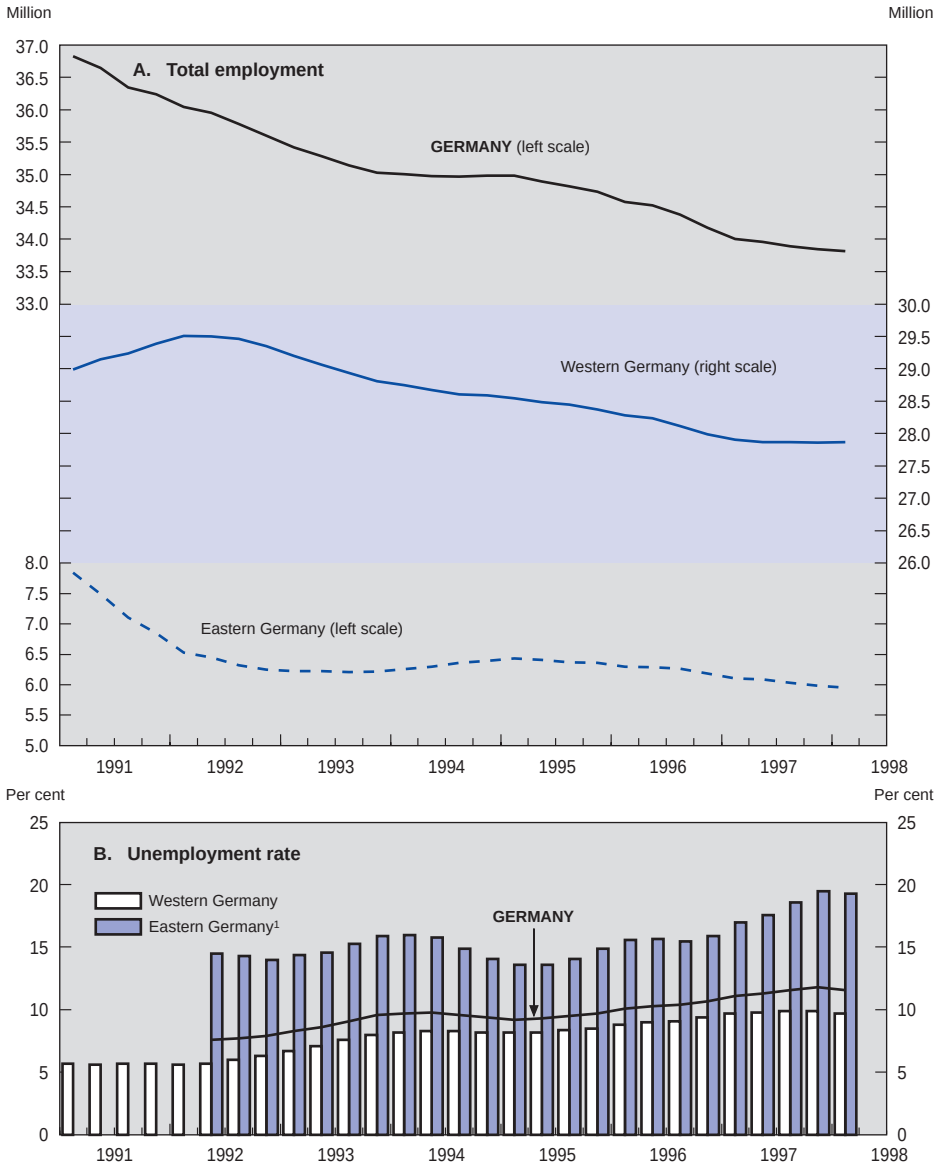
Lower employment and falling inflation

Unemployment reached an historically high level of 11.6 per cent at the start of 1998 (12.6 per cent not seasonally adjusted), due in equal measure to a continuing run-down in the numbers on job creation programmes and retraining measures, and to a further reduction of employment, especially in the troubled construction sector in the new Länder. In contrast to developments in the eastern states, the unemployment rate has been stable in western Germany since the summer of 1997 and employment has been showing signs of recovering (Figure 3). In Germany as a whole, significant increases in employment occurred in private services, in particular those oriented to business services, but these were insufficient to compensate for the adverse employment development in manufacturing, construction and in several other sectors oriented to domestic demand (Chapter IV).

The 1997 wage agreements followed the moderate course established the previous year, with an increase in wage rates of 1½ per cent in the west and just under 2 per cent in the east. Social security contributions also rose, increasing labour costs by around a further ¼ per cent. Actual wage rates, however, developed more modestly than the agreed rates, in particular in eastern Germany. Over the past few years, in response to the poor labour market situation, the trend has been for wage agreements to incorporate greater flexibility both in agreed rates of pay and in negotiated working hours (see Chapter IV for details). The introduction of a ''payment corridor'' in the west German chemical industry, which can be used from January 1998 onwards, is the most recent example of this trend. This makes it possible to lower the wages and salaries of all the employees of an enterprise up to a certain amount and for a limited period, in order to re-establish or maintain competitiveness and secure employment. Wage moderation has made an important contribution to declining unit labour costs in the business sector in both eastern and western Germany, although the greatest contribution has arisen from the marked increase in labour productivity, especially in the manufacturing sector.

Figure 3. **EMPLOYMENT AND UNEMPLOYMENT RATES**

Quarterly seasonally adjusted figures



1. Before Q2 1992 data are not available for eastern Germany.

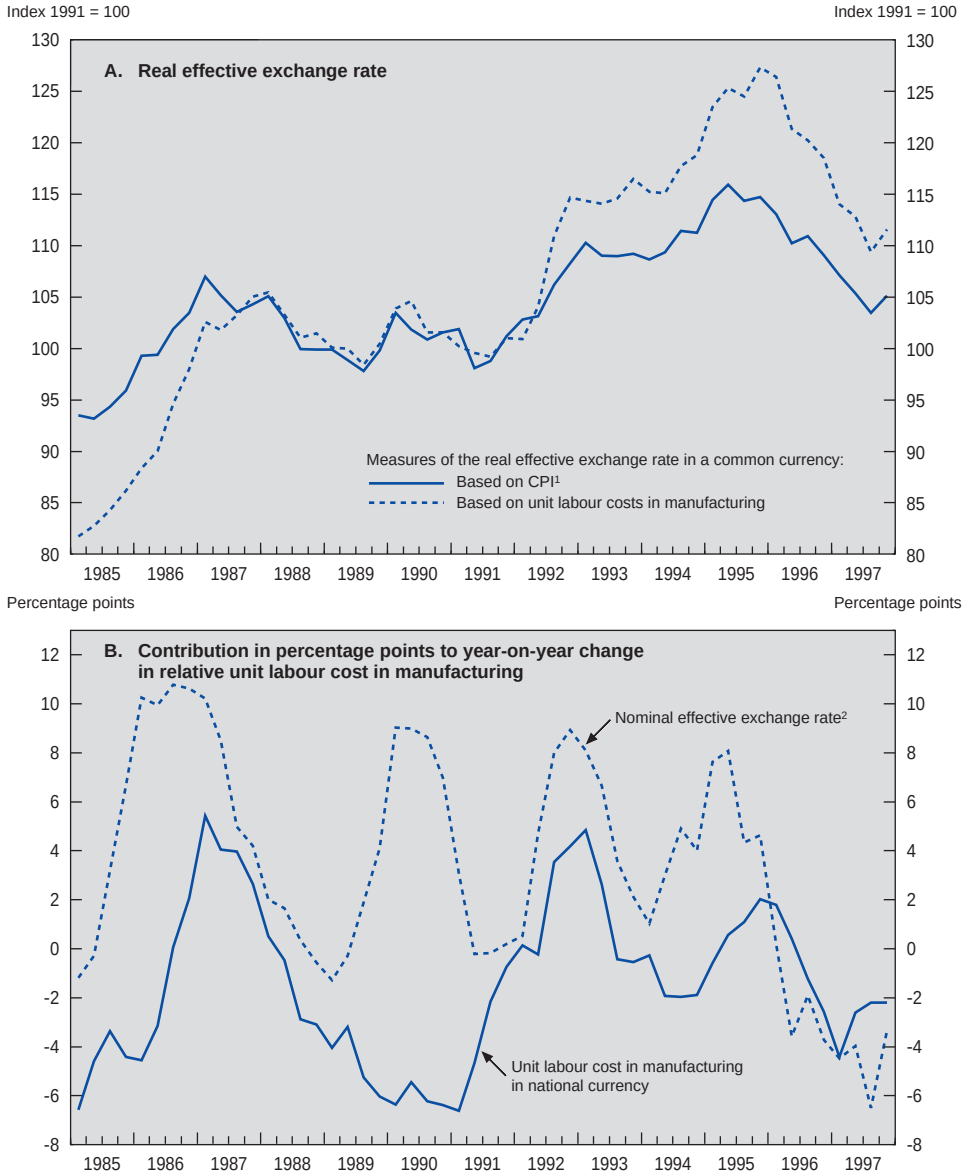
Source: Deutsche Bundesbank.

Partly as a result of wage moderation, the underlying inflation rate (*i.e.* after allowing for changes in administered prices and seasonal factors) has fallen to around 1½ per cent which, in view of likely measurement errors of around ¾ per cent per annum (Chapter II), implies virtual price stability. The consumer price index rose by 1¾ per cent in 1997, compared with 1½ per cent in 1996, but a significant part of the pick up in prices (around 0.7 percentage point) was due to changes in administered prices, including a change in the way health payments were charged. By the start of 1998, lower mineral oil prices had brought the rate of consumer inflation down to an annual rate of 1 per cent and weak commodity prices continued to dampen consumer price rises throughout the first half. Inflation picked up in April following the increase of the VAT general rate from 15 to 16 per cent, but the immediate pass-through to prices was smaller than expected. Abstracting from these temporary effects, underlying inflation probably remained around 1½ per cent. Due in part to falling commodity prices, import prices have also fallen, so that the underlying rate of producer price inflation is now around 1 per cent.

Prospects for a broadening of the expansion

Continuing strong exports and strengthening business investment in machinery and equipment should underpin growth of around 2¾ per cent in 1998 – with some acceleration in 1999 to around 3 per cent – although the contribution from net exports should decline over the period (Table 1). Export growth is being sustained by the improved competitiveness of recent years: the real effective exchange rate measured in terms of unit labour costs in manufacturing has declined by some 13 per cent since its peak in 1995 (Figure 4) and with a long-run price elasticity of exports of a little over unity, this expansionary influence should continue in the immediate future given the time lags involved. The sharp falls in the exchange rates of some Asian countries have not led to any significant appreciation of the deutschemark in effective terms on a trade-weighted basis, so that competitiveness remains strong (see below). Concern to preserve this advantage, together with market opening in protected sectors such as telecommunications and energy, will serve to place downward pressure on price inflation, while the combination of moderate wage growth and significant productivity gains should ensure further falls in unit labour costs. Despite the crisis in Asia, business sentiment regarding current and prospective conditions, which had improved

Figure 4. **INTERNATIONAL COMPETITIVENESS**

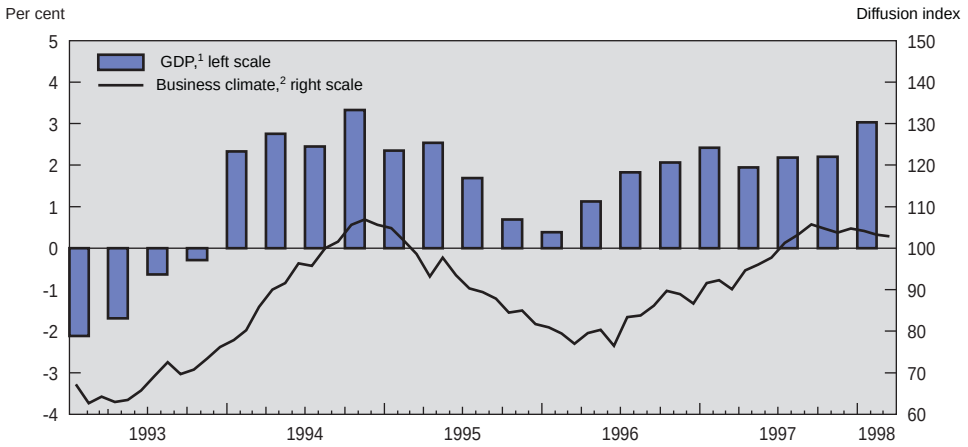


1. Weighted real external value of the DM against currencies of 34 industrial countries.
2. Weighted external value of the DM against currencies of 34 industrial countries.

Source: OECD.

throughout 1997, has remained buoyant (Figure 5). Reports from trade associations indicate that orders from Asia are declining but that this is being offset by strengthening orders from Europe and by rising domestic demand, as the prospects for a more balanced demand pattern improve.

Figure 5. **THE BUSINESS CLIMATE**



1. In 1991 prices. Percentage change over year before.
2. Western Germany. Weighted average of present and future (six months ahead) business situation. A level of 100 corresponds to "normality".

Source: IFO Institut; OECD.

Real disposable incomes should begin to expand following the period of fiscal retrenchment, leading to a gradual pick-up in personal consumption. The favourable outlook for inflation, in combination with moderate output and employment growth, is expected to result in continuing supportive monetary conditions in the immediate future. EMU convergence is assumed to result in only a limited shift of monetary policy toward a less accommodative stance (see Chapter II). Long-term interest rates have fallen substantially, partly due to influences associated with financial turbulence in Asia, and subdued inflation expectations are expected to sustain a low level of long-term rates throughout the projection period, despite a narrowing output gap. The general government deficit is projected to decline by only 0.10 percentage point of GDP in 1998 and by a further ¼ percentage point in 1999, so that the fiscal policy stance will tighten

less than in recent years (Chapter II). Employment should start to pick up in the course of the year, leading to a stabilisation and eventual decline in the unemployment rate, but the growth projected for 1998 would not be strong enough to lead to a marked improvement in labour market conditions.

There are, however, substantial uncertainties and risks associated with the projections. Instability in emerging markets, including any spreading of turbulence to eastern Europe, represents a general risk. More specific risks for the projection are associated with the run up to European Economic and Monetary Union: following decisions on membership of the Union in May, pressures could mount for increases in Bundesbank policy rates, in order to reposition the monetary policy stance from a European perspective. Wage growth in 1999 could also be stronger than projected if traditional patterns of bargaining return, which would lead to lower employment growth. While there is scope for real incomes to rise, business confidence could be adversely affected if the structural changes which have led to higher productivity and a more flexible labour market were to prove transitory. The extent to which structural factors are playing a role in the current expansion is examined in the next section.

The impact of structural adjustment

Improvements in competitiveness

The significant improvement in international competitiveness which has been experienced since 1995 has been due both to the change in the nominal effective exchange rate – an exogenous factor which could just as easily reverse in the future as it has in the past – and to a significant improvement in unit labour costs in domestic currency terms. The competitive pressure for enterprises to reduce costs has to be seen against the background of the cost disadvantage which opened up after reunification. In terms of absolute cost-competitiveness, by the early 1990s Germany had become a relatively high-cost producer as relative unit labour costs increased sharply. As a result, the business sector, and manufacturing in particular, has needed to embark on a major programme of structural adjustment in order to lower unit labour costs. The adjustment has helped to restore German price-competitiveness to its early-1990s level and its world export market share increased in 1997.

Although labour productivity has increased rapidly in the new Länder, competitiveness remains weak: in 1997, aggregate labour costs per nominal unit of value added were around one-quarter higher than in the old Länder.[1] Despite the marked overall reduction of the gap between labour costs compared with 1991, virtually no further progress has been made since 1994 in terms of this key cost variable for the competitiveness of east German enterprises. However, the situation varies significantly between sectors. In industry, unit labour costs declined by some 8 per cent in 1997 and labour costs per unit of nominal value added are now only 4 per cent higher than in western Germany – and exports have started to grow rapidly in some branches. Similarly, unit costs in the construction sector are now only 2 per cent above those in the west. On the other hand, in the service, trade and transport sectors unit labour costs are around a third higher than in the west.

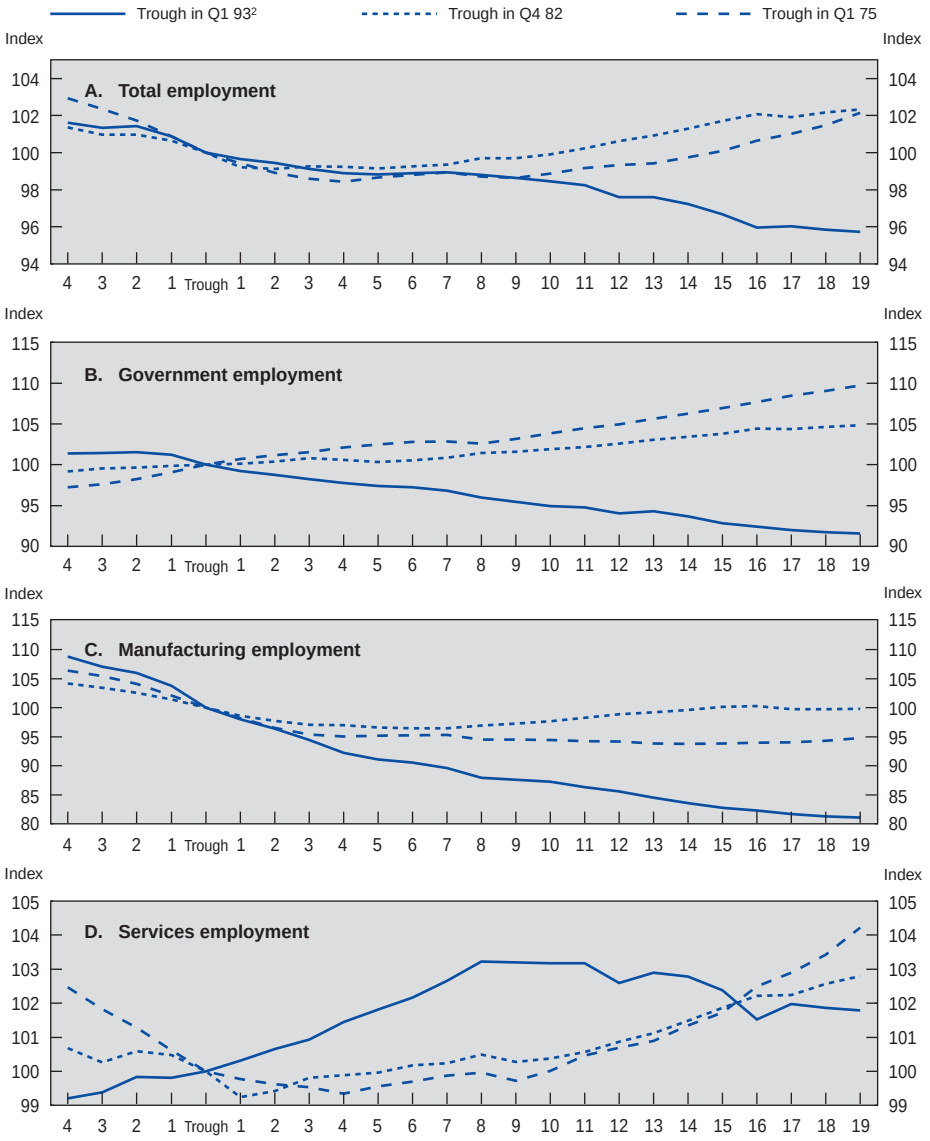
Greater efficiency in the use of capital and labour

Employment in manufacturing has fallen sharply

The substantial improvement in unit labour costs has been associated with a major shakeout of employment in the manufacturing sector, where labour productivity has risen at a rate of around 8 per cent in each of the past two years. Although reunification renders comparisons difficult, the development of employment in manufacturing appears to be quite different in comparison with previous cycles (Figure 6). Service sector employment has grown rapidly, due in part to the contracting out of business related services from the manufacturing sector, but more recently the growth rate of service sector employment has declined, so this has not been the only factor at work (Chapter IV). Econometric analysis suggests that the decline of business sector employment (*i.e.* service- plus goods-producing sectors) has been much greater than could have been expected on the basis of past behaviour, and productivity growth significantly faster (Figure 7).[2] Both have been associated with substantial innovation and restructuring at the enterprise level. Since the beginning of the 1990s firms have preferred to invest in rationalisation projects (Figure 8) and they have paid more attention to innovation, as reflected in the importance attached to product innovation in investment decisions. Capacity-widening investment has remained of little overall importance in recent years.

Figure 6. **EMPLOYMENT GROWTH IN DIFFERENT RECOVERIES**

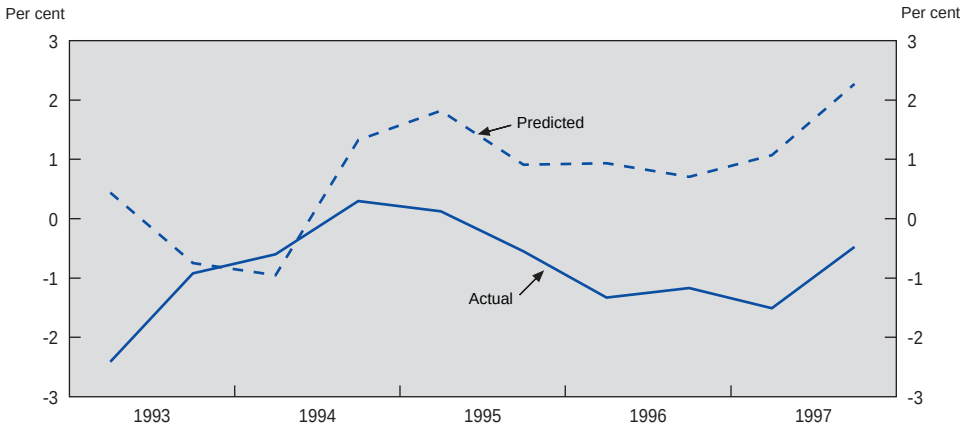
Index "trough" = 100[1]



1. The "trough" is defined as the lowest GDP growth.
2. Overall Germany. The nineteenth period corresponds to 1997 Q4.

Source: OECD.

Figure 7. **ACTUAL AND PREDICTED GROWTH OF BUSINESS SECTOR EMPLOYMENT**[1]



1. See Annex I for details of estimates and simulations.
Source: OECD.

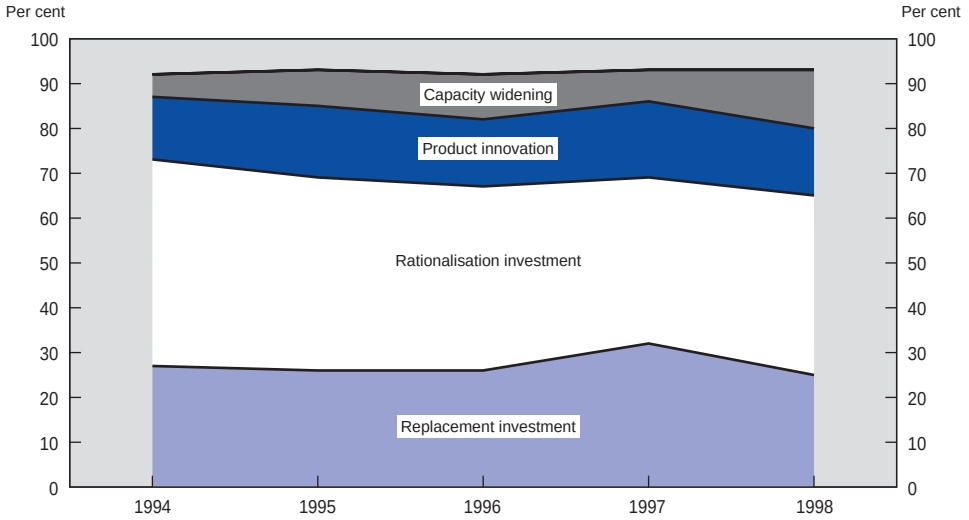
Investment is lower than in previous cycles

The changed composition of investment demand has also been associated with a low level of investment activity and, overall, investment has grown less rapidly in this recovery than during the previous ones. Figure 9 compares the current recovery with the two preceding cycles, the trough in real GDP serving as the reference point. In comparison with the past, both construction and public investment, after being relatively supportive for most of the recovery, are now declining sharply. The fall in construction investment towards more normal levels of building activity reflects the weak construction demand in the east and the cut-back in building promotion programmes. The recent fall in public investment was due to deficit-reduction measures to meet the Maastricht criteria. However, what is striking in this business cycle is the weak pattern of investment in plant and machinery.

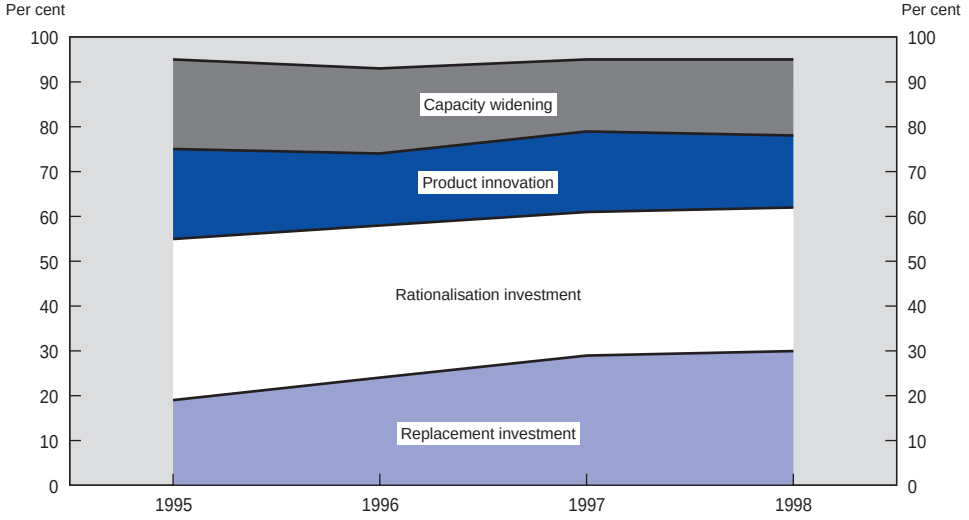
As with employment, the out-of-sample performance of the investment equation since 1995 indicates an over-prediction of capital and investment, implying that the capital stock is lower than could have been expected on the basis of past behaviour (Figure 10). Some of this apparent weakness could be due

Figure 8. **MAIN MOTIVES FOR BUSINESS FIXED INVESTMENT**[1]

A. Industries in western Germany



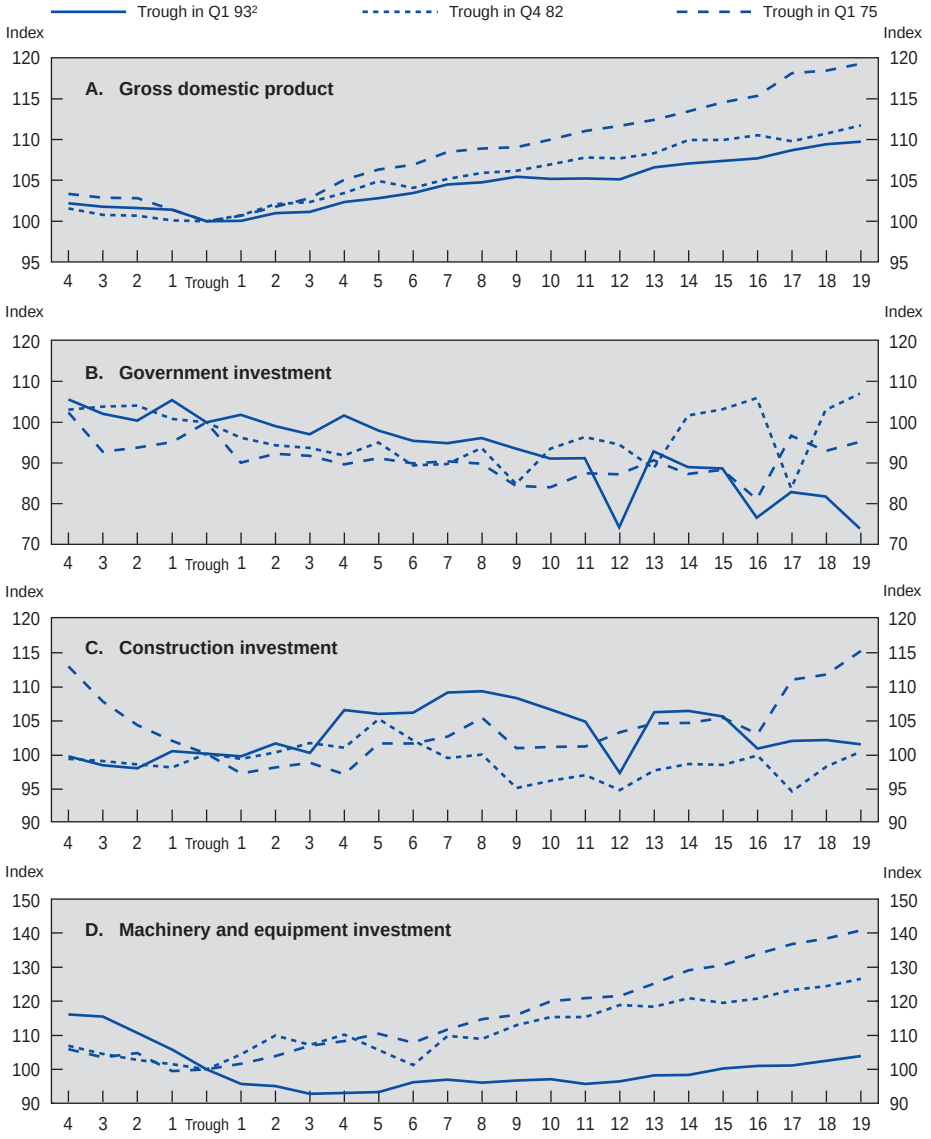
B. Industries in eastern Germany



1. Motive for investment in the reference year according to surveys conducted in autumn of previous year.
Source: Deutscher Industrie- und Handelstag.

Figure 9. **RECOVERY PERIODS COMPARED**

Index "trough" = 100[1]

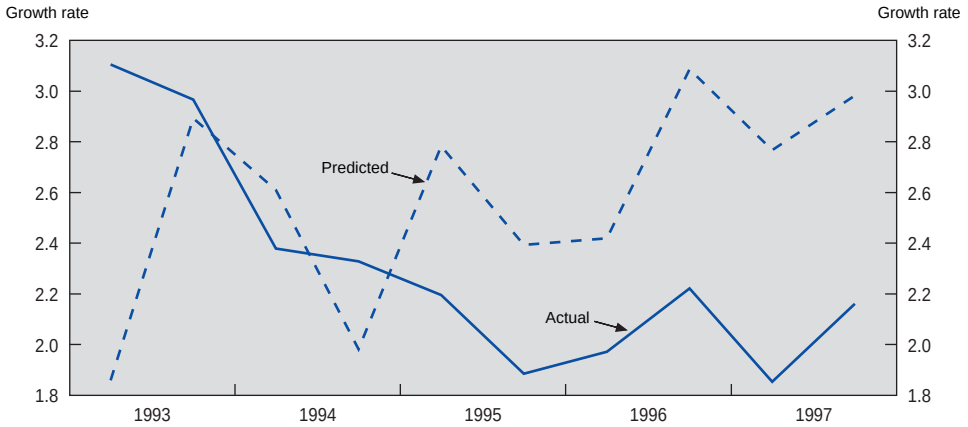


1. The "trough" is defined as the lowest GDP growth.
2. Overall Germany. The nineteenth period corresponds to 1997 Q4.

Source: OECD.

Figure 10. **ACTUAL AND PREDICTED GROWTH OF INVESTMENT IN THE BUSINESS SECTOR**[1]

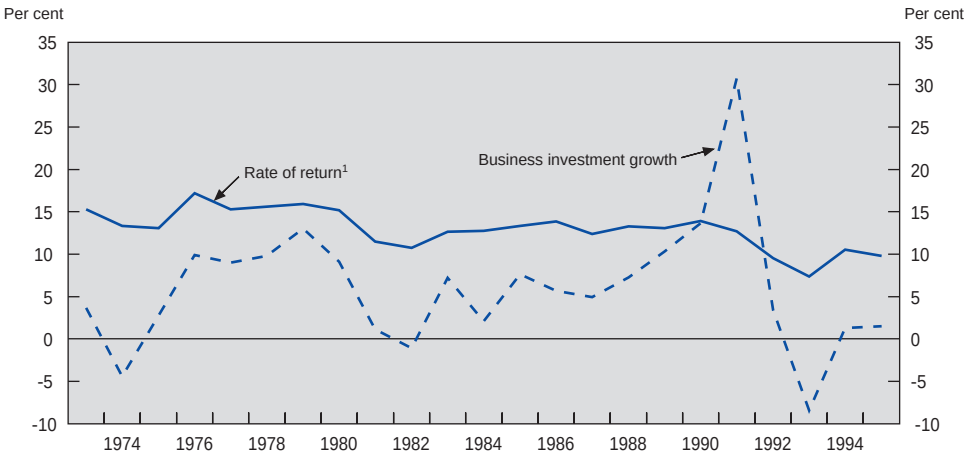
Per cent



1. See Annex I for details of estimates and simulations.
Source: OECD.

to measurement errors in the price of investment goods: prices need to be adjusted for improvements to the quality of capital goods and if prices are overstated, real investment is underestimated.[3] Moreover, costs associated with the high level of patent activity (Chapter IV) are not classified as investment. The weakness in investment cannot be due to the cost of finance, since the long-term interest rate adjusted for the rate of inflation has been well below its level of much of the 1970s and 1980s. It could be due to greater uncertainty, which has remained high during this recovery, as a result of shifts in the real effective exchange rate and the failure of the tax reform. Both may have affected expected profitability, inducing cautious behaviour by firms in taking investment decisions. Looking at profits themselves, Figure 11 shows that there has traditionally been a close relationship between the growth rate of business investment and profitability. The rate of return on capital was, indeed, rather low up to 1995. It has subsequently recovered to relatively high levels and this is a factor behind the expected pick up. It is, however, important to evaluate profit performance in the light of the fact that investment options in Germany are increasingly having to

Figure 11. **PROFITS AND GROWTH OF BUSINESS INVESTMENT**



1. Total income less total costs as a share of own funds plus provisions for pensions.
Source: Deutsche Bundesbank; OECD.

compete with options in other locations. From this perspective it is important to note that inward foreign direct investment has remained weak while outward direct investment, which amounted to some DM 57½ billion in 1997, remains similar to other countries as a proportion of GDP.

Perhaps most significantly, there is some evidence that firms have been making better use of existing capital stock via more flexible running times and this could, in the short term, both lower investment and weaken the relationship with capacity utilisation.[4] There has been a positive relationship historically between business investment intensity and capacity utilisation – and an even closer relationship with capacity-expanding motives for investment.[5] Capacity utilisation at the beginning of the present recovery was higher than at the beginning of the previous one, and with investment growing less rapidly than in previous recoveries, the same is still true. The IFO business survey indicates that capacity utilisation has increased since 1996 and is now at a level which, in the past, corresponded to higher investment growth than the one actually realised.

Total factor productivity has risen

The fact that both employment and investment equations are over-predicting suggests that during the current business cycle the story is, outside the eastern Länder, not simply one of factor substitution, but rather of increased total factor productivity. This development has been driven by the need to preserve competitiveness and increase profitability, and has been facilitated by important structural reforms which are detailed in Chapter IV. Labour markets have become more flexible, allowing machine running times to increase[6] and labour to be more flexibly and effectively utilised; overtime and sickness have both been reduced. Raising total factor productivity would have lowered investment activity in the short term while more flexible labour time and labour costs might also have lowered the demand for workers, while raising hours of work. Indeed, illustrative analysis of the macroeconomic effects of productivity shocks using the OECD's Interlink model points towards complex interactions during the adjustment phase as labour released from the manufacturing sector exceeds the capacity of the service sector to expand in the short term.[7]

Medium-term perspectives

Prospects for the medium term turn on two considerations: on the nature of the productivity boost and the extent to which it is ongoing, and on the degree to which behaviour in the labour market has changed but needs to change further. In recent years, trend (labour)[8] productivity growth has made by far the greatest contribution to underlying growth (Table 4). An important question is whether this rapid rise in productivity represents a more or less one-off increase or whether it reflects an increase in the trend growth rate of labour productivity, especially in the manufacturing sector, due to the unleashing of new dynamic forces. It is too early at this stage to arrive at a definitive judgement. On the one hand, it would seem that the process of cost-cutting has restored German competitiveness and the profitability of exports, so that, to the extent that productivity gains have come from labour-shedding, these will diminish. But the trend growth rate of labour productivity could have risen in part because of new technology, and this process is likely to continue, stimulated by international competition. Continuing adjustment of firms to new possibilities for more efficient capital utilisation means that total factor productivity growth (which is equivalent to

Table 4. **Contributions to German potential output growth**

Average annual rate

	1971/1980	1980/1990	1992/1995	1995/1999[1]
Potential output	2.5	2.2	2.3	2.3
of which (percentage points):				
Trend labour productivity	1.7	1.2	1.6	1.7
Trend hours	–0.5	–0.4	–0.2	–0.3
Capital stock	1.3	0.9	1.0	0.9
Trend participation rate	–0.1	0.1	–0.1	–0.1
Working-age population	0.3	0.5	0.2	0.2
Change of the NAWRU	–0.3	–0.2	–0.1	–0.1
Memorandum item:				
Real GDP	2.7	2.2	1.1	2.3

1. Including projections for 1998 and 1999 by the OECD.
Source: OECD.

trend labour productivity in the model of the OECD) is likely to remain strong for some time, as capital productivity continues to rise. OECD estimates which filter productivity developments for their underlying trend indicate that potential output growth could currently be in the region of 2½ per cent a year, compared with 2¼ per cent for most of the 1990s (Figure 12). If trend labour efficiency growth were to increase by 1 percentage point, the growth rate of potential would improve by 0.6 percentage point to a little over 3 per cent. If the recent productivity shift represents only a one-off improvement, the underlying growth rate would remain unchanged, but the level of potential output (together with the current output gap) would both have increased by some 0.6 percentage points.

With respect to the other key factor behind potential output growth, the growth rate of the capital stock, if this were to recover to previous levels of around 3.5 per cent per annum (1 percentage point higher than at present), the growth rate of potential output would be raised by one-third of a percentage point per annum. As cost-competitiveness has been restored, this capacity-widening component of GDP growth is projected to play a more important role. However, the key medium-term question concerns the development of (un)employment in the face of continuing productivity gains. For higher growth to translate into higher employment the functioning of labour-market institutions is crucial. The structural rate of unemployment – the non-accelerating wage rate of

Figure 12. **KEY DETERMINANTS OF POTENTIAL OUTPUT GROWTH**[1]

1. Break in 1991 for reunification after which calculations are for Germany as a whole. From 1997 data are estimates.
Source: OECD.

unemployment (NAWRU) (see Annex II) – is currently estimated by the OECD at around 9½ per cent. However, the structural rate is a function of the difference over time between real wage growth and productivity increases. Hence, if the level of potential output has risen because of higher productivity and this remains permanently unreflected in higher wages because of the increased degree of labour-market flexibility, the NAWRU would have fallen (to around 8½ per cent of the labour force on the basis of a potential output shift of ¾ per cent). The biggest underlying risk or uncertainty for the medium term is that changes in labour market behaviour could prove only transitory, and once labour market conditions start to improve, old patterns of reaction could re-emerge, with wages recouping past gains in productivity.[9] As underlined by recent research,[10] the rate of structural unemployment is influenced by a number of policy interventions, including changes in the social security system, which affect wage negotiating behaviour, and these may be needed to reinforce the measures already taken.

Special problems still attach to the new states and the process of convergence has yet to attain a satisfactory pace. Recent high labour productivity growth mainly reflects capital/labour substitution in response to the excessive level of wages rather than gains in total factor productivity, and there are few grounds for optimism that the growth rate of potential output in the medium term is either high or has risen. Although a high proportion of the capital stock is modern (50 per cent for the economy as a whole, higher than in the west), the capital/labour ratio is only about two-fifths of that in the west and incentives for capacity-widening investment are weak because profitability is lacking across wide sectors of the economy. Moreover, to a far greater extent than in the west, the ability to create employment depends upon a significant reallocation of resources: construction accounts for 15 per cent of GDP but only 5 per cent in the old Länder. The construction industry needs to decline substantially, resources being reallocated in part to manufacturing which is still relatively small (15 per cent of GDP compared with 25 per cent in the west). For such a reallocation to occur, profitability needs to rise. But while labour-market institutions in the east are adapting to this problem, the pace of adjustment is insufficient relative to the scale of the problems faced.

Overall, it would seem that the German economy is in a transitional phase towards greater efficiency and lower structural unemployment. The short-term effect has been to raise unemployment, and to affect consumer confidence

adversely. The near-term outlook is for these negative influences to dissipate gradually, with growth accelerating somewhat this year and in 1999 and domestic demand strengthening as unemployment comes down slowly. Structural reforms already undertaken have raised potential output and introduced an extra degree of flexibility into the labour market which should have lowered the structural unemployment rate. However, the unemployment rate will remain high, especially in the eastern Länder where the gap between wages and labour productivity is large, so that further structural reforms are called for. Realising the improved potential of the economy in the medium term will require, *inter alia*, resource reallocation in the economy from the industrial to the service sector. These policy issues are taken up in Chapter IV, following a discussion of monetary and fiscal issues in Chapter II.

II. Monetary and fiscal policies

Overview

The background to macroeconomic policies has been shaped to an important extent by the run-up to European Economic and Monetary Union (EMU). Fiscal policy has focused on meeting the Maastricht budget deficit criterion, which has required severe curbs on expenditure in the face of unusually weak tax revenues in 1997. German monetary policy has been supportive of growth. Worries about a weakening of the deutschemark and a changed inflation perspective prompted the Bundesbank to make a small upward adjustment to interest rates in October, but policy-controlled rates have subsequently been stable and monetary conditions have continued to provide a favourable background for the expansion. The success of German monetary policy in shaping market sentiment about the level of short-term interest rates and inflation at which EMU convergence will occur is reflected in a firming of the currency and bond markets, where yields have fallen to record low levels, assisted by a ''flight to quality'' in the wake of the Asian crisis. With inflation having settled at levels which approach price stability, monetary conditions should continue to provide support to the expansion, even if they are increasingly geared to European considerations, especially with the formal decision in May to proceed with EMU. However, fiscal policy faces more severe challenges. Further fiscal consolidation is an essential element both in meeting the requirements of the Stability and Growth Pact and in attaining the government's objectives for a leaner and more effective public sector and a more efficient tax system. Tax reform is a priority, but has so far been blocked by the Länder.

Monetary policy

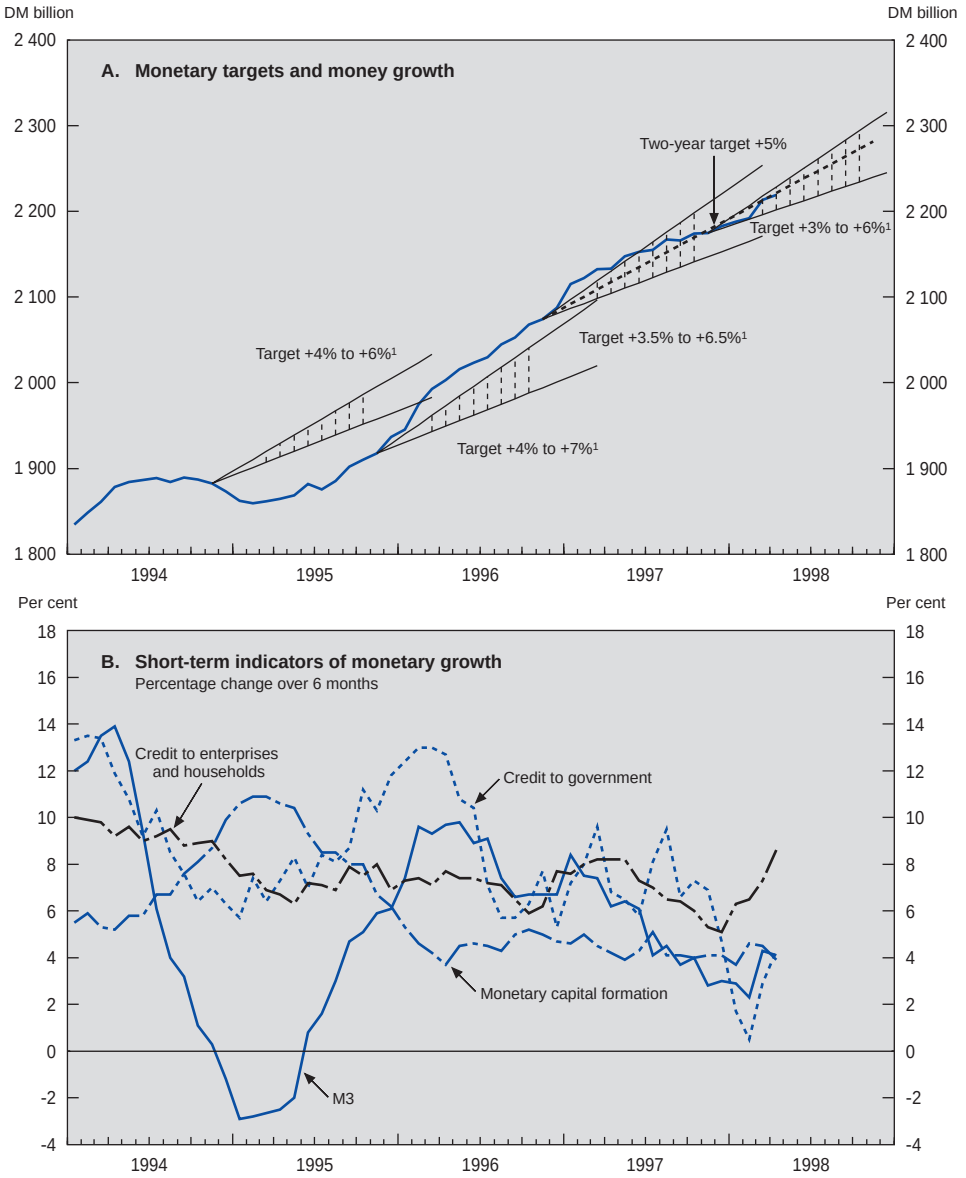
Money and credit aggregates: slowing money growth

Having expanded significantly faster than targeted in 1996 and early 1997, the Bundesbank's preferred monetary aggregate, M3, decelerated through the remainder of 1997 and into 1998, due both to slowing domestic credit expansion and to a continuing significant capital outflow. High money growth rates at the start of 1997 meant that the money supply remained above the target band of 3½ to 6½ per cent until the spring, and for 1997 as an average the 6.2 per cent expansion of M3 was faster than the Bundesbank had envisaged. But monetary growth decelerated in the course of the year. In the fourth quarter of 1997 the money stock M3 exceeded its average level of the fourth quarter of 1996 by 4.7 per cent. The monetary target was thus reached and the entry into the 1998 target corridor was fairly unproblematic. In April, M3 exceeded the level of the fourth quarter of 1997 by an annualised rate of 4.7 per cent, well within the target corridor of 3 per cent to 6 per cent (Figure 13). Credit growth decelerated in 1997 due to a slowing of bank borrowing by all major borrower groups. The growth of housing loans which had been buoyant since 1993 returned to more normal levels. Loans to the manufacturing sector were also, by and large, stagnant during 1997 despite a gradual strengthening in investment activity. However, total lending to enterprises and individuals is continuing to expand at rates much above those observed during the long period of economic growth in the 1980s, so that the rate of credit growth should not be a concern for policy. With the exception of December 1997 when net external assets of the banking system increased by some DM 43 billion due in part to turbulence in Asia, net external outflows have continued to be the significant dampening influence on the growth of domestic money supply. The high outflow of funds was primarily due to purchases of foreign securities by German investors.

Policy rates: a stability-oriented stance

Since the previous *Survey*, policy-controlled rates have changed only once, in October 1997 when the repurchase rate was raised by 30 basis points to 3.30 per cent (Figure 14). While the inflation environment has been generally benign, the move was taken against the background of a sharp depreciation of the deutschemark exchange rate to nearly DM 1.90 per dollar, and there was market

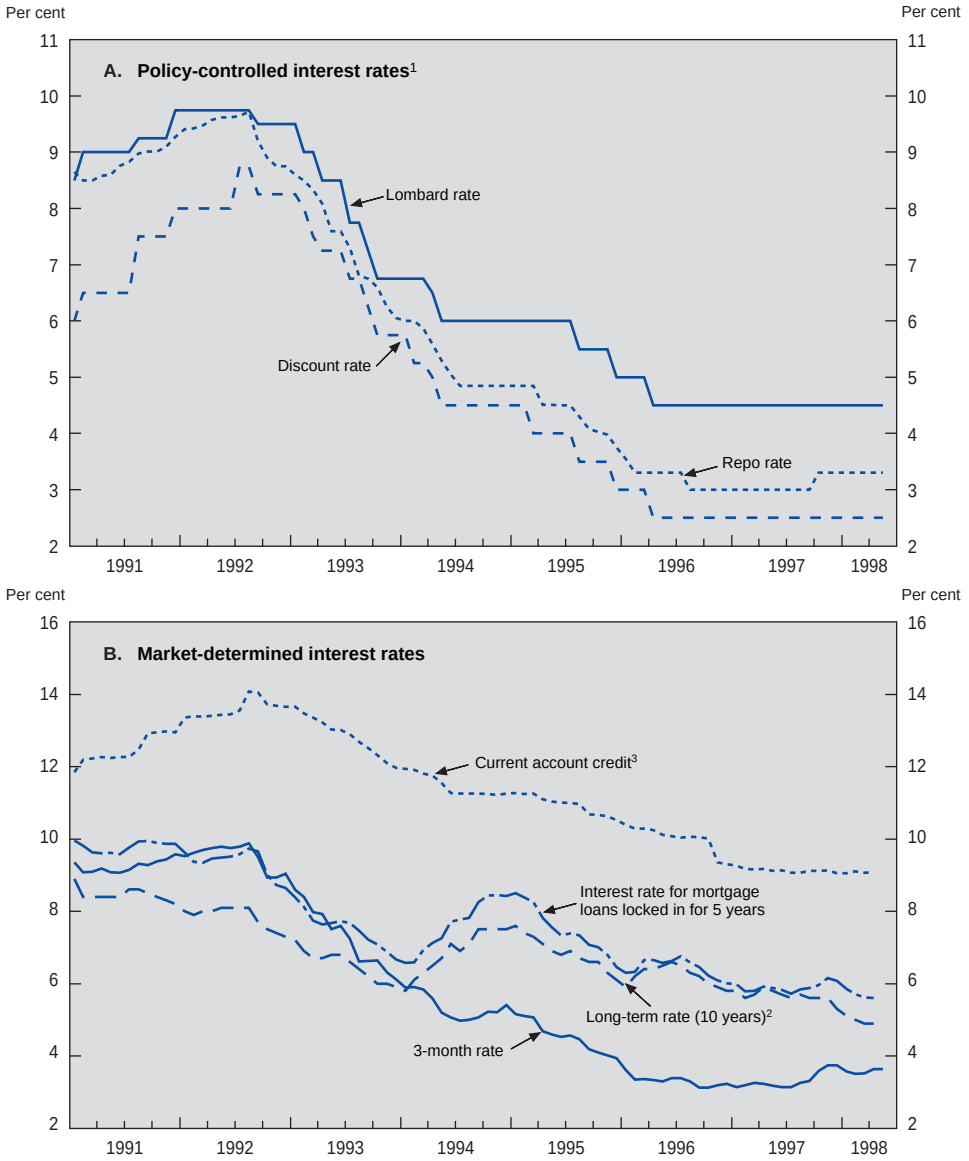
Figure 13. **MONETARY TARGETS AND MONEY GROWTH**



1. Between the 4th quarter of the preceding year and the 4th quarter of the current year. M3 is normally subject to major random fluctuations around the turn of the year.

Source: Deutsche Bundesbank.

Figure 14. **INTEREST RATE DEVELOPMENTS**



1. End of period.
2. Yield on bonds outstanding.
3. Credit of less than DM 1 million.

Source: Deutsche Bundesbank.

sentiment at the time that the rate could even go significantly above DM 2.00 per dollar. At the same time, dollar import prices were rising rapidly (Figure 15), in part as a result of prior increases in raw material prices. The consumer price index was also edging up following a sharp increase in co-payments for pharmaceuticals. With the money supply also continuing to grow rapidly, and lying in the upper half of its range, the Bundesbank warned in July that a reduction in the growth rate of M3 was desirable. Taken as a whole, the situation led the Bundesbank to state on a number of occasions that inflation had probably bottomed out. In addition, market sentiment as reflected in forward rates was indicating an expectation that short-term rates in Europe would converge at levels significantly above those prevailing in October 1997 (*i.e.* at around 4.5 per cent in comparison with an actual rate of around 3⅓ per cent).

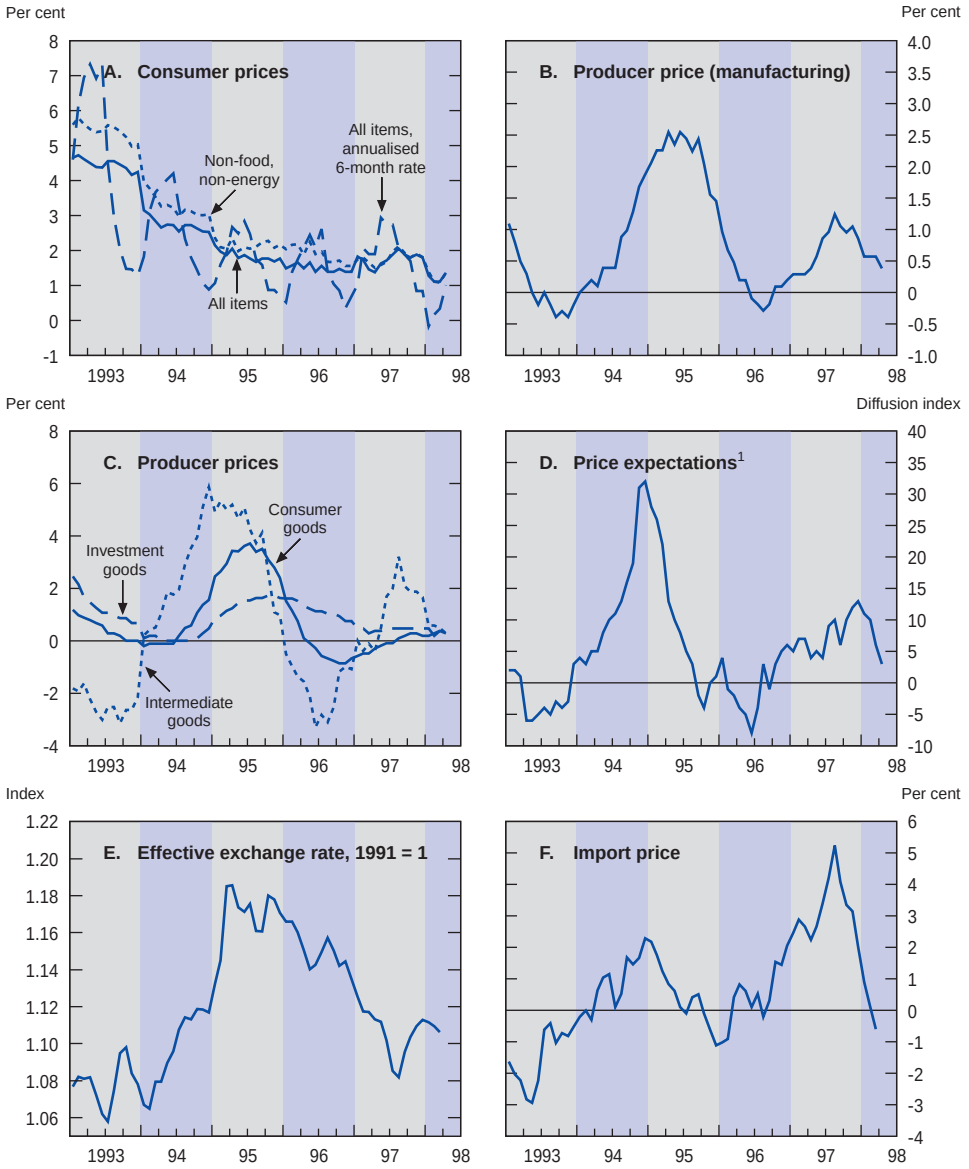
Following the interest rate move, concern about inflation subsided and policy shifted to, and has remained with, a wait-and-see stance, as the repo rate has been kept unchanged. This attitude has aimed at anchoring expectations ahead of the establishment of EMU and has been accompanied by greater market confidence that EMU convergence will not imply significant rate rises. At an operational level, to help stabilise market expectations, repurchase agreements since October have once again been conducted through a fixed-rate (''volume'') tender with pre-announced conditions. In the period leading up to the increase in the repurchase rate, the Bundesbank announced terms only on the day of tender, a move signalling a potential change in stance to the market and which had the effect of raising market rates. In general, in an environment of historically low short-term rates, the Bundesbank has been quite sensitive to the possibility that misjudgements about liquidity demand could result in a rise in the tender rate which would give misleading signals to markets, so that it has not used its once-preferred variable-rate tender since February 1996.

Long-term rates fall to low levels

During the period of monetary easing, up to early 1996, monetary policy was successful in inducing a decline in long-term rates while maintaining stable expectations as to inflation and to the future course of monetary policy. But the yield curve remained relatively steep, indicating that markets still saw potential risks for price stability in the future. Following the rise in policy rates in October, short-term market rates rose but the yield curve flattened (Figure 16, Panel A).

Figure 15. **THE INFLATION ENVIRONMENT**

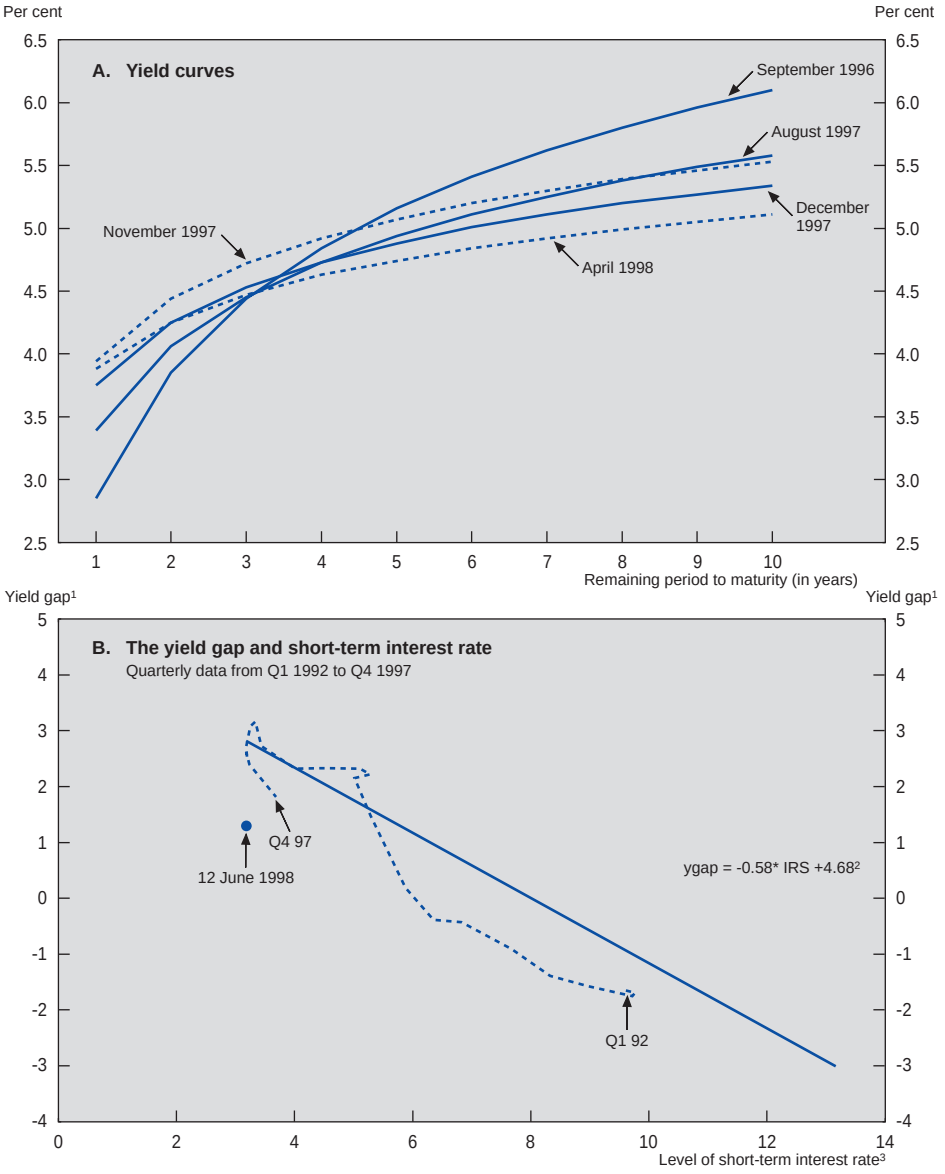
Percentage changes over 12 months



1. Diffusion index. Balance of higher and lower expectations. Seasonally adjusted.

Source: Deutsche Bundesbank, Monatsberichte; OECD, *Main Economic Indicators.*

Figure 16. **YIELD CURVES AND THE YIELD GAP**



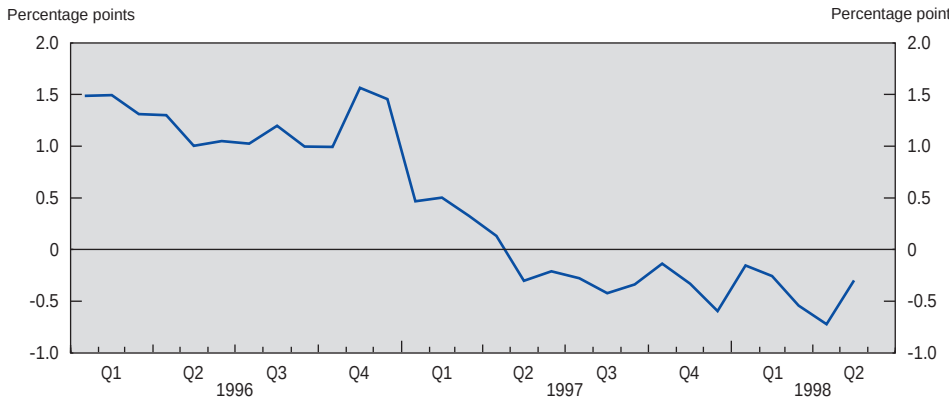
1. Yield to maturity for benchmark government bonds (10 years) minus 3-month interest rate.
2. Slope of the regression line estimated using quarterly data from Q1 1980 to Q4 1997.
3. 3-month rate.

Source: Deutsche Bundesbank; OECD.

Subsequently, there has been a marked fall in long-term rates, which have declined nearly a full percentage point to under 5 per cent (June 1998), while short-term market rates have remained relatively stable. In the process, the yield gap has narrowed (Figure 16, Panel B). A number of factors could underlie the decline in long-term rates. Inflation expectations seem to have fallen further with the decline in inflation, while the international trend in interest rates has been downward. Also, there has been a firming of financial-market expectations about the likely course of European monetary policy. The latter influence is, however, difficult to prove. Perhaps of greater importance has been the general ''flight to quality'' in the wake of turbulence in Asia. Capital inflows have been strong, pushing up both bond and share prices, reaching record levels during the first quarter of 1998. As a result, real long-term interest rates, which had been significantly higher than those in the United States despite weaker activity, have been somewhat lower since mid-1997 (Figure 17).

Nominal yields on German bonds have been below their US equivalent since early 1996, the differential reaching 1 percentage point in the spring of 1997 (Figure 18). The widening gap between German and US rates was

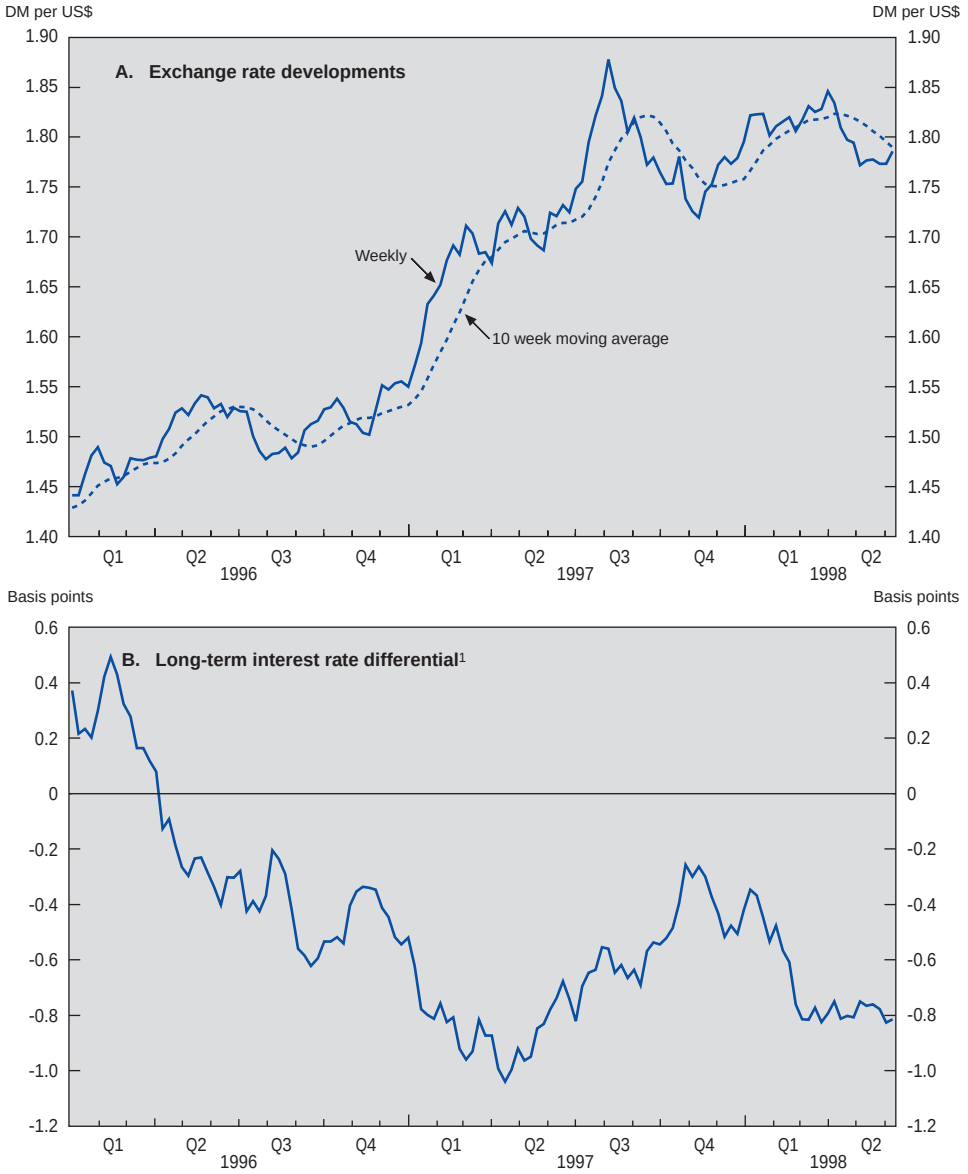
Figure 17. **REAL LONG-TERM INTEREST RATE DIFFERENTIAL BETWEEN GERMANY AND THE UNITED STATES**[1]



1. Real interest rate for Germany minus real interest rate for the United States. The underlying long-term interest rates are the following:
Germany: yields on listed federal securities with residual maturities of 9 to 10 years;
United States: the benchmark government bond (10 years).

Source: Deutsche Bundesbank; OECD, *Main Economic Indicators.*

Figure 18. **EXCHANGE RATE AND LONG-TERM INTEREST DIFFERENTIALS**



1. Yields to maturity for benchmark government bonds (10 years). Interest rate for German bonds minus interest rate for US bonds.

Source: OECD.

associated with a gradual depreciation of the DM in terms of the dollar, the 1997 *Survey* linking this combination of factors to evolving views about the strength of the US economy compared with Germany and the expected future course of monetary policies. However, DM depreciation continued as the excess of US over German rates narrowed in the middle two quarters of 1997, and the possible inflationary implications were a factor behind the October repo hike. Following the increase in the repo rate the deutschemark strengthened as the differential between Germany and the United States narrowed, but weakened again during the first quarter of 1998 as it began to widen once more (Figure 18).

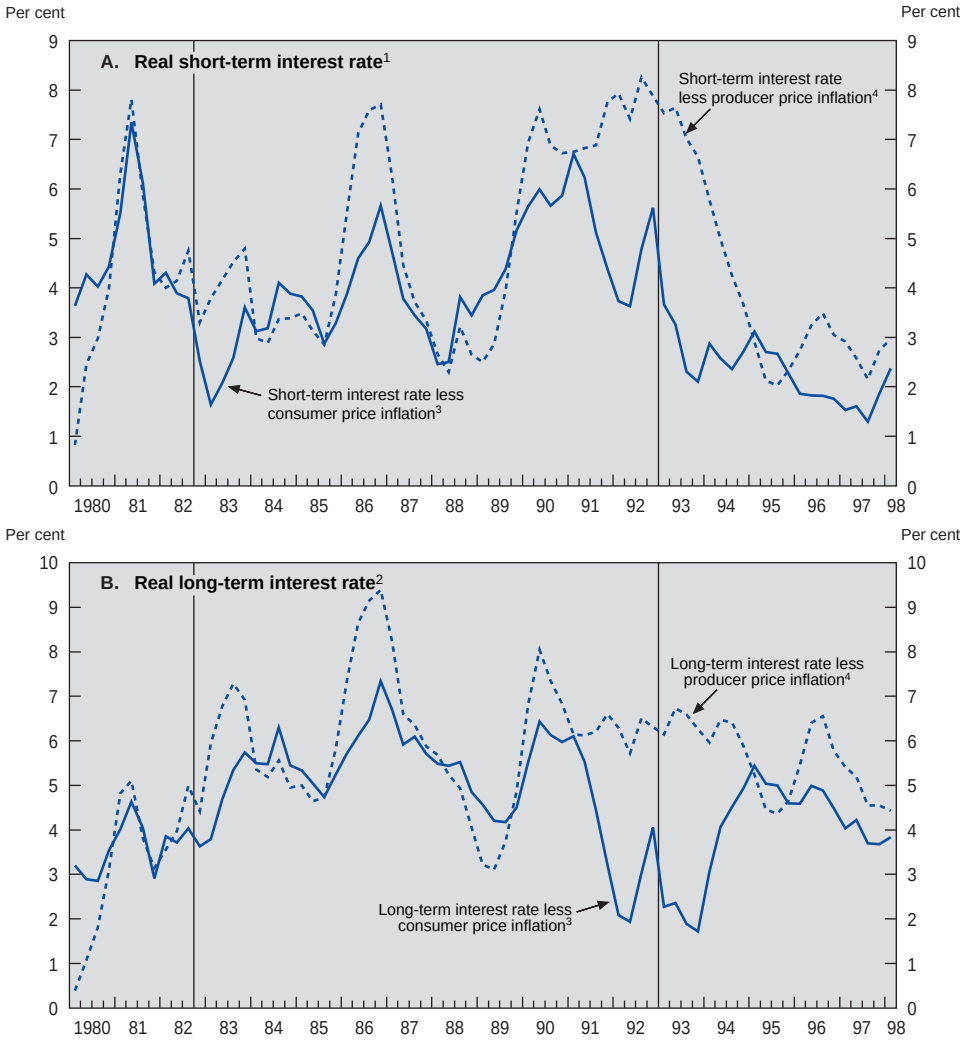
Assessment of monetary conditions

Monetary conditions over the past year have constituted a favourable basis for faster growth. Although the money supply has recently decelerated, there is no evidence that the economy is credit-constrained. Despite the rapid slowing of inflation, real interest rates are now at low levels which, since the early 1980s, have only been seen during periods of accelerating inflation (Figure 19). Moreover, the OECD's so-called ''monetary conditions indicator'' suggests that the overall stance is as supportive to growth as at any time since the early 1980s (Figure 20). This indicator would show an even more favourable picture if long-term interest rates and asset values were to be taken into account. Exchange rate depreciation has made the greatest contribution to the calculated easing of overall monetary conditions. The ''Taylor rule'' also suggests that monetary conditions have been expansionary with the actual interest rate lower than the ''warranted'' rate (Figure 20).

For 1998, the last year of an independent German monetary policy, the Bundesbank has maintained its pre-announced money supply growth target for both 1997 and 1998 of 5 per cent per annum.[11] The target corridor has been lowered by ½ percentage point to 3-6 per cent from the fourth quarter of 1997 to the fourth quarter of 1998, the 3 percentage point band width being retained to take account of the short-term volatility of money growth, which continues to be rather high. The lowering of the target band reflects the Bundesbank's judgement that liquidity supply has been ample, and that a slowing of the money growth rate was possible (and necessary) without affecting activity. It also reflects the desire to avoid creating any monetary disturbances in the run-up to EMU so that the new European Central Bank will be able to take over its responsibilities smoothly.

Figure 19. **REAL INTEREST RATES**

Quarterly average in per cent

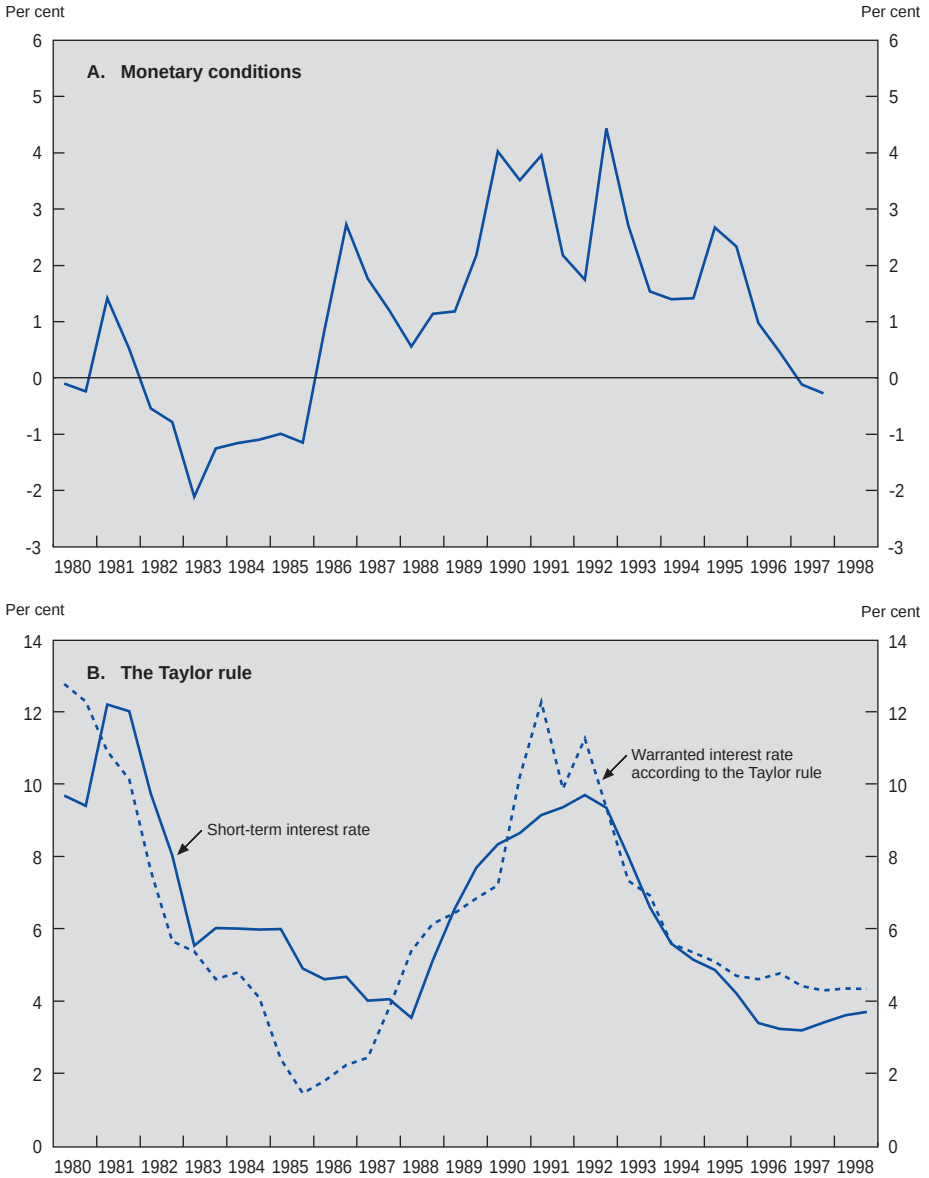


The vertical lines indicate the troughs (see reference in Figure 6).

1. 3-month rate.
2. Until 1990 7-15 year German public sector bonds, from 1990 yields on listed federal securities with residual maturity of 9 to 10 years.
3. As measured by the 4-quarter percentage change in the consumer price index excluding food and energy.
4. As measured by the 4-quarter percentage change in the producer price index.

Source: Deutsche Bundesbank; OECD, *Main Economic Indicators.*

Figure 20. **INDICATORS OF MONETARY CONDITIONS**[1]



1. See *Economic Survey Germany*, 1997 for details of computation, notes 33 and 34.
Source: OECD.

An issue which has come increasingly to the fore concerns the appropriate stance of monetary policy in a low inflation environment, especially where inflation may be mismeasured. The debate has been given added impetus by the publication for Germany of a study showing that over time the average measurement bias in the consumer price index could have been around ¾ percentage point per annum.[12] If so, Germany is characterised currently by very low rates of inflation, which could be classified as virtual price stability. In this case, there do not appear to be grounds for lowering inflation objectives still further, and indeed this is the approach which the Bundesbank has implicitly taken.

The more immediate issue is whether, in view of the favourable inflation environment and moderate growth, the evolution of monetary policy will be fully appropriate to German needs now that it is being increasingly conditioned by European-wide considerations. In the former high-inflation members, the process of converging to German short interest rate levels could result in a monetary stimulus which would push up rates in general (*i.e.* convergence will take place at levels higher than in the present low interest rate countries). Forward rates have priced-in such views, although, as noted, the expected level of rates in the second half of 1998 has fallen since the beginning of the year. In principle, however, the OECD analysis supports the proposition that inflation expectations are converging to a level which will allow interest rates in the Euro area to settle near to German rates. In this case, it would seem that the current configuration of interest-rate and monetary parameters are fully consistent with the stability-oriented stance which has hitherto characterised monetary policy in Germany. With activity picking up, some further increase in policy rates could, however, be required in the course of the year.

Fiscal policy

The budget deficit came in at 2.7 per cent of GDP in 1997, below the 3 per cent Maastricht ceiling (Table 5). Gross debt increased from 60.4 per cent of GDP to 61.3 per cent, a little above the Maastricht threshold but the emphasis with respect to this criterion has been on convergence with the reference value, rather than its current level. However, the unification process and the absorption into public debt in 1995 of many associated outstanding liabilities have obscured the underlying trend (Table 6). Progress was also made towards the medium-term

Table 5. **Public sector financial balances**

DM billion

	1992	1993	1994	1995[1]	1996	1997	1998[2]
General government							
(Maastricht basis)	**–80.7**	**–101.1**	**–79.5**	**–113.0**	**–120.2**	**–96.5**	**–97½**
(Per cent of GDP)	(–2.6)	(–3.2)	(–2.4)	(–3.3)	(–3.4)	(–2.7)	(–2½)
of which:							
Territorial authorities	**–115.8**	**–137.8**	**–116.4**	**–109.4**	**–118.9**	**–95.9**	**–78**
Federal government	–39.3	–66.9	–50.6	–50.5	–78.5	–63.4	–57
State government west[3]	–15.9	–22.5	–24.7	–28.7	–31.4	–26.7	–21
Local government west	–9.4	–8.9	–5.9	–12.4	–4.6	–4.2	–5½
State government east[3]	–15.1	–19.9	–19.9	–16.4	–14.8	–12.9	–11½
Local government east	–7.5	–4.4	–4.8	–1.9	–2.2	–1.6	–1½
German Unity Fund	–22.4	–13.5	–3.0	2.3	12.6	13.1	18½
Other funds[4]	–6.2	–1.7	–7.5	–1.8			
National accounts adjustments[5]	**37.3**	**29.3**	**30.0**	**7.5**	**9.3**	**–5.4**	**–34**
Social security	**–2.2**	**7.4**	**6.9**	**–11.1**	**–10.6**	**4.7**	**14½**
Memorandum items:							
Treuhandanstalt[6]	–29.7	–38.1	–34.4	. .	. .	. .	. .
Public enterprises[7]	–25.7	–20.5	–20.2	. .	. .	. .	. .
Extended public sector	**–136.1**	**–159.7**	**–134.1**	**–113.0**	**–120.2**	**–96.5**	**–97½**
(Per cent of GDP)	(–4.4)	(–5.1)	(–4.0)	(–3.3)	(–3.4)	(–2.7)	(–2½)
(Cyclically-adjusted, per cent of potential GDP)	(–5.9)	(–4.6)	(–3.7)	(–2.8)	(–2.6)	(–1.8)	(. .)

1. 1995 figures do not include (as a capital transfer item) the takeover by the federal government of debits of the Treuhand (DM 204.6 billion) and of the East German Housing Fund (DM 30 billion). Inclusion of these items would imply a general government deficit in 1995 of the order of 9 to 10 per cent of GDP.
2. Government estimates. The federal budget includes a deferral of transfers to the German Unity Fund of DM 1.5 billion.
3. West without Berlin; east including Berlin.
4. Credit Fund, Economic Recovery Programme Fund, Financial Equalisation Fund, Compensation Fund, Railway Fund, Coal Fund and Inherited Debt Fund.
5. Including lending operations, privatisation receipts, timing adjustments, and, until 1995, profits paid by the Bundesbank into a debt sinking fund.
6. Wound up at the end of 1994. The successor organisation is financed directly from the federal budget.
7. Until 1995 included Post/Telecom and railway companies in the west and in the east; Post and Telecoms were incorporated in 1995 pending privatisation while subsidies are now paid directly to the railways by both the state and federal governments.

Source: BMF *Finanzbericht;* Deutsche Bundesbank, *Monthly report;* submission from the Ministry of Finance; OECD projections.

objective of restoring public spending to its pre-unification level of 46 per cent of GDP. This was achieved against the background of a deteriorating tax revenue/GDP ratio – to below its 1990 level of 23.5 per cent. The aim of lowering the overall social security contribution rate to under 40 per cent of gross wages has been more difficult. Meeting this objective while complying with the Stability and Growth Pact will imply persistent pressure to squeeze public

Table 6. **Public debt by government level**[1]

DM billion

	1993	1994	1995	1996	1997[2]
Federal government	685	713	754	833	899
Länder west	391	409	438	468	498
Länder east	37	51	65	79	88
Communities west	134	136	139	142	141
Communities east	18	23	26	28	29
Special funds of the Federal government	217	291	530	522	516
of which:					
Unity Fund	88	89	87	84	80
Credit Fund	101	103	. .	. .	. .
ERP Fund	28	28	34	34	34
Railways Fund[3]	. .	71	78	78	77
Inherited debt Fund[4]	. .	. .	329	324	322
Total – territorial authorities[5]	**1 499**	**1 545**	**1 976**	**2 094**	**2 191**
(Per cent of GDP)	(47.4)	(46.4)	(57.1)	(59.1)	(60.2)
Maastricht definition	**1 519**	**1 672**	**2 006**	**2 141**	**2 232**
(Per cent of GDP)	(48.0)	(50.2)	(58.0)	(60.5)	(61.3)
Financial liabilities	**1 585**	**1 671**	**2 083**	**2 219**	**2 303**
(Per cent of GDP)	(50.1)	(50.2)	(60.2)	(62.7)	(63.2)

1. The Ministry of Finance calculation does not correspond to that published by the Bundesbank and which is reported in the *Economic Outlook* (financial liabilities). There are differences in valuation methods and in netting intra-government holdings.
2. Estimates.
3. 1994: reform of railways.
4. 1995: debts of Treuhandanstalt, Credit fund, eastern German public housing sector.
5. Including indebtedness of municipal special-purpose associations and municipal hospitals.

Source: Ministry of Finance; Deutsche Bundesbank; OECD.

spending in the medium term, while requiring a more efficient and equitable tax system. Following a discussion of the 1997 budget outturn and the prospects for consolidation in 1998, this section focuses on structural reform issues, including tax reform, health-care management and improvements in administrative efficiency needed for the completion of the consolidation process in the medium term.

Meeting the Maastricht deficit criterion

Achieving a deficit of 2.7 per cent of GDP in 1997 represented an unexpectedly large consolidation of 0.7 percentage point of GDP (Table 5). Moreover, the consolidation was achieved against the background of unexpected shortfalls in tax revenue amounting to some DM 25 billion (see Box 1), together with a

Box 1. **Low and unpredictable tax revenues**

Over the last few years, tax revenues have been substantially lower than forecast by the semi-annual official tax estimates. Various explanations have been put forward. A slower-than-expected recovery from the 1993 recession, including high levels of unemployment and a heavy reliance on exports, account for a considerable part of the unexpected shortfalls in tax receipts, notably with respect to the VAT and the wage tax.[1] Frequent and profound revisions in the tax legislation, which have occurred since Germany's reunification, appear to have contributed in a major way to the difficulty in forecasting revenues and, more importantly, to the low level of taxes in relation to economic activity. It appears that once allowance is made for changes in the tax laws – and for one-off effects – revenues are not greatly out of line with activity and revenues may be stabilising, albeit at a low level in relation to GDP.

Income and corporate taxes have been particularly difficult to forecast and are low by past standards. This is only partly attributable to conjunctural factors. Between 1991 and 1997, assessed income tax receipts (income tax excluding wage and withholding taxes) fell from DM 41.5 billion to only DM 5.8 billion (the bulk of income tax receipts is not assessed). Only part of the decline can be attributed to shifts in favour of the unassessed wage tax and the interest source tax. Corporate tax receipts declined from DM 31.7 billion in 1991 to DM 18.1 billion in 1995, recovering to DM 33.3 billion in 1997, hardly more than the level in 1991 despite a higher level of profits. Annual revenues from the assessed income tax and the corporation tax are affected by three sets of factors which in sum are particularly difficult to assess and have been strongly affected by changes in legislation: prepayments for the current year, retrospective payments and refunds by the tax authorities for past years, and deductions determining the current tax liability (*e.g.* investment allowances) or tax base (*e.g.* depreciation allowances). In 1997, repayments amounted to a massive DM 46 billion for income tax and DM 1.1 billion for corporate tax.

One reason for the large repayments and for the uncertainty surrounding tax estimates is the possibility for enterprises and businesses subject to income tax to carry losses forwards and backwards. Carrying losses backwards leads to tax refunds for the last two years. Carrying forward losses leads to a reduction in the tax base in future years and is allowed without time limit. In 1992, the year for which the most recent statistical information is available, accumulated tax loss carry-forwards amounted to more than DM 250 billion and a significant proportion of these might be associated with the acquisition of enterprises in the new Länder.[2] Low profits after 1992, and especially during the recession year of 1993, might have made the situation worse. Regarding changes in tax legislation, investment bonuses and special depreciation allowances for personal income tax and corporate taxes have been used to foster investment in the new Länder. The recent shortfall of tax receipts is in part attributable to a greater than expected take-up of special depreciation allowances, although complete and up-to-date

(continued on next page)

(continued)

statistical information is lacking. High levels of depreciation allowances have, it is believed, often led to income tax for previous years needing to be reassessed when losses have been carried backwards. Successive reductions in corporate tax rates for retained and distributed profits have also contributed to important revenue shortfalls since transition provisions have given companies room for manoeuvre with respect to the tax treatment of previously retained profits for several years.[3] Recent refunds of corporate taxes are in part driven by the expiration of the transition period in December 1998.[4]

The predictability of tax revenues should start to improve with the expiration of some special tax allowances, but the level of tax revenues is likely to remain low due to large loss carry forwards. Since these frequently formed the basis for business decisions in the past, a retro-active change in the law would be inappropriate. Although a shifting of capital income abroad has occurred, it does not appear to be the principal reason for low tax revenues. The importance of tax expenditures in lowering the tax base does, however, underline the need for a far reaching tax reform.

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1. Econometric and other evidence to support this conclusion is presented in J. Ratzinger ''Steueraufkommen 1997: Das zweite Halbjahr muß es bringen'', *ifo Schnelldienst*, 22/1997 and A. Boss, ''Untersuchungen zur Bemessungsgrundlage und zum kassenmäßigen aufkommen der Steuen vom Umsatz'', *Kieler Arbeitspapiere*, 844, 1997.
2. See Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, *Jahresgutachten 1997/98*, Stuttgart 1997.
3. Rates have been lowered from 56 and 36 per cent for retained and distributed profits at the end of the 1980s, respectively, to 45 and 30 per cent.
4. By the end of 1998, retained profits which had been taxed at a rate of 50 per cent, the rate in force until 1993, will be subjected to the new reduced rate of 45 per cent if they continue to be retained. If they have been distributed, or will be distributed in the following months, they are taxed at the reduced rate of 30 per cent. In both cases, refunds of corporate taxes occur, the magnitude of which depends on the tax behaviour of the companies. Offsetting to some extent tax repayments will be higher revenues from capital gains tax.

further rise in unemployment which contributed to additional pressure on expenditures. Serving to offset these slippages was across-the-board spending restraint by all levels of government, which led to a decline in investment by around 9.5 per cent and which also contributed to a stagnation in public consumption (Table 7). However, an important role was played by mandated increases in social security charges, which led to a turnaround in the finances of the social security system of some DM 15 billion, while the decline in the deficit of the territorial authorities (on a financial basis) was in the order of DM 23 billion (Table 5).

Table 7. **Appropriation account for general government**

Including social security

	1995	1996	1997[1]	1998[2]	1999[2]
	DM billion				
Receipts					
A. Direct taxes	387.9	365.2	359.2	368.0	388.0
B. Social security contributions	670.1	702.1	728.3	743.7	754.2
C. Other current transfers received	46.0	48.4	47.6	49.0	50.7
D. Indirect taxes	447.4	449.5	459.9	479.2	503.8
E. Property and entrepreneurial income	41.9	39.6	39.4	40.5	38.0
F. Current receipts, total	1 593.4	1 604.7	1 634.3	1 680.4	1 734.7
(Per cent of GDP)	(46.1)	(45.3)	(44.9)	(44.5)	(44.1)
Disbursements					
G. Government consumption	686.6	704.8	705.2	719.3	736.7
of which: Wages and salaries	358.9	363.0	363.6	367.3	373.0
H. Property income payable (interest on public debt)	131.1	131.8	135.9	140.7	145.3
I. Subsidies	71.5	70.0	66.9	67.6	68.9
J. Social security outlays	619.4	623.1	630.6	643.7	654.8
K. Other current transfers paid	110.4	119.1	124.5	132.0	137.9
L. Current disbursements, total	1 618.9	1 648.7	1 663.0	1 703.3	1 743.5
(Per cent of GDP)	(46.8)	(46.6)	(45.7)	(45.1)	(44.3)
M. Saving (F – L)	–25.5	–44.0	–28.7	–22.9	–8.9
N. Gross investment	83.1	76.7	69.5	70.2	71.6
O. Net capital transfers received	–37.8	–31.4	–30.4	–36.3	–43.6
P. Consumption of fixed capital	25.6	26.3	26.7	27.3	27.6
R. Net lending (M – N + O + P)	–120.9	–125.8	–101.9	–102.1	–96.5
	Per cent of nominal GDP				
Memorandum items:					
Net lending	–3.5	–3.6	–2.8	–2.7	–2.5
Net lending, Maastricht basis	–3.3	–3.4	–2.7	–2.6	–2.3
Cyclically-adjusted net lending (potential GDP)	–3.0	–2.8	–2.0	–2.0	–2.0

1. Provisional estimates.
2. OECD projection.

Source: Statistisches Bundesamt; OECD.

The 1997 federal budget aimed for a deficit of DM 53.3 billion but in the light of more difficult fiscal conditions than expected, a supplementary budget was presented to parliament in late autumn, raising the deficit ceiling to DM 70.9 billion. With the revised deficit target exceeding planned investment, which was declining sharply, the government was forced to waive the ‘‘golden rule’’, laid down in the constitution, that the federal deficit (on a financial basis) should not exceed investment, except when prevailing conditions constitute a

disturbance of ''macroeconomic equilibrium''. The unexpectedly strong rise in unemployment was seen as constituting such a disequilibrium. In the event, the deficit came in at DM 63.4 billion (Table 8). An important success of federal fiscal policy in 1997 was a reduction in spending by DM 13.7 billion in comparison with the outcome for 1996. Across-the-board savings generated by the global spending controls are estimated to have reduced spending by some DM 5 billion and a substantial reduction in active labour market measures (Chapter IV) generated about DM 2.5 billion in the form of reduced transfers to the Federal Labour Office. Low interest rates and lower than expected transfers to the Federal Railways also contributed to containing federal spending, as did land sales which serve to lower registered investment expenditures.

Table 8. **The Federal budget**

DM billion

	1994 Outcome	1995 Outcome	1996		1997			1998 Budget
			Budget	Outcome	Budget	Revised budget	Outcome	
Expenditure	**471.2**	**464.7**	**451.3**	**455.6**	**439.9**	**444.8** [1]	**441.9**	**456.8** [1]
(Per cent of GDP)	(14.2)	(13.4)	(12.7)	(12.9)	(12.0)	(12.2)	(12.1)	(12.1)
of which:								
Consumption	90.6	91.5	93.1	92.0	91.7	92.5	90.6	92.0
Interest payments	53.1	49.7	53.4	50.9	54.4	53.7	53.4	56.5
Investment	12.0	12.3	12.4	12.1	13.0	12.8	12.2	13.7
Transfers and lending	315.5	311.2	292.6	301.0	282.9	293.1	285.8	295.5
of which:								
To other administrations	96.4	89.6	93.7	92.3	85.9	81.5	81.1	81.1
Others	219.2	221.6	198.9	208.7	197.0	211.6	204.7	214.4
Revenues	**420.6**	**414.1**	**391.2**	**377.0**	**386.5**	**373.9**	**378.4**	**400.3**
(Per cent of GDP)	(12.7)	(12.0)	(11.1)	(10.6)	(10.5)	(10.3)	(10.4)	(10.6)
of which:								
Taxes	379.0	366.1	351.2 [2]	338.6	345.7	330.2	331.1	331.8
Other	41.6	48.0	40.0	38.5	40.8	43.7	47.4	68.6
Financial balance	**–50.6**	**–50.5**	**–60.1**	**–78.5**	**–53.4**	**–70.9**	**–63.4**	**–56.4**
(Per cent of GDP)	(–1.5)	(–1.5)	(–1.7)	(–2.2)	(–1.4)	(–1.9)	(–1.7)	(–1.5)

1. Total expenditure is smaller than the sum of the individual spending items by DM 7.3 billion in 1997 and DM 1.0 billion in 1998. This is due to overall expenditure restrictions not having been allocated to an expenditure category.
2. Includes a reduction of DM 21.2 billion due to the treatment of child benefits from 1996 as reduced tax revenues rather than social expenditures.

Source: Ministry of Finance; OECD.

The Länder were also severely affected by the unexpected losses in tax revenues and this was compounded by the failure (contrary to expectations) of increases in the inheritance and real estate taxes to compensate fully for the revenues foregone by the abolition of the Länder's wealth tax at the beginning of 1997. In the course of the year, several Länder introduced spending controls which, together with a continued reduction in personnel and a further curbing of investment spending, reduced their combined deficit from DM 46.2 in 1996 to DM 39.6 billion in 1997. While this exceeded the original target agreed in the Financial Planning Council in autumn 1996, it was lower than expected toward the end of 1997 due to a late surge in local tax revenues.

Following two years of deficits exceeding DM 10 billion, the social security system recorded a surplus of some DM 5 billion in 1997. The improvement reflected a mandated increase in pension contribution rates of 1.1 per cent of the wage base needed to rebuild financial reserves, together with a further increase in the surplus of the new long-term care insurance *(Pflegeversicherung)*. The overall improvement in the social security accounts was larger than expected and could be attributable to the effectiveness of past measures aimed at curbing the generosity of labour market-related benefits as well the favourable impact of new health reforms (see following section).

Fiscal policy in 1998: a slower rate of fiscal consolidation

Fiscal consolidation is set to continue at a slower pace in 1998, with the authorities projecting a decline of around a quarter point of GDP in the general government deficit ratio (Table 5). The debt/GDP ratio is expected to decline marginally. The surplus in the social security system will be some DM 15 billion (0.4 per cent of GDP), while the territorial authorities are expected to incur a combined deficit of around 3 per cent. Reflecting rising public pressures, government consumption and investment are budgeted to expand by 1 to 2 per cent, and for investment this will be the first real increase since 1994. The growth rate of tax revenues is expected to recover to around 3 to 4 per cent, while a large number of tax measures are being introduced, including a 1 percentage-point rise in the general rate of VAT in April which is intended to avoid the necessity of once again raising the contribution rate for pensions. Although further tax shortfalls cannot be excluded, incorporating the fiscal assumptions outlined in Box 2 the OECD deficit projections are close to the official ones (Table 7). General government debt could fall by some ½ percentage point of GDP, assisted by the proceeds of privatisations.

Box 2. **Fiscal assumptions underlying the OECD projections**

Despite the failure to implement a full tax reform in 1997 (see below), the 1998 budget incorporates a number of tax measures, including some which were deferred from 1997 because of the need to remain within the Maastricht constraints. The basic income tax allowance has been increased and the ''Solidarity'' income tax surcharge reduced from 7½ to 5½ per cent, which will incur revenue losses of some DM 2.5 billion and DM 7 billion respectively compared with the 1996 tax law. The business capital tax has been finally abolished with net revenue losses of some DM 4 billion, but corporate depreciation allowances have also been cut in order to fully offset these revenue losses. While the abolition of the business capital tax mainly affects the budgets of the Länder and communities, the federal government has compensated the loss by ceding a greater share of VAT receipts to them (see also Chapter III below). In an important shift of tax policy, the VAT has been increased from April 1998 by 1 percentage point to 16 per cent to cover increased transfers to the pension system and thereby to fulfil the government's commitment to stabilise the rise in social security contributions. Social security contribution rates are thus assumed to remain constant in 1998.

While public sector wage growth is projected to accelerate somewhat in 1999, the existing policy to restrict current expenditures is expected to continue, with general government consumption growing by around 1 per cent in real terms on an annual basis over the projection period. Further cuts in staffing levels are therefore assumed. The growth rate of social transfers is projected to remain modest due to the impact of recent changes to entitlements and to declining unemployment in 1999. Privatisation and asset sales of some DM 42 billion and DM 17½ billion are projected to reduce gross general government liabilities by the same amount in 1998 and 1999 respectively. Budgetary policy is, however, taking place against the background of increased spending pressures. In the face of a continuously high level of unemployment, the government has decided to expand active labour market measures again by an estimated DM 4.5 billion and this is reflected in an increase in transfers to the Federal Labour Office (see Chapter IV). Responding to the high level of unemployment in the building sector, construction spending is to be increased by about DM 1 billion (9½ per cent over the 1997 outcome), and investment grants have been increased by DM 4 billion (10 per cent). In addition, federal transfers to the public pension system are set to rise by some DM 3 billion.

Further progress in consolidation is also likely to occur at the Länder level. On current budget plans, the Länder intend to reduce their deficit from DM 39.6 billion in 1997 to DM 34 billion in 1998. For all but one of the states, planned expenditure increases fall significantly short of the 2 per cent medium-term upper limit that has been agreed in the Financial Planning Council

(Finanzplanungsrat), and some Länder, notably in the east, plan to reduce their spending even in nominal terms. The financial position of the new Länder and their local governments will continue to be underpinned by transfers which are scheduled to increase, after a decline in 1997 (Table 9). Higher transfers from the federal government to the pension system in combination with recent reforms in social and health related benefits systems should ensure that the social security system will achieve a substantial surplus in 1998.

Table 9. **Public financial transfers to eastern Germany**

DM billion

	1991	1992	1993	1994	1995	1996	1997	1998[1]
1. Gross transfers[2]	**139**	**151**	**167**	**169**	**185**	**187**	**183**	**189**
Federal government transfers to the eastern Länder and communities[3]	75	88	114	114	135	138	131	139
Western Länder and communities[4]	5	5	10	14	10	11	11	11
Pension scheme[5]	. .	5	9	12	17	19	18	18
Transfers of the Federal Labour Office[6]	25	38	38	28	23	26	26	28
German Unity Fund borrowing[7]	31	24	15	5	. .	. .	. .	. .
Transfers from the EU[8]	4	5	5	6	7	7	7	7
2. Receipts	**33**	**37**	**39**	**43**	**45**	**47**	**47**	**48**
of which:								
Additional federal tax receipts from the east[8]	31	35	37	41	43	45	45	46
Additional federal administrative receipts from the east	2	2	2	2	2	2	2	2
Net transfers (1 – 2)	**106**	**114**	**128**	**126**	**140**	**140**	**136**	**141**
Per cent of:								
All-German GDP	3.7	3.7	4.0	3.8	4.0	4.0	3.7	3.7
Western German GDP	4.0	4.1	4.5	4.3	4.6	4.5	4.2	. .
Eastern German GDP	51.5	42.9	39.6	34.4	35.2	33.9	32.2	. .

1. Government projections; GDP as projected by OECD.
2. Adjusted for the double counting of the federal payments to the Federal Labour Office.
3. 1995 including DM 35 billion federal tax losses caused by the revision of the fiscal equalisation measures.
4. 1993 and 1994 including payments to the German Unity Fund; 1994 including tax losses caused by the revision of the fiscal equalisation measures.
5. Excluding payments of the federal government.
6. Total deficit east, including federal payments.
7. Excluding payments of the federal government and Länder.
8. Estimates.

Source: Ministry of Finance; OECD.

Fulfilling the medium-term programme

In conformity with the Stability and Growth Pact (Box 3), which comes into force in 1999, the German convergence programme plans a reduction in the general government deficit to 2 per cent of GDP in that year and 1.5 per cent

Box 3. **The Stability and Growth Pact**

The European Union's Pact for Stability and Growth, finalised at the Amsterdam Summit in June 1997, clarifies the Maastricht Treaty's provision for dealing with ''excessive deficits'' and provides an institutional framework for its enforcement, in part through strengthened surveillance and co-ordination of economic policies via the annual review of national stability programmes. The Pact also calls on participants in the Economic and Monetary Union (EMU) to adopt budgetary balance (or even a surplus) as a medium term objective.

For countries participating in EMU, the Pact considers a general government deficit above 3 per cent as excessive unless the European Commission judges it to be temporary or there are special circumstances. By temporary is meant that, according to the projections of the Commission, the deficit would fall beneath the 3 per cent threshold in the following year. In the case of an excess budget, the Council of Ministers will, on the recommendation of the Commission, propose a course of action for the country which should be followed by effective measures within four months. The Council will monitor the measures and if they are found to be inappropriate they will make further more detailed proposals, which will be published. If corrective measures have not been implemented within ten months, sanctions will be imposed. These would initially take the form of non-remunerated deposits, with a fixed component equal to 0. 2 per cent of GDP and a variable component rising in line with the size of the excessive deficit. Such deposits are limited to a maximum of 0.5 per cent of GDP per year, but would accumulate each year until the excessive deficit is eliminated. Provided the excessive deficit is corrected within two years the deposits are returned to the country, otherwise the deposits will become non-refundable. In the event that the measures taken by a country are ineffective, the whole process will be restarted but sanctions will be imposed within three months.

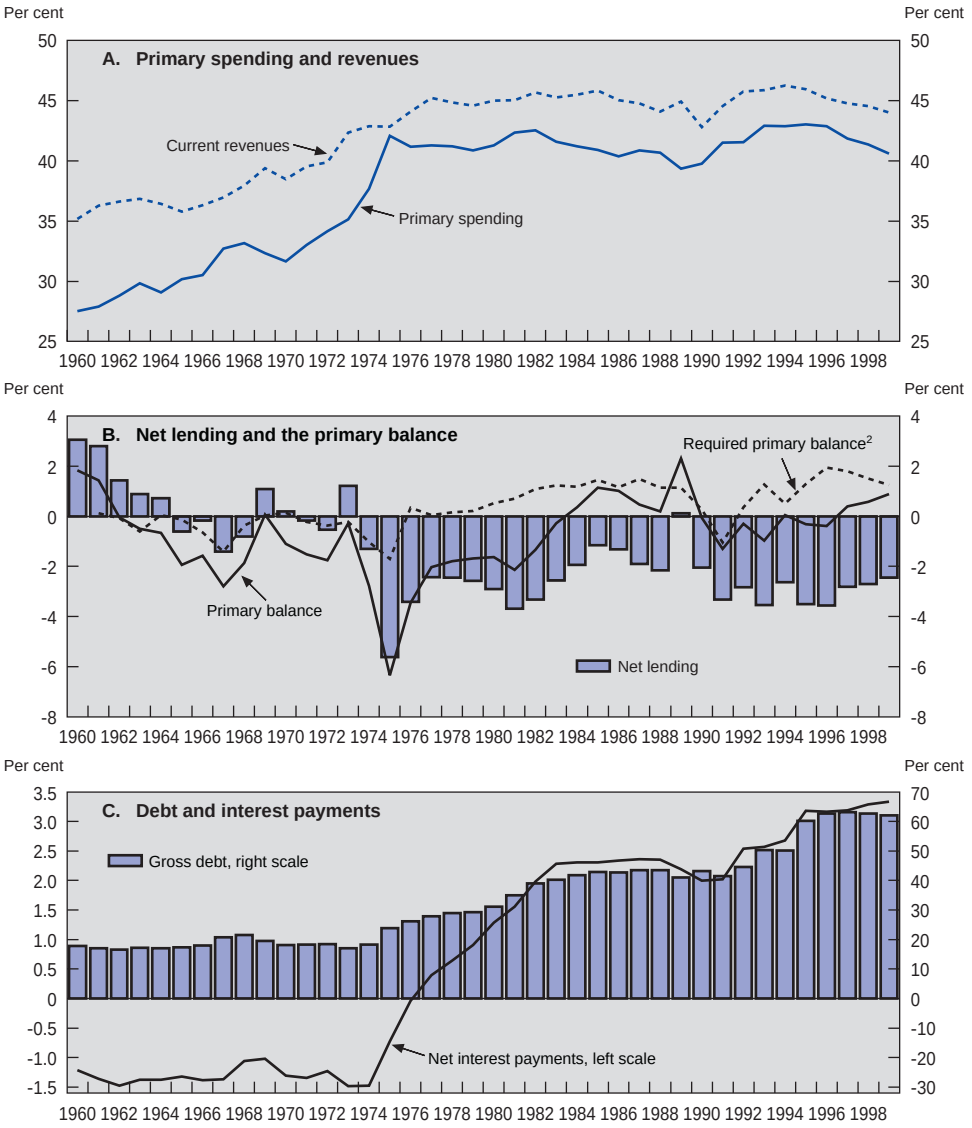
When a country is judged to be in recession, which is defined as an annual fall in real output (GDP) of at least 0.75 per cent, the Pact will be enforced in a differentiated manner. If economic output in a member country declines by 2 per cent or more – and provided the deficit is temporary – exemption from the procedure is granted automatically. In the event GDP falls by between 0.75 per cent and 2 per cent, exemption can be granted in special circumstances by the Council of Ministers. The country would need to convince the Council that the economic decline was ''exceptional'' in terms of its abruptness or in relation to past experience.

in 2000, which would be sufficient to reduce the level of debt below 60 per cent of GDP in 2001. However, on the basis of current tax rates and expenditure trends the OECD projects that the general government deficit will only stabilise at around 2¼ per cent of GDP in 1999, the cyclically-adjusted deficit remaining at around 2 per cent, unless the high surplus in the social security funds (which is reduced in the projections by some DM 10 billion or almost 0.3 per cent of GDP) is maintained (Table 7). Under this scenario, improvements in the primary balance, although significant, would hardly be sufficient to stabilise net government debt as a percentage of GDP (Figure 21). The achievement of the government's deficit and debt objectives thus requires further discretionary action.

It should also be noted that the deficit targeted for 1999 and 2000 is very near to the absolute minimum needed to ensure that the deficit remains below the 3 per cent ceiling specified under the Stability and Growth Pact, making it vulnerable to any unforeseen cyclical weakening. Simulations by the OECD suggest that a 1 per cent increase in the output gap would increase the general government deficit by 0.5 per cent of GDP. For Germany the mean value of the output gap at the trough of recessions since 1975 has been 2.8 per cent. This suggests that from a historical perspective, a structural deficit target of less than 1¾ per cent would be needed to avoid breaching the 3 per cent deficit ceiling specified under the Stability and Growth Pact in the case of a ''normal recession'' – or the need could arise to resort to pro-cyclical fiscal restraint to avoid overshooting. From this perspective, and given the difficulties in enforcing the Pact in a federal state (discussed in Chapter III), a lower deficit level is called for in 1999.

Meeting the government's long-standing medium-term objectives of reducing public expenditures to their pre-unification level of 46 per cent by 2000 while lowering the tax burden, including a reduction of the overall social security contribution rate to under 40 per cent of gross wages, will be difficult to meet unless a more ambitious fiscal programme is adopted. Indeed, achieving sustainable fiscal consolidation and meeting policy targets requires that more permanent fiscal measures be implemented. Repeated resort to global spending caps by the federal and other levels of government over the past three years has contributed to a significant decline of public sector investment at the federal level and in the old Länder and has narrowed the scope for further application of this *ad hoc* instrument of budget control. What is now required are additional measures to

Figure 21. **THE BUDGET BALANCE AND GENERAL GOVERNMENT DEBT**
Per cent of GDP[1]



1. 1998 and 1999 OECD projections.
2. Primary surplus needed to stabilize the net debt ratio.

Source: OECD.

improve public sector efficiency as well as to control more effectively the growth of social spending in the medium term, and in the longer term when pensions are particularly important. In view of the fact that major budgetary pressures have arisen in recent years through an erosion of the tax base, reforms are also necessary with respect both to the tax system, as well as to the subsidy system more generally. Moreover, the new fiscal framework which will become fully operative in 1999 sets other challenges, particularly the need to allocate responsibility for fiscal consolidation efficiently and effectively in a federal state. These issues are taken up below.

Structural policies for effective medium-term consolidation

Towards a more efficient state

Plans to lower public sector employment and to cut administration costs frequently run the risk of dealing with symptoms rather than the cause of low public sector efficiency. The government has therefore embarked on a programme to develop a leaner state *(Schlanker Staat)*. To this end, it has started a review of the extent to which services for ministries, which up to now have been produced within the public administration, could be provided by private firms. A number of pilot projects has been started, covering fields such as printing, the servicing of buildings, the provision of travel services and medical services. Since none of the services under consideration constitutes a public good in the economic sense there is no obvious reason why public provision should be warranted. The programme is an important step to lower the cost of public administration and should be extended to the other layers of government.

At all layers of government the potential for the participation of private firms in the planning and financing of infrastructure projects is currently under review and there has already been some experience of this, particularly at community level. Outsourcing and private financing could contribute to increasing public sector efficiency. However, certain financial arrangements run the risk of producing adverse effects on public finances. For example, experience elsewhere shows that leasing-back by the public sector of privately financed infrastructure projects may not reduce the government's spending obligations in present value terms. Rather, it tends to reduce present budgetary pressures at the expense of greater obligations in the future and reduced transparency of fiscal policy. Moreover, with spending restraints less binding, fiscal discipline in the government sector

could suffer. Hence, ensuring that budget planning takes account of the longer-term balance-sheet consequences of present decisions should have a high priority on the agenda for further public sector reform. This could also be accompanied by some increased flexibility; an amendment of the federal budget law came into force in January which allows for a greater transferability of funds between consecutive fiscal years and limited experience in Germany suggests that it could lead to overall savings.

The search for public sector efficiency – defined in the broadest sense to include the impact on the economy at large – needs to be extended in other directions. In particular, attention needs to be given to examining the structure of government with a view to identifying those incentive structures (administrative and economic) which might lead to public sector inefficiency in the broadest sense. Indeed, the debate in Germany about the role and effects of federal economic arrangements is tending in this direction (Chapter III). Defining the issue even more widely, the whole system of regulatory responsibilities needs to be re-examined to determine whether public administrative involvement is necessary or not. In some circumstances it may, indeed, be detrimental to the economy.[13]

Public health provision: first experience with reform

The statutory health insurance system provides an important case study in the need for reform and was analysed in the 1997 *Survey*. With a view to controlling not only the growth of public consumption but also to contain the increase of health contribution rates and labour costs, the government took further important steps in the reform of the health system during 1997. The measures, which were reviewed in detail in last year's *Survey*, aimed to increase competition between statutory health funds, raise co-payments for medicines and increase responsibility of consumers more generally, and to further the introduction of performance-related payments in the hospital sector. The *Survey* concluded that the measures were welcome, although they did not deal with a number of fundamental issues. In particular, the competitive operating environment of the health funds needed to be clarified and a basis established for them to evolve into purchasers of health services, either as a group or individually. In addition, the integration of the health system, especially between the stationary and ambulatory sector, needed to be facilitated. Initial experience with reforms during the last year has served to underpin this judgement, while other issues have become apparent.

The most immediate success of the reforms to date has been a fall in expenditures of the statutory health insurance system by some 2½ per cent in 1997, with marked declines in the second half, when most measures became fully effective. Following a deficit of nearly DM 4 billion in the first half of 1997, a turnaround occurred in the second half, so that for the year as a whole there was a surplus of DM 1.1 billion. The improvement has been due in part to the reduction in expenditures on pharmaceuticals and on visits to spas following a large increase in co-payments[14] – although in the latter case the sharp fall in visits has led to some recent easing of policy. Of greater importance, however, has been the drop in expenditure on sick pay by some 22 per cent. Some of the decline can be attributed to the 20 per cent cut in the statutory rate of sick pay, but equally important has been the decline in the level of sickness. As noted in Chapter IV, this is in part attributable to tighter control by firms and to difficult labour market conditions. But it is also the case that health funds are now controlling the incidence of sick leave more carefully because they have an incentive to do so: since the new law came into force at the start of 1997, individuals have been exercising their right in large numbers to change their health fund to those with lower contribution rates.[15] It is thus crucial for funds to control their costs in order to compete and attract new members and, indeed, administrative expenditures fell in 1997 even though the responsibilities of the health funds have increased.

Health insurance in eastern Germany has continued to present problems, with the system incurring a deficit in 1997 of DM 300 million, the loss being covered by bank loans. This situation is legally untenable and would normally have forced a marked increase in the already high level of contribution rates. In addition to relatively high costs of medical facilities in the new Länder, revenues of the health funds are weak, due to lower average wages than in the old Länder and the high rate of unemployment. In response to the situation, the federal government and the states agreed to incorporate the new Länder temporarily into the risk equalisation fund for the two years 1999 to 2001 after which there will be a new pooling arrangement. This should result in a transfer of DM 1.2 billion to the new Länder, which will be financed by health funds in western Germany.

The 1997 *Survey* noted the possibility of adverse spill-over effects between the private and the statutory health funds and this problem has started to materialise: doctors are said to be seeking to raise their charges for private patients, to

compensate for tighter budget restraints imposed by the statutory health funds.[16] In response, there have been proposals in the parliament to reduce the legally-permitted maximum difference between public and private charging or to regulate charging in other ways. The logic of health reforms would be to allow the private health funds to negotiate directly, even with individual doctors, with respect to fees.

Key aspects of the 1997 reforms were moves to strengthen performance-related financing of hospitals and to encourage experimentation with new organisational structures of health service provision. With respect to hospitals, full implementation of performance-related payments has been delayed, with a global budget constraint to be abolished in 2000. Moreover, negotiations for financing appear not to have changed, with good and bad hospitals still being treated in an undifferentiated manner. With respect to experimentation, there are currently some 35 model projects which seek to improve performance in the ambulatory sector – often through establishing networks of doctors – which should cut the number of hospital admissions. Other projects involve rebates of contributions to insured who do not claim. The requirement that innovations be agreed with the association of doctors would seem to have limited the projects which the statutory health funds have been able to introduce, even though in some cases groups of individual doctors wished to participate. Moreover, little has been achieved to date with respect to further integration between hospitals and the ambulatory sector.

Overall, the initial experience of health reforms has been positive, with expenditures having been brought down and health funds starting to compete aggressively for members by seeking to control their costs and thereby maintaining low contribution rates. However, holding costs down for a short time is relatively easy. The challenge is to turn these into lasting savings. For this to be achieved, the statutory health funds will need to tackle aggressively the cost levels in the hospital sector and in the ambulatory sector, which are the two largest cost elements. As noted in last year's *Survey*, greater freedom in dealing with health suppliers will be necessary and first experience has served to underline this conclusion.

Controlling pension finance

As documented in previous *Surveys*, the greatest potential pressure on public spending will arise in the longer run from public pension obligations. A great

deal has been achieved in placing the pension system on a more financial footing during the past year, a new system being scheduled to become effective in 1999. Key to the reform is the incorporation of a demographic life-expectancy adjustment into the benefit formula which determines the future level of pensions. As a result, the reference replacement rate will be lowered gradually from about 70 per cent of average net wages (after 45 years of contributions and average earnings) at present to 64 per cent in 2030. Existing pensions for reduced capability to work *(Rente wegen geminderter Erwerbsfähigkeit)* and for invalidity *(Berufsunfähigkeitsrente)* will be combined into a single disability pension. The degree to which the ability to work is diminished will be assessed solely on health grounds, without reference to labour market conditions as is the case at present. Experience in Germany and in other countries shows that, unless labour market conditions are excluded as qualifying criteria, invalidity pensions tend to function as substitutes for other early retirement programmes.[17] Hence, as has been argued in previous *Surveys*, conditioning eligibility for invalidity pensions exclusively on health grounds is a necessary condition for curbing the overall inflow into early retirement. With respect to determining the level of pension claims, time spent out of the workforce raising children will be treated for up to three years like periods of employment by increasing their valuation from 75 per cent at present to 100 per cent.

The reduction of spending which results from lowering the average replacement rate by 6 percentage points and the other already-established amendments to the pension scheme will not suffice on their own to cope with the fiscal burden implied by population ageing (see *Economic Survey of Germany, 1997)* and burdens have been actually raised by expanding pension rights for periods of child care. Hence, while the reform will lead to a significant short-run easing of the fiscal burden, a considerable increase in contribution rates is still expected to be needed from the second decade of the next century onwards, even if the increase in the tax-financed transfer from the federal budget is taken into consideration. Further pension reforms will thus be required, and options for developing additional private coverage still need to be considered. The Third Financial Market Promotion Law is a first step in this direction and needs to be extended (see Chapter IV).

Tax reform: an unfinished agenda

The need for tax reform to put the public finances on a firmer – and more equitable – footing has been emphasised by the recent revenue shortfalls. More generally, the tax code is characterised by numerous tax breaks introduced for diverse economic and social policy reasons – especially the desire to support the new states – so that the system is complex and opaque. Indeed, as noted in Box 1, it is even difficult to form reliable tax estimates, and tax expenditures have certainly been higher than anticipated (Chapter IV). In addition, the personal income tax schedule is characterised by high marginal effective rates at the low income end and by very high marginal rates at the high income end. In combination with special tax treatment for business income, there are important opportunities for tax avoidance and incentives to tax evasion, particularly for high income groups which are subject to relatively high statutory tax rates. Tax reforms which lower statutory rates and reduce tax exemptions would foster tax compliance.

The government's proposals for a wide-ranging reform of both the personal income and corporate tax systems through a reduction in statutory tax rates and a broadening of the tax base were rejected by the second chamber of parliament *(Bundesrat)* in 1997 after two mediation attempts. As with most tax reforms in the OECD area, a major point of disagreement has revolved around distributional issues and especially the static and dynamic consequences of lowering the top marginal rate of income tax. Despite the setback for overall tax reform, there has been some progress in individual areas and the government has finally been able to implement two measures which have long been promised: the ''Solidarity'' income tax surcharge has been reduced by 2 percentage points and the business capital tax finally abolished. The problems inherent with partial tax reforms have, however, become apparent. Abolition of the business capital tax was widely demanded since it was seen as an impediment to growth and investment as it was based on capital and not on earnings (see OECD *Economic Survey of Germany*, 1996, 1997). But many small and medium-sized companies that were exempt from the tax are now affected by the reduction in depreciation allowances introduced to compensate for revenue losses. Moreover, the general issue of tax reform involves questions of federal-state responsibilities which are quite complex and which are discussed in the following chapter.

III. Towards more efficient government: reforming federal fiscal relations

Introduction

In a federal state, the drive to improve public sector efficiency and to produce a durable reform of the tax system naturally tends to bring fiscal relations between the various levels of government to the fore of the policy debate. In the current German context, there are several reasons why this focus is important. First, the integration of the new states is producing greater financial strains than anticipated, and as it has become increasingly evident that convergence will take some time, the danger has emerged that the incentive structure of existing financial relationships may be perverse: the system of inter-governmental transfers aimed at equalising revenue levels leaves little return to a state's efforts to improve its finances and two of the German Länder have threatened to appeal to the Constitutional Court if a political agreement on reform is not reached on the grounds that the high degree of redistribution of taxes is unconstitutional. Second, the Stability and Growth Pact at the European level has raised questions as to how the international commitments to fiscal consolidation can be efficiently and equitably allocated between different levels of government. Third, there is the more general resource allocation issue of whether a leaner and more efficient public sector could be achieved through greater conformity with the constitutional principle of subsidiarity, whereby public spending would be more closely matched with state and local tax responsibilities. In formally assigning tasks to the different levels of government – the federal government *(Bund)*, the states *(Länder)* and the local communities *(Gemeinden)* – and stipulating budgetary independence of the Bund and the Länder, the German constitution *(Grundgesetz)* broadly follows the subsidiarity principle. However, in practice, taxing powers are largely centralised and there is only limited regional and local fiscal autonomy. The theory of fiscal federalism suggests that more effective resource

allocation could be reached if the federation allowed lower levels of governments more responsibility both for decisions as to what goods and services to provide and for the financing of these.

This chapter provides an appraisal of the major features of German federal fiscal relations from the viewpoint of public sector efficiency. The outline of the chapter is as follows: the first section highlights the principal features of German federal-state relations and the diffusion of responsibilities among the component governments. The second section assesses the redistributive function of the inter-state financial equalisation system and its inherent disincentive aspects. The following section then considers the implications for the equalisation system which arise with the Stability and Growth Pact, in the context of the need to cap borrowing at federal and state level. Finally, the chapter considers more fundamental questions related to the allocation of spending and taxing powers over the various levels of government, where greater autonomy of sub-central layers of government and a closer matching of tax and spending responsibilities may be called for if the objectives of a smaller and more efficient public sector are to be achieved.

Inter-governmental relations: consensus and co-ordination

An important motivation behind Germany's constitution was to provide a federal framework in which the states would enjoy a high degree of autonomy, but the historical outcome has been rather different. In most areas the primacy attached to federal legislation leaves very little room for individual state autonomy. But at the same time the states are collectively allowed an important say in federal decision-making, so that German federal relations are based upon inter-governmental co-operation and the co-ordination of fiscal policies. Policy co-ordination between the levels of government is institutionalised in a number of councils, the most important of which is the Financial Planning Council *(Finanzplanungsrat)*, comprising the ministers of finance of the federation and the Länder as well as the Federal Minister of Economics and representatives of the communities. The council exchanges assumptions about the economic and financial development and makes recommendations on the co-ordination of budgets and financial plans of the federal, Länder, and communal governments. However, since the constitution stipulates that all territorial authorities are autonomous with respect to their budgets, the council cannot make any binding decisions.

The states are also directly involved in the federal legislative process. The Länder are represented in the second chamber of parliament, the *Bundesrat*, where they each have between three and six votes depending on their size. Direct involvement of the states in the federal legislative process contrasts with other federal countries such as the United States where the Senate typically does not represent state governments as such. Federal laws which affect directly the Länder financially or administratively require the consent of the *Bundesrat* and the trend toward tax sharing has *de facto* widened their right to consent, which now covers some 60 per cent of federal legislation. The *Bundesrat* may also veto legislation which does not require its approval, but such vetoes can be rejected by a majority of all members of the first chamber of parliament *(Bundestag).* The *Bundesrat* may also submit its own bills to the Bundestag.

The strong centralisation of the German federal fiscal system is thus characterised, *de facto,* by institutions which promote co-operation and consensus. While in the German constitution, co-operation and independence receive equal emphasis, in practice the area of state autonomy remains highly restricted, largely based on the constitutional principle of creating a uniformity of living conditions throughout the country. This has led to complex inter-governmental arrangements, which foster co-ordination but benefit neither from the discretionary flexibility of unitary states, nor from the potential for efficient resource allocation which is usually thought to come from devolved public-sector decision making.

Expenditure responsibilities overlap

Public expenditure is characterised by shared responsibilities for implementation and finance which make for a system of quite complex overlapping competencies. The federal government is in principle responsible for matters considered to be of relevance for the whole country, as well as for the redistribution of income across the federation. Since the writing of the constitution in 1949, justification of federal legislation in these terms has become quite broad. The tasks comprise, *inter alia,* federal public transport, formulation of economic policy, supervision of competition policy, and transfers to the social security system. The Länder are responsible for all tasks which are not explicitly attributed to the federal government. Major competencies comprise the education and university systems, health services and the provision of law and order. The

communities are attributed the right to control local matters such as road construction, school building, health services, public transport and energy supply, as well as the provision of social assistance. The functional segregation of responsibilities is not as strict as the above dichotomy suggests, however, and the system is now characterised by overlapping responsibilities. For example, all three territorial authorities and the social security system perform tasks with respect to health care: the Bund sets the pertinent legal conditions, the statutory health funds finance operating costs for hospitals, the Länder finance hospital investment and regulate hospital capacity, and the communities provide local health care services. Investment in transport and aid to enterprises are other examples where spending involves all three layers of government. Administrative responsibilities belong to a considerable degree to the Länder.[18]

Shared responsibilities also involve co-financing between the federal and the state governments. The constitution recognises three types of Länder expenditures which can be co-financed by the Bund. For each of them, the Bund takes on between 50 and 60 per cent of the associated costs (Table 10). First, ''Joint Fiscal Tasks'' *(Gemeinschaftsaufgaben)* comprise those projects which are jointly planned by the Bund and the Länder, covering the fields ''investment in universities'', ''improvement of the regional economic structure'', and ''improvement of the structure of the agricultural sector and coast protection''. Decisions about co-financing and project planning are taken jointly with equal voting rights for the Bund and the Länder (combined). Co-funding the programme ''improvement of the regional economic structure'' is conditional on the recipient state having financing capacity below the federal average or its economy being subject to structural change such that disadvantages for it could be expected. Second, Financial Aid *(Finanzhilfen)* can be granted for major Länder or communal investments for the purpose of averting economic disequilibrium, equalising differing economic capacities or promoting growth. Finally, grants for ''Disbursement of Funds'' *(Geldleistungsgesetze)* support social transfers by the Länder and may be implemented at the request of the federation.

Inter-governmental planning and co-financing of public activities have increased in importance. Mixed financing and co-determination between levels of government was formally introduced into the constitution in 1969. Between the end of the 1970s and reunification federal outlays devoted to cofinancing of Länder tasks – Joint Fiscal Tasks, transfer for Länder Disbursements and

Table 10. **Länder expenditures co-financed by the federal government**

	Joint Fiscal Tasks	Financial Aid for Investment	Disbursement of Funds
Major projects or disbursements	University construction Regional economic structure Agricultural structure and coast protection	Communal road construction Construction of social housing Modernisation of residential buildings Flood protection Energy provision and saving	Education benefits Residence benefits
Financial contribution of the federal government			
Share	50 per cent or more	50 per cent or more	50 per cent or more
Volume in 1990			
Billion DM	8.3	7.6	13.8
(Per cent of GDP)	(0.3)	(0.3)	(0.6)
Volume in 1997			
Billion DM	13.0	13.3	16.0
(Per cent of GDP)	(0.4)	(0.4)	(0.4)
of which: To the new Länder	4.8	7.3	2.4

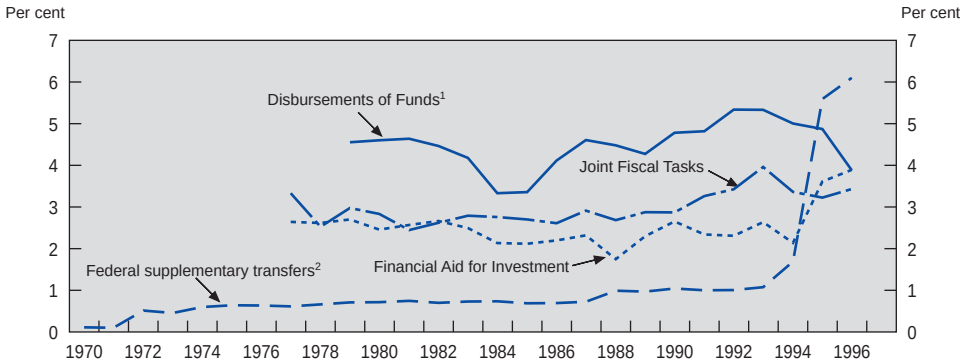
Source: Ministry of Finance; OECD.

Financial Aid for Investment – increased by some 15 per cent in real terms. Measured as a share of total federal revenues, federal support for Disbursements and Joint Fiscal Tasks increased over the 1980s (the former after a significant drop in the first half of the decade) while Financial Aid for Investment declined only a little. Federal grants combined surged after reunification, mainly to aid investment in the new Länder (Figure 22). In addition, supplementary federal transfers *(Bundesergänzungszuweisungen)*, which are not earmarked, increased slowly but steadily as a percentage of federal government revenues after the beginning of the 1970s and surged after unification.

Increasing reliance on tax sharing...

The federal revenue system has relied increasingly on tax sharing (Box 4), with the fraction of own taxes, accruing exclusively to a particular tier of government, declining at each level (Table 11).[19] Given that the percentage shares in VAT receipts of Bund and Länder can be adjusted by simple legislation,

Figure 22. **FEDERAL CO-FINANCING OF SUB-CENTRAL FISCAL PROJECTS**
Grants by federal government as per cent of total federal receipts



1. The drop in 1996 is dominated by the restructuring of child benefits for whose provision the Länder are no longer responsible.
2. These are not earnmarked. Between 1990 and 1994 additional federal transfers were channelled through the German Unity Funds to the new Länder, but are not included here.

Source: Ministry of Finance; OECD.

the vertical distribution of VAT revenue has quite often been adjusted; rising spending pressure, in particular due to reunification, and transfers to the new states by the old Länder led to the Länder share in VAT receipts being raised to its present level.[20] The importance of tax sharing will increase further with the termination of the local business capital tax in 1998 since VAT sharing is being extended to the communities as compensation (2.2 per cent share in 1998).

Since the beginning of the 1960s the Länder have received an increasing allocation of the territorial authorities' total tax receipts, raising their portion by almost 10 percentage points to some 40 per cent. While the portion allocated to the communities fluctuated at around 13 per cent, that received by the Bund declined accordingly to some 48 per cent (Table 11).

... and declining tax autonomy

In line with the rising importance of tax sharing, the degree of tax autonomy of the sub-central levels of government, defined as the right to vary either tax bases or tax rates, has declined. The most fundamental change occurred in 1955,

Box 4. **Tax sharing and revenue equalisation in Germany**

The German system of tax revenue allocation comprises two distinct processes: the first involves the division of shared taxes between the territorial authorities – the Bund, sixteen Länder and the communities – according to agreed criteria, while the second adjusts the distribution of tax revenues to achieve distributional goals.

The primary system of tax sharing

About three quarters of overall tax revenues in Germany are shared vertically between different layers of government (shared taxes), the remainder accruing to one particular tier only (own taxes). Shared taxes mainly comprise VAT and income taxes (personal and corporate), which account for 30 and 40 per cent of total tax revenues, respectively. Both are subject to federal legislation, but tax collection is administered by the Länder. The percentage shares of the Länder and the federal government in income and corporate tax revenue are stipulated in the constitution such that any reallocation requires a constitutional change. For VAT, on the other hand, the percentage shares of the Bund and the Länder can be adjusted by simple legislation, subject to the consent of the Länder in the Bundesrat. According to the constitution, VAT revenues should be allocated such that the coverage of expenditures by revenues be even between the Bund and the Länder. Accordingly, the vertical allocation of VAT receipts has been regarded as the main flexible element in the overall distribution of tax revenues between the Bund and the Länder. For 1998, 3.64 per cent of VAT receipts have been assigned to the Bund for transfers to the pension system. The communities receive 2.2 per cent of the remaining VAT revenues. Of the remainder, 50.5 per cent accrues to the Bund and 49.5 per cent to the Länder. With respect to income tax receipts (net of the solidarity tax surcharge) 42.5 per cent accrue to the Bund and the Länder respectively and the remaining 15 per cent accrue to the communities. Other shared taxes are the business trade tax and the mineral oil tax. Major own taxes are tolls, the insurance tax, the tobacco tax and the ''solidarity'' income tax surcharge accruing to the Bund, the inheritance and the automobile tax accruing to the states, and the real estate tax and consumers taxes accruing to the communities.

Horizontal apportioning of income-tax revenues between states proceeds according to the residence principle, *i.e.* income tax accrues to the tax-payer's resident state.[1] Corporate tax revenue is divided among states by a formula which is based on plant location, taking into consideration that companies may have branches in different states. Another difficult horizontal assignment problem concerns VAT revenue. Neither the source nor the residence principal can be applied in a meaningful way. As a pragmatic solution, VAT revenue is simply distributed to states on a per capita base.

The secondary system of revenue equalisation

This primary system of tax revenue allocation is transformed by a second system which redistributes revenues between the territories based on the constitutional objective

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to create broadly equal living conditions across regions. Länder which are financially weaker in terms of their primary tax receipts receive both horizontal transfers from financially stronger Länder and vertical transfers from the federal government to enable them to finance their fiscal tasks. There is a similar redistribution of revenues within the Länder for financially weaker communities.

Financial equalisation of Länder tax receipts is conditional on the primary distribution of tax revenues over the Länder and communities and proceeds in three stages, the first two of which are horizontal while the third involves vertical transfers from the Bund. At the first stage *(Umsatzsteuervorwegausgleich)* up to 25 per cent of the value-added tax receipts of the Länder is redistributed in favour of Länder which are endowed with relatively low-tax revenues after the primary tax allocation. At present, redistribution at this stage exclusively benefits the new Länder (Table 12, Column 1). Equalisation at the second stage *(Länderfinanzausgleich im engeren Sinne)* is conditional on the Länders' tax revenue endowment after stage one, including half of its local communities' tax receipts. For each Land the resulting finance endowment *(Finanzkraft)* is compared with its ''financial needs'' *(Finanzbedarf)*. Unlike in many other federal systems, the concept of ''need'' is not related to expenditures but simply refers to per capita tax income for the whole country. However, expenditure needs are taken into account to the extent that effective population is regarded as 35 per cent higher than actual population for the three city states: Hamburg, Bremen and Berlin. Revenues are redistributed from states whose financial endowments exceed their ''financial needs'' (or reference income level) to the states for which the opposite is true (Table 12, Column 2). For contributing states, the surplus of financial endowment over financial needs is transferred to the receiving states at a progressive rate which increases up to 80 per cent. At this stage financially-weaker Länder reach 95 per cent of their ''financial needs''.

At the third stage of the equalisation system, the financial endowment of the financially weaker Länder is lifted to at least 99.5 per cent of their ''financial needs'' by supplementary grants of the federal government *(Fehlbetrags-Bundesergänzungszuweisungen)* (Table 12, Column 3). In addition, the Bund makes supplementary grants for special needs *(Sonderbedarfs-Bundesergänzungszuweisungen)* for the purpose of compensating some states for special burdens (Table 12, Column 4). Special supplementary grants benefit in the first place the new Länder, which obtain a lump-sum transfer of DM 14 billion per year until 2004.[2] Special supplementary grants are also paid to small Länder to compensate them for higher administrative costs and to western Länder as a compensation for higher fiscal burdens due to reunification (digressive over ten years). Two Länder obtain special supplementary transfers as federal aid for their debt servicing obligations.

Although the equalisation system was redesigned to include the new Länder from 1995 onwards, its features changed little from the former system, which remained in force until 1994 for the old Länder. The restructuring was necessary because inclusion of

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the eastern states into the old system would have implied substantial transfers from the western to the eastern Länder and an effective elimination of transfers to western states. As a result, the volume of supplementary grants was substantially increased. In the interim period between 1990 and 1995, funds were redistributed to the new states mainly via the German Unity Fund, which was financed via contributions of the Bund (31 per cent) and the old Länder (10 per cent) as well as by borrowing on the capital market (59 per cent). Between 1990 and 1994 the German Unity Fund redistributed DM 160.7 billion to the new Länder (on average about 1 per cent of GDP annually).

1. This rule has been criticised by the city states, Berlin, Bremen and Hamburg, which lose tax revenue from workers living in neighbouring states.
2. The level of the special supplementary transfers to the new Länder will be re-examined in 1999.

when the Bund took over the legislative responsibility for the income tax, and the recent abolition of the local business capital tax has increased centralisation still further.[21] Sub-central governments have relatively little tax autonomy. Even for unshared taxes, the right to revenues and the right to change the tax legislation often diverge: major Länder taxes are determined at the federal level and major local taxes at the Länder level. A notable exception is the local business tax *(Gewerbesteuer)* whose rates are determined by the communities within certain limits. The business tax accounts for some 40 per cent of the communal tax revenues after tax sharing. Legislation with respect to the income and the value added taxes rests exclusively at the federal level. By contrast, in several OECD countries regional or local governments have discretionary power over income and/or value added or sales taxes which are the most important taxes in the OECD. This applies for the United States, Canada, Denmark, Switzerland, Finland, and Sweden.

Up to unification the Länder on average were able to finance about 65 per cent of their total spending by tax revenues (own and shared), but this changed rapidly after reunification (Figure 23): the low taxing capacity of the new Länder and weak tax revenues in the wake of the economic downswing in 1993 have led to a decline in the tax-to-spending ratio to below 60 per cent, though with some increase more recently.[22] The communities finance around 35 per cent of their

Table 11. **Tax accruals by layer of government**

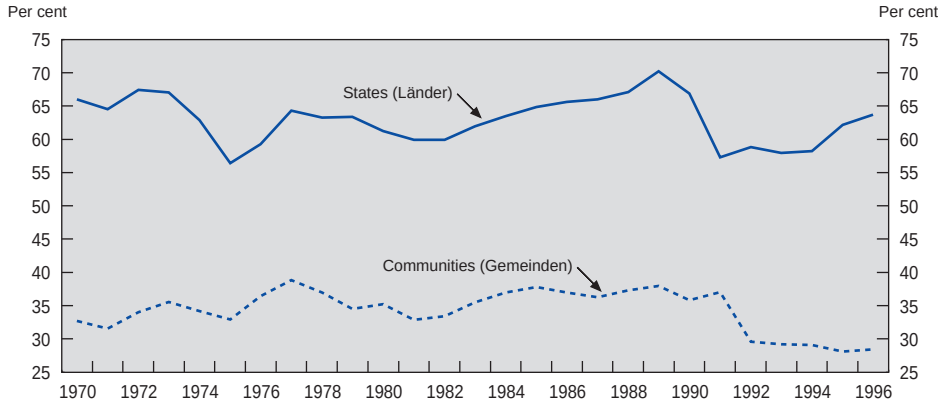
Per cent of respective total

	1951		1960		1970		1980		1989		1996	
Federal government[1]												
Portion of total tax revenues	58.0		54.9		54.8		50.1		48.7		47.8	
of which:												
Unshared federal taxes		89.0		76.6		32.8		23.4		24.0		16.1
Taxes shared with other layer of government		11.0		23.4		67.2		76.6		76.0		83.9[2]
Länder												
Portion of total tax revenues	28.0		30.6		33.1		35.4		36.9		39.8	
of which:												
Unshared Länder taxes		17.0		22.2		18.9		12.8		12.8		12.7
Taxes shared with other layer of government		83.0		77.8		81.1		87.2		87.2		87.3[2]
Communities												
Portion of total tax revenues	14.0		14.5		12.1		14.5		14.4		12.4	
of which:												
Unshared communal taxes		100.0		100.0		26.8		16.4		12.9		17.2
Taxes shared with other layer of government		0.0		0.0		73.2		83.6		87.1		82.8[3]
Total	100.0		100.0		100.0		100.0		100.0		100.0	

1. Excluding taxes which accrue to the EU.
2. Personal income tax including wage tax, corporate income tax, source tax on interest, VAT, business tax, mineral oil tax.
3. Personal income tax including wage tax, source tax on interest, business tax.

Source: Ministry of Finance; OECD.

Figure 23. **SHARE OF STATE AND COMMUNITY SPENDING FINANCED BY TAXES**[1]



1. Taxes as a percentage of total spending according to finance statistics. Taxes of Länder are net of federal supplementary transfers (Bundesergänzungszuweisungen).

Source: Ministry of Finance, Finanzbericht, various issues; OECD.

total spending by tax receipts, with the degree of tax sufficiency slightly increasing until the beginning of the 1990s. At present, tax receipts cover less than 30 per cent of total spending.[23]

The fiscal equalisation system

Revenues are equalised across states

The primary system of revenue sharing described in Box 4 is transformed by a secondary system based on the principle of burden sharing. Indeed, the constitutional principle that each layer of government has to bear the costs for the fulfilment of its public tasks has, in fact, been supplanted by the principle of revenue equalisation arising from the constitutional objective to broadly equalise living conditions. Therefore, financially weak Länder receive transfers from wealthier states (horizontal transfers) and transfers from the federal government (vertical transfers) to enable them to finance their responsibilities (Box 4). There is a similar redistribution of revenues within the Länder to financially weaker

communities. While the Länder equalisation system is fixed by federal law, there are different arrangements for the equalisation system at the communal level which are governed by legislation of the respective Länder. This section focuses on the design of the equalisation system for the Länder and its associated incentive structure.[24]

The fiscal equalisation system yields a substantial redistribution of income in favour of the financially weaker Länder (Table 12, Columns 5, 6, 7 and 8). In 1996, the per capita tax revenues of the poorest Land, including communities, amounted to 80.1 per cent of the average prior to redistribution; after redistribution and supplementary transfers it exceeded the average by 8.7 per cent. The coefficient of variation for per capita revenues even slightly increases as a result of the process. On the other hand, taking a long term perspective the fiscal equalisation system appears to have had little success in achieving convergence of the Länder's economic performance. Between 1970 and the beginning of transitory equalisation arrangements in 1990 only one state ceased to be a recipient. More importantly, among the old Länder, the dispersion of real GDP per capita has declined little, the standard deviation remaining at around one fourth of the mean over the last 27 years (Figure 24). Similarly, over the same time span, the dispersion of unemployment rates remained roughly constant. With active labour-market measures, like training and job-provision programmes, netted out, the disparities in states' unemployment are even higher. While this development also reflects the lack of regional differentiation in collective bargaining outcomes, it also indicates that there was little or no convergence in the Länders' employment capacity.

Disincentive effects of high marginal effective tax rates on the Länder

In contrast to other OECD countries, the inter-state equalisation component has a high weight in the overall redistribution, so that marginal increases in tax revenue accrue only partially to the states involved and are automatically redistributed, raising questions about the structure of incentives facing policy-makers and the electorate. The same problem is repeated at lower levels with the communal redistribution system, which equalises income of the communities within each state. With per capita revenue levels more equalised, the system has been repeatedly criticised for providing disincentives for the Länder to raise their own economic performance and tax bases.

Table 12. **Redistribution of tax revenues in the Länder financial equalisation system**

1996

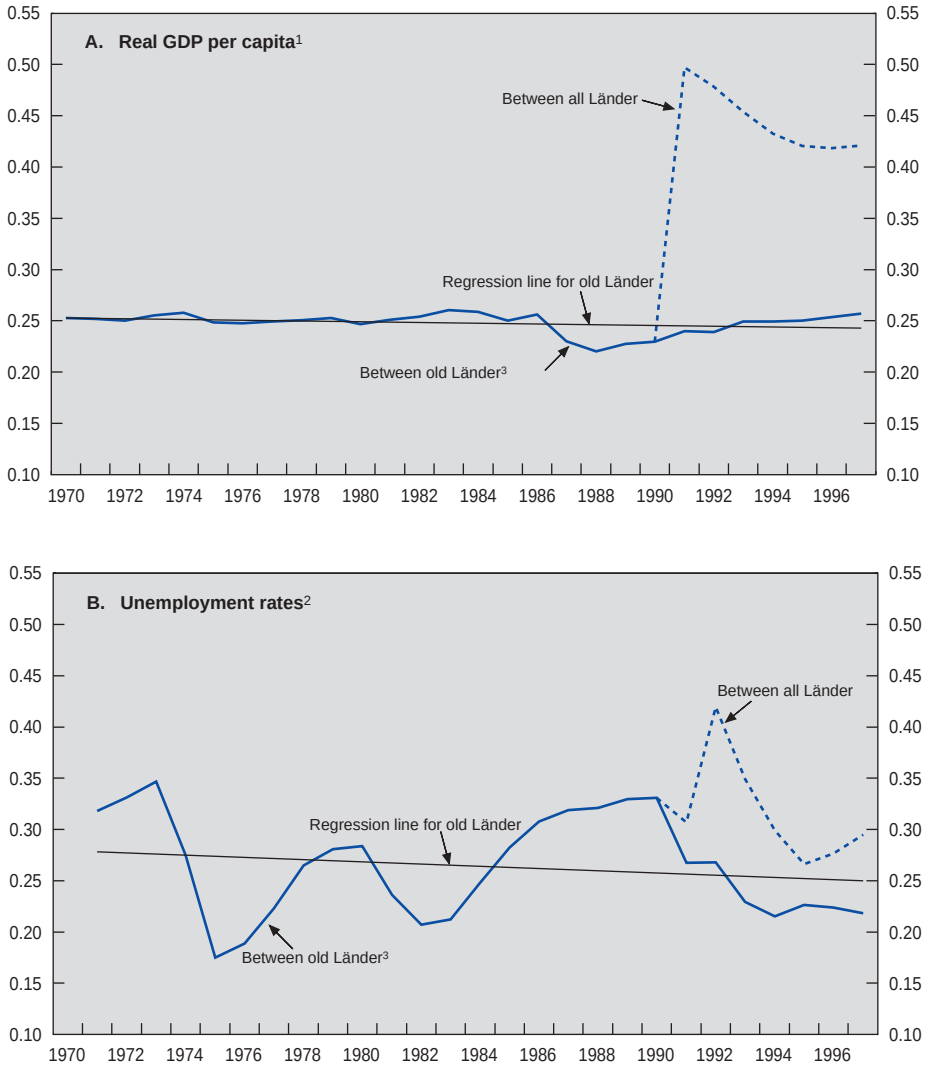
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	VAT equalisation	Horizontal state equalisation	Federal supplementary transfers	Federal special supplementary transfers	Taxes prior to redistribution[1]	Revenues after redistribution[2]	Taxes per capita prior to redistribution	Revenues per capita after redistribution
	DM million						DM	
North Rhine-Westphalia	–3 669	–3 125	0	0	92 115	85 321	5 132.4	4 753.9
Bavaria	–2 462	–2 862	0	0	61 416	56 092	5 099.3	4 657.3
Baden-Württemberg	–2 119	–2 521	0	0	53 082	48 442	5 116.6	4 669.3
Lower Saxony	–1 597	+553	830	456	35 225	35 467	4 507.3	4 538.3
Hesse	–1 233	–3 240	0	0	33 769	29 296	5 602.7	4 860.5
Saxony	+4 186	+1 965	846	3 658	17 747	28 402	3 904.1	6 248.1
Rhineland-Palatinate	–816	+231	352	625	18 259	18 651	4 564.2	4 662.1
Saxony-Anhalt	+2 821	+1 241	506	2 372	10 517	17 457	3 861.4	6 409.5
Schleswig-Holstein	–559	+16	24	368	12 733	12 582	4 643.2	4 588.2
Thuringia	+2 635	+1 127	462	2 172	9 367	15 763	3 760.2	6 327.7
Brandenburg	+2 194	+1 035	471	2 149	10 125	15 974	3 963.9	6 253.7
Mecklenburg-Western Pomerania	+1 820	+856	337	1 643	6 989	11 645	3 845.9	6 408.1
Saarland	–1	+234	201	1 825	4 655	6 914	4 293.1	6 376.6
Berlin	–710	+4 336	859	2 881	15 876	23 242	4 589.9	6 719.6
Hamburg	–350	–482	0	0	12 126	11 294	7 099.7	6 612.6
Bremen	–139	+635	120	1 998	3 502	6 116	5 166.5	9 023.1
Total	**±13 656**	**±12 229**	**5 008**	**20 148**	**397 503**	**422 658**	**. .**	**. .**
Average	0	0	313	1 259	24 844	26 416	4 696.9	5 819.3
Coefficient of variation	. .	. .	. .	. .	. .	. .	0.18	0.21

1. The figures include the partial reallocation of mineral oil tax revenues from the Bund to the Länder which was introduced in 1996.
2. (6) = (5) + (1) + (2) + (3) + (4).

Source: Ministry of Finance; OECD.

Figure 24. **VARIABILITY OF PER CAPITA INCOME AND UNEMPLOYMENT BETWEEN LÄNDER**

Coefficient of variation



1. Population measured end of year. For 1997 population as 31 June 1997.
2. Unemployment as a percentage of dependent employment.
3. Excluding west Berlin from 1991 onwards.

Source: Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder"; Statistisches Bundesamt; OECD.

By design, the financial equalisation system (in a more narrow sense, net of federal special-needs supplementary transfers) should leave the financial ranking of the contributors unchanged, but taking special transfers into consideration it changes the ranking in terms of per capita revenues.[25] This aspect has come in for some criticism. But, disincentive effects become apparent by considering the effective outflow of additional tax revenues generated by the Länder which is implied by the rules of tax sharing and subsequent redistribution. For individual states, an additional DM 1 million in income tax receipts – either personal or corporate – generates only between DM 80 000 and DM 290 000 in extra tax income, depending on the Land (Figure 25).[26] The remainder is allocated to both the Bund and the other Länder. For wealthy states the equalisation system transfers part of the additional revenues after tax sharing to poorer states. However, for less wealthy states, including those in eastern Germany, additional revenues after tax sharing reduce the eligibility for equalising vertical and horizontal transfers. With effective taxation of additional income tax revenues amounting to between 70 and 90 per cent, the system produces few incentives for the Länder to create conditions which attract companies and which increases their GDP.

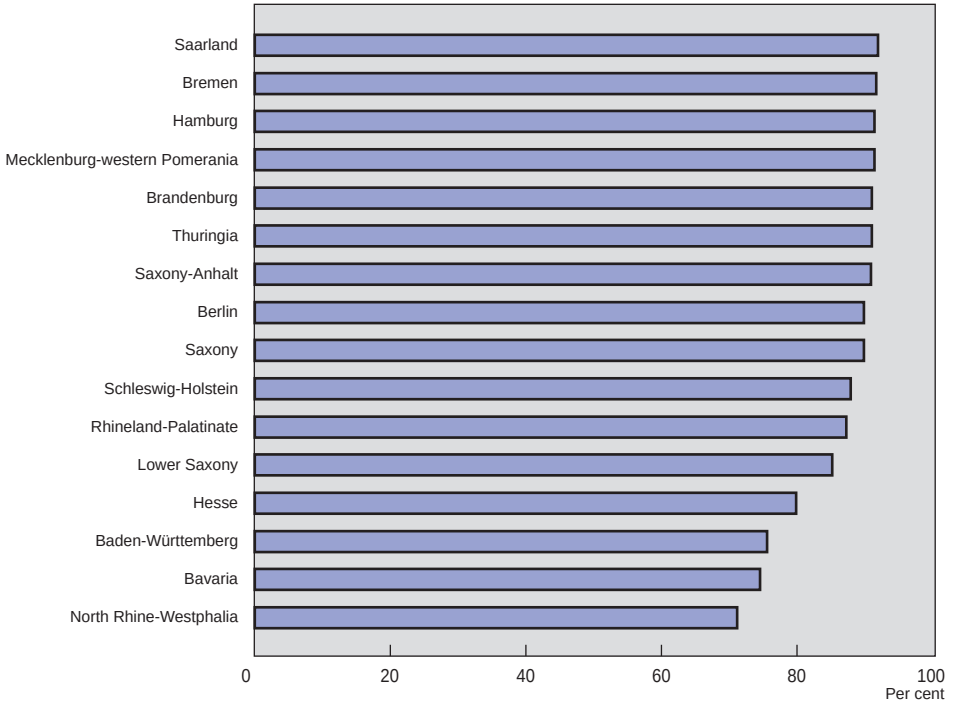
Low rates of tax auditing by the Länder may also be due to the fact that although they bear the cost of tax administration, only a small fraction of additional tax revenues accrues to them, so that it hardly pays from the individual Länder perspective to strengthen audits. A recent study evaluating auditing statistics concluded that the average time between company tax audits amounted to between seven and twelve and a half years, depending on the state. For small enterprises, the average period between audits extends up to several decades.[27] A low incidence of tax auditing creates opportunities for enterprises to reduce their effective tax burden. Indeed, ‘‘tax competition’’, driven by the political incentive to seek to increase employment, could take the form of differential state interpretation of the provisions of federal tax law.

Reform requirements: ensuring support while reducing disincentives

Given that the existing system of financial equalisation appears to discourage the development of taxable activity, while not helping to unwind existing disparities in economic capacity, the aim of equalising living conditions among the states appears to have quite harmful economic side-effects. Conversely, a

Figure 25. **OUTFLOW OF LÄNDER TAX REVENUES DUE TO INTERACTION OF TAX SHARING AND FINANCIAL EQUALISATION**[1]

Per cent

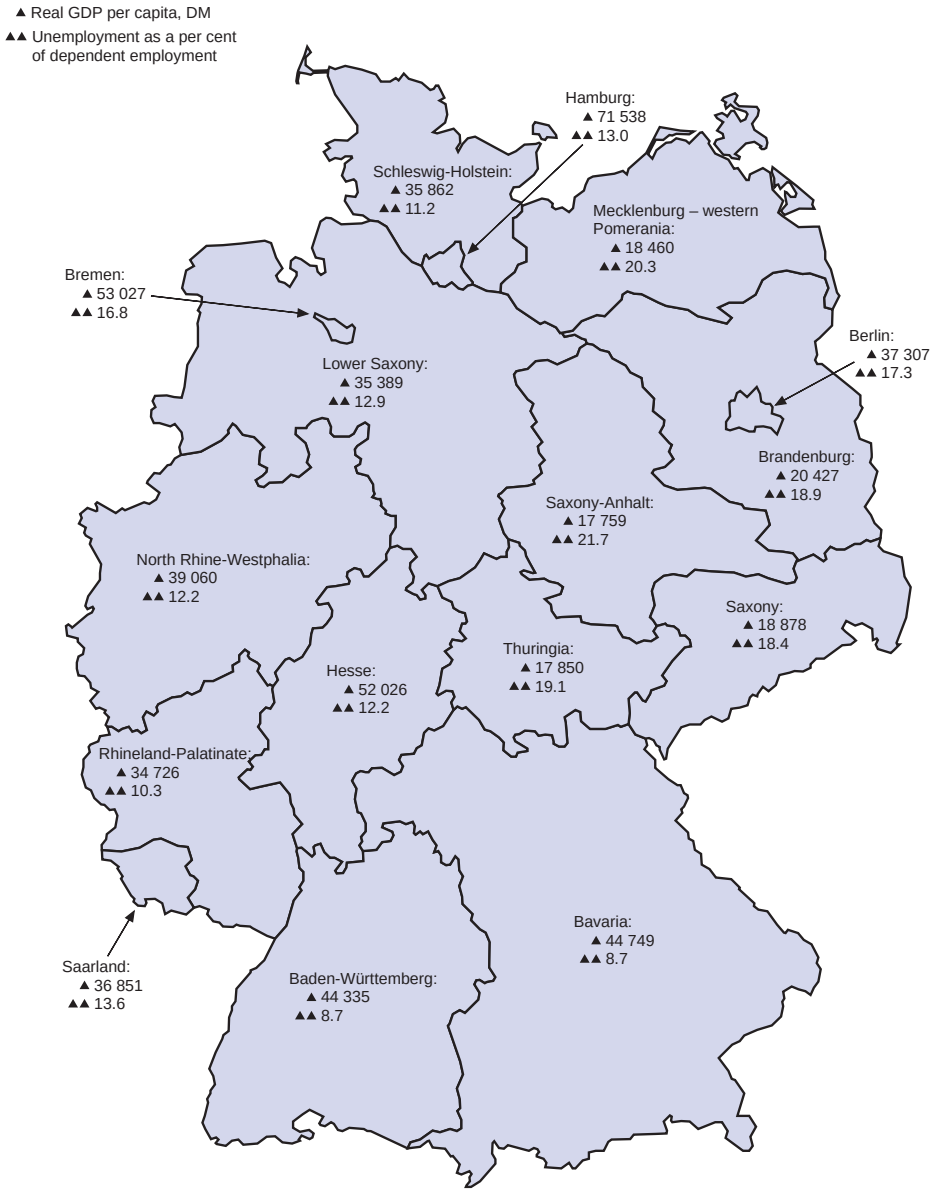


1. In 1996. Percentage outflow arising from an assumed DM 1 million increase in the Land's income tax receipts.

Source: Huber, B. and K. Lichtblau, mimeo, Germany's Federal Financial System – An Analysis and a Reform Proposal, Munich, 1998.

system which reduced the marginal taxation of Länder revenues significantly and increased the transparency of the system would be likely to improve public sector efficiency without necessarily compromising the basic aim of the fiscal equalisation process. Indeed, there may be areas where a more incentive-based system could be more effective in achieving this objective than is currently the case. The latter aspect is of particular importance with respect to the new Länder which command a revenues base which still falls dramatically short of the taxing capacity of the western Länder (Figure 26). In view of the fact that the financial equalisation system in the narrower sense (excluding special-needs and

Figure 26. **REAL PER CAPITA INCOME AND UNEMPLOYMENT IN THE LÄNDER**[1]



1. In 1997.

Source: Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder"; Statistiches Bundesamt; OECD.

supplementary grants) involves transfers accounting for less than 10 per cent of the Länders' and communities' tax revenues, the reform task would appear to be a manageable one.

One such a reform, which is consistent with ambitious redistribution objectives, could be realised by implementing a two-stage system which allocates lump-sum payments, fixed over a multi-annual period, to poorer states at stage one and redistributes taxes between states at the second stage conditional on their financial capacity as it prevails after stage one. In such a system, lump-sum transfers to particular states would achieve a minimum endowment of the Länders' financial resources over a multi-annual horizon in accordance with political objectives. After the reference period has passed, the allocation of lump-sum transfers would be reviewed and new allocations decided. In principle, the transfers at the first stage could be financed by contributions of both the Bund and the Länder. Given that the financial endowment of the poor Länder would have been already raised at the lump-sum stage, the transfer rate for redistributing revenues between the Länder at the second stage could be reduced to a fraction of the rates prevailing in the present system. For transparency and administrative simplicity, a single linear transfer rate would need to be applied and the entire equalisation system should be confined to two stages only. It is important to note that such a system could be introduced in a way that leaves the financial endowment of the parties involved unaffected in the transition phase. Thereafter, all parties would gain from improved efficiency of the system.[28]

Fixing lump-sum payments over a multi-annual horizon is a necessary condition for reducing the effective marginal taxation of the Länders' tax revenues in the redistribution process, which in turn generates incentives for the Länder to strengthen their own revenue base. If instead payments at stage one were frequently revised in accordance with the evolution of the Länders' revenue raising capacity, the equalisation process would effectively produce disincentives similar to those of the present system.

Implementing the Stability and Growth Pact: the need for a domestic equivalent

The issue of fiscal incentives and Länder responsibilities overlaps with the debate on the implementation of the Stability and Growth Pact. Germany is

obliged under the terms of the Stability and Growth Pact to observe a general government deficit limit and to accept sanctions including fines in the event of non-compliance (see Box 3), and this has created a need to reconsider aspects of the current federal system. Arguing that under present conditions the Länder could run excessive deficits at the expense of the Bund which would have to bear the fines under EU regulation, the government has proposed to determine legally binding allocations of the Maastricht deficit limit both vertically between the Bund and the states, including the social security system, and horizontally across the Länder. Such budget caps would only apply in the event of an excessive budget deficit for Germany as defined in the Maastricht treaty. It is proposed that violations of deficit limits would determine the allocation of EU fines among Bund and Länder. The emerging debate has concerned questions regarding the extent to which binding deficit allocations are required; what fraction of the total deficit should be allocated to the Bund and the states; how allowances should be shared between the Länder, and to what extent fines should be borne by governments which exceed the allowance. Most fundamentally, the issue of penalising individual Länder stands in potential contradiction to the whole principle of fiscal equalisation as presently set up and argues for the introduction of an incentive system which allows a certain amount of extra-state budgetary autonomy. This section considers important aspects of the debate.

According to the constitution deficits of the federal government must not exceed federal investment spending (see Chapter II). This ''golden rule'' is also contained in some Länder constitutions. While the purpose of the rule is to prevent deficit spending which is not considered to be economically useful, it is not designed as an instrument to secure deficit limits in absolute terms. In contrast, allocating deficit allowances to the different layers of government appears to be the most suitable way to translate the requirements of the Maastricht treaty into an internal stability arrangement.[29]

In principle, supplementing the Maastricht provisions by internal arrangements would also be beneficial with respect to the stabilisation of output over the business cycle. Since the beginning of the 1980s Germany has not in any major way considered fiscal policy an instrument for counter-cyclical policy. Rather, fiscal consolidation is seen as an important condition for fostering economic growth while the fiscal response to conjunctural fluctuations is largely left to the working of automatic stabilisers. Internal deficit allocations can support this

approach: to the extent budget caps for the territorial authorities foster the reduction of the general government structural deficit they simultaneously reduce the risk that the working of automatic stabilisers conflict with the 3 per cent deficit limit.

It is important that internal arrangements do not allow for a loosening of the definition of ''excessive deficits'' over the provisions of the Maastricht treaty. Negotiability of deficit limits could raise general doubts about how binding the caps would be, and might undermine consolidation efforts.[30] Moreover, agreement between all parties might be difficult to reach, and negotiations could give rise to strategic behaviour which would be counterproductive with respect to fiscal consolidation. Hence, the distribution of deficits should not be negotiable but strictly fixed, taking into consideration the higher cyclical sensitivity of the federal budget. The latter is attributable *inter alia* to the fact that the Bund and the social security system bear the bulk of unemployment related benefits which provide a major element of social and fiscal stabilisation at the Länder level. There are at present two formula being negotiated covering the broad vertical allocation of the deficit limit: one foresees equal deficits for the states and the federal budget including the social security system, while the other, proposed by the states, places the deficit allocation at a ratio of 60/40 in favour of the states.

Regarding the allocation of deficit limits between states, financially weaker Länder (especially those in the east) have argued for an allocation in line with past deficits, motivated by the perception that high deficits reflect high financial needs which would not be taken care of adequately by other allocation rules. Indeed, the deficit of the new Länder (financial basis, including the communities and including Berlin) averaged 5.1 per cent of own GDP in 1995 – far above the limit deficit/GDP ratio for the states as a whole of between 1. 5 and 1.8 per cent depending on the vertical allocation rule chosen. Moreover, above average deficits in the east do coincide with higher state investment but also with higher outlays for personnel (Table 13).

Allocating deficit allowances on the basis of past deficits has not been welcomed by low deficit states, which believe that it would effectively imply a punishment for a policy of fiscal consolidation in the past and add to existing disincentives for economical usage of resources. However, although a per capita allocation would be preferable on efficiency grounds, it would imply that the deficits of the new Länder would have to fall significantly, given that the eastern

Table 13. **Spending and deficits of the Länder**

1995

	Deficit/ GDP[1]	Investment/ GDP	Investment per capita	Spending on personnel/ GDP	Spending on personnel/ per capita
	Per cent		DM	Per cent	DM
North Rhine-Westphalia	–1.3	2.4	1 034.8	7.9	3 401.0
Bavaria	–0.9	3.7	1 802.2	7.5	3 643.9
Baden-Württemberg	–1.0	2.7	1 299.0	7.8	3 730.5
Lower Saxony	–1.9	2.9	1 127.0	9.2	3 575.1
Hesse	–1.2	2.1	1 187.7	6.8	3 762.3
Saxony	–2.9	11.8	2 985.4	13.6	3 446.1
Rhineland-Palatinate	–1.6	2.9	1 091.8	8.5	3 251.6
Saxony-Anhalt	–5.5	10.7	2 633.1	15.6	3 817.1
Schleswig-Holstein	–1.5	3.0	1 167.5	9.2	3 625.3
Thuringia	–4.4	10.7	2 595.0	14.7	3 559.1
Brandenburg	–4.5	9.2	2 472.6	13.4	3 600.2
Mecklenburg-Western Pomerania	–5.2	10.9	2 748.4	14.1	3 550.3
Saarland[2]	0.5	2.2	872.7	8.8	3 560.9
Berlin	–7.0	4.6	2 000.6	12.4	5 449.2
Hamburg	–1.2	1.5	1 198.0	6.2	4 876.4
Bremen[2]	0.1	2.2	1 254.4	9.0	5 114.7

1. Balances according to finance statistics, including purely financial transactions (privatisations and credits). Communities are included.
2. Balances of the Saarland and Bremen include transfers from the federal government to be used for debt servicing. Net of these transfers the deficit amounts to –3.2 per cent of GDP for the Saarland and –4.5 per cent of GDP for Bremen.

Source: Statistisches Bundesamt; Ministry of Finance; OECD.

states account for 21½ per cent of the population but account for 35 per cent of the overall Länder deficit in 1997. Hence, to avoid the necessity of strongly curtailing investment outlays, which are necessary to catch up with the western states in terms of infrastructure, a transitory arrangement could be made which allows the new Länder to converge gradually from their past deficit positions to the population-based deficits. Alternatively, the allocation of deficit allowances could be made on both investment and population shares.[31] To the extent that the above-average investment of the new Länder will level off with their catching up with the west, the two rules should produce similar allocations of allowances in the medium term.[32]

According to the government's plans, deficit violations would be sanctioned by allocating the EU fines between the Bund and the Länder. However, for sanctions to be effective, the implied financial burden should not be allowed to

trigger burden sharing or equalising transfers from other governments. It is questionable to what degree this condition could be met in Germany in view of the fact that, based on a ruling of the constitutional court, two states are presently receiving federal supplementary transfers for the purpose of supporting their debt servicing. Hence, legal conditions would need to be created which prevent any bailing-out. Similarly, allowing for relaxation of deficit allowances in cases of exceptional interest burdens and budgetary emergencies would run the risk of being counter-productive since it would relieve the pressure for state compliance. Instead, the opportunity should be taken to increase the degree of state accountability in fiscal matters.

Matching expenditures and revenues: the case for greater state autonomy

Since fiscal consolidation on all levels of government is fostered by efficient arrangements for spending and financing, increasing state fiscal autonomy would seem to be a key condition for the smooth functioning of an internal stability pact, as well as reducing the inefficiencies inherent in the present redistribution system. At the same time, as noted in Chapter II, all levels of government will remain under considerable pressure to improve their efficiency and from this viewpoint reforms would need to aim at establishing a greater degree of congruency between spending and the responsibility for financing by level of government. As noted above, responsibilities are at present diffuse, with co-financing in particular blurring the lines. Clarifying lines of responsibility inevitably raises the question of improving taxing rights for lower levels of government. These questions are taken up in this section.[33]

Co-financing goes too far

Co-financing and co-operation reflect the weight that Germany attaches to realising broadly equal living conditions across the federation, but in practice they probably also provide incentives to expand spending. The economic theory of federal fiscal relations suggests that co-funding is warranted to the extent that externalities exist between governments because leaving the financing exclusively to the government which provides the goods could lead to underprovision due to free-riding of consumers in other states. Maintaining and upgrading energy

and water supply, coast protection, and the environmental aspects of communal road construction could, for example, be seen as requiring inter-governmental co-operation and funding. Also, since different transport systems at the communal level can imply different environmental burdens (road versus subway), co-financing of environmentally preferable systems by the Bund might be justified.[34] Similarly, participation of the Bund in funding university construction might be justified since universities convey positive externalities to other states which can benefit from the education provided (Table 10). It is nevertheless important to note that the existence of externalities does not in every case require the involvement of the federal government. For example, part of university construction costs could effectively be transferred to the benefiting states by introducing fees for services.

If not justified by the existence of spill-over effects, federal grants run the risk of leading to overspending by the recipients and to efficiency losses, since the true costs of providing goods and services are not reflected in the Länder budgets. This allows sub-central governments to provide their electorates with services and benefits while passing on part of the associated costs to a wider community of tax payers. As discussed in the previous *OECD Economic Survey of Germany,* investment in hospitals serves as an example that a lack of congruency between spending and financing obligations on the side of the Länder can lead to overspending. Hospital capacity is regulated by the Länder which are also responsible for investment expenditures but not for financing operating costs or covering deficits. The lack of financial responsibility of the Länder implies a strong incentive to expand hospital capacity, thereby creating local employment.

Overall, it appears that a large share of projects covered by co-determination or investment aid is in fact relevant only for individual states.[35] Hence for a large part of such programmes, spending decisions and financing could be left entirely in the domain of the Länder. After reunification the government reduced its regional aid in the west and concentrated it on eastern Germany.[36] This policy should be considered an opportunity to examine the degree of co-financing and focus it more narrowly, taking into consideration equity goals.

Over the last few decades social benefits might have expanded less vigorously into new areas if the budgetary consequences for their funding had been concentrated in a single entity. A review of social spending to determine which benefits could be allowed to vary across regions would be important.

Responsibilities for financing could be given entirely to the Länder (and communities) if they are judged to be supplementary at the regional level, or if complete standardisation of provision is considered to be desirable, to the federal government. The restructuring of family benefits in 1996 is an example since child benefits have been deleted from the list of co-financed transfers.[37] If some regional discretion over benefit levels is desired, the Bund could fully finance a basic provision whose topping up could be left to the Länder or communities.

At present, federal law standardises the salaries for civil servants across the country, prohibiting state-specific salary adjustments. In view of the fact that outlays for personnel, including pensions, account for about 45 and 40 per cent of current spending of the Länder and the communities, respectively, considerable scope for the Länder and community finances could be gained if this regulation were relaxed, allowing regional pay settlements.

Also, new demands and changes in the competitive environment suggest that in some fields services could be provided more efficiently if governments co-operated or were combined into larger administrative units. For example, deregulation in the energy sector implies that demarcation and exclusivity conditions for local suppliers will be dropped (see *OECD Economic Survey of Germany 1996* and Chapter IV below). Securing the competitiveness of local utilities *(Stadtwerke)* without subsidisation will require combination across communes of generators into larger units. This, in turn, requires co-operation among the communes which are legally responsible for local energy suppliers.

Taxing powers need to be devolved

Incentives to expand spending, which are inherent in mixed financing, are strengthened by the restricted taxing powers of the Länder and communities. The limited – and decreasing – scope to finance additional spending out of own revenues reinforces demands for higher transfers from the federal budget and for increasing tax shares. This tendency is further supported by the strong position the Länder have in the federal legislative process. The fact that all laws which affect the Länder financially or administratively require approval from the upper chamber of parliament can be used for strategic behaviour to increase the states' share in overall revenues. Lack of tax autonomy also increases the risk that the Länder object to general fiscal measures which may have repercussions on federal government transfers. For example, although the proceeds from the

‘‘Solidarity’’ income tax surcharge are not earmarked and accrue exclusively to the federal budget, the Länder initially rejected the government’s plan to reduce the surcharge because they feared that transfers to eastern Germany might be curbed.

The tax autonomy of sub-central layers of government could be increased by allowing the states and/or communities some discretion with respect to income taxation. This goal could be achieved even if the tax base remains federally defined. Indeed, experience in other OECD member countries indicates that achieving the benefits of decentralised income-taxing while preserving the benefits of an administratively simple tax system requires that core elements of the tax codes be harmonised. In Canada, provinces have managed to preserve a single tax administration while exercising a considerable degree of taxing autonomy. It is evident, however, that allowing tax bases to differ between the various layers of government complicates the administration of taxes and increases the taxpayers’ compliance costs. Although Canadian provinces are in principle free to vary their income tax base, preserving administrative efficiency requires a close harmonisation between provincial tax bases and the federal base, which for most provinces is supported by formal agreements. As a consequence, tax competition manifests itself primarily in the provinces’ choice of income tax rates.[38] Similarly, in the United States a substantial majority of states define their taxable income bases along the lines of the federal definition. In Switzerland, where compliance costs are potentially high due to a lack of harmonisation in cantonal tax laws, a federal tax harmonisation law came into effect in 1993 giving the federal government a mandate, *inter alia*, for working towards uniform definitions of tax bases.

In Germany, with a given uniform tax code for the Federal Republic, an element of tax autonomy with respect to income taxation could be introduced by incorporating a Länder-specific surcharge into any reformed – and correspondingly reduced – tax schedule.[39] While compliance costs might be higher than in a completely harmonised system, this setting would require virtually no additional expertise on the side of the tax payer and minimise administrative costs on the side of the tax authorities.[40]

It may be argued that regional tax discretion and ensuing tax competition could lead to an erosion of the tax base of poor states. However, as noted above, in the present system states compete already to attract companies by offering

various types of costly aid, often cofinanced by the federal government, or – arguably – favourably interpreting the federal tax code. Enhancing regional tax autonomy, and the greater tax competition which could ensue, would generally entail a search for savings and increase the transparency of policies. Equity considerations could be incorporated by restricting the maximum size of admissible state-specific surcharges by federal law.[41] On the other hand, in an economy where the ratio of taxes and contributions to national income is already high, enhanced regional tax autonomy should not be taken as implying scope to raise taxes. Indeed the state surcharge in a reduced tax schedule would, as noted, be consistent with the overall aim of reducing the tax burden.

Similarly, incentives to reduce the tax burden and balance the costs and benefits of publicly-provided goods would be strengthened by a shift from taxation to a system which relied more strongly than at present on charging fees for services. Most importantly, this matches public service provision to local choice as customers tend to demand more tailored services when they pay for them directly.[42]

Summing up

While co-operation and budgetary independence are simultaneously emphasised in the German constitution, in practice the federal fiscal set-up has evolved towards consensus and co-operation, where equalisation of living standards takes precedence over public choice and economic incentives. While the states and communities do compete with each other to attract economic activity, the scope for expressing regional and local preferences with respect to public spending patterns remains highly restricted. The scope for reducing the burden of taxation by evaluating the marginal tax burdens against spending benefits is even more restricted. Shared taxes and the low reliance on own taxes make for a relatively inefficient control over public spending. In public finance terms, the German system thus does not achieve the allocational benefits which should accrue to a federal fiscal system.

At the same time, the system may be criticised for not achieving economic convergence among the states. The difference in regional living standards is smaller than in some other economies, but the fiscal system introduces a high degree of stationarity into the economy where the revenue equalisation system

offers no incentives to expand the tax base and may even promote tax avoidance. There has been almost no change in the dispersion of GDP per capita between the states over the last three decades. Greater dynamism could be achieved with a less confiscatory equalisation system and a higher degree of tax autonomy and both of these should form part of any balanced tax and expenditure reform.

IV. Implementing structural reform: a review of progress

The previous *OECD Economic Survey of Germany* noted that the government had introduced an impressive range of legislation covering product and factor markets, going broadly in the direction of the recommendations of the OECD *Jobs Study*. Implementing the programme – especially those aspects concerning labour cost and employment flexibility – has not been easy, and generous ''grandfathering'' provisions mean that it will take some time for the effects of several of the policy changes to become apparent (such as the increase in the early retirement age, or loosened dismissal protection). Some of the measures, such as the reduction of legally-required sick pay, represent changes in the framework for bargaining by the social partners[43] and their effect will continue to depend on how the partners adapt their behaviour to new circumstances. Chapter I of the present *Survey* concludes that the recent dramatic changes in capital and labour efficiency, which are important for the overall sustainability of the economy's performance, are an indication that firms are already taking advantage of the new-found flexibility arising from structural reforms and changed labour-market behaviour. The adjustment has contributed to a decline in employment: structural policy reforms are typically associated at first with short-term adjustment costs, with the full benefits gradually becoming apparent in the form of higher employment and increased productivity.

The progress made should, however, be seen in the perspective of the problems which need to be resolved. Service sector employment has been growing for some time, but not enough to compensate for the fact that manufacturing employment has been declining rapidly. Certain sectors of industry remain heavily subsidised and the situation in the new Länder remains serious, with severe labour market adjustments accompanying the restructuring of the economy away from construction. Long-term unemployment has continued to increase and

remains a particular problem among the old and less qualified. Looking ahead, pressures for further labour market adjustment are increasingly apparent: international competition from lower cost producers is likely to continue to intensify, and the start of European Economic and Monetary Union in 1999, in combination with completion of the single market to encompass previously protected sectors, places increased pressures on economic institutions, and on the labour market in particular. These tendencies are expected to continue, so that the challenge for economic policy will be to facilitate both the restructuring of employment and the reabsorption of the unemployed, probably for the most part in the service sector. This chapter discusses these issues in more detail, noting the impact effects of policy changes while pointing to how employment performance might respond in the recovery. The first section reviews the recent trends in employment and unemployment in some detail, while the following sections describe policy initiatives over the past year and the agenda for further action.

Dimensions of the employment and unemployment problem

Employment has continued to decline in both old and new Länder, even when adjustment is made for the policy-induced reduction in subsidised jobs during 1997. The fall in manufacturing employment has been marked in both east and west, even, until recently, in branches experiencing high export growth. Employment in construction fell by some 5 per cent during 1997 while employment in mining fell by 13 per cent in eastern Germany. These sectoral developments of employment have been unfolding over the last few years, but the effect on aggregate employment has been partially offset by steady growth in the service sector, with the exception of transport, communications and banking and insurance, which have all experienced declining employment in response to rising competition. It is therefore of some concern that the growth of general service employment *(Sonstige Dienstleistungen)* has shown signs of slowing markedly. Indeed, in the new Länder employment in this branch fell in 1997. There are a number of factors which might be at work: in the new Länder there appear to be deficiencies in the provision of manufacturing-related services rather than in the supply of personal services,[44] while in the old states there remains a deficiency in the provision of consumer-related services.[45]

In an accounting sense, the continued rise in unemployment has been due in almost equal measure to the further decline in employment and to the policy-induced reduction in retraining courses (not counted as unemployed) and in subsidised jobs (regarded as employment) (Table 14). At the same time, the composition of unemployment has continued to change: despite active labour market measures, long-term unemployment has risen further, as has the share of the unemployed accounted for by older workers (Figure 27). On the other hand, the share of younger workers without employment has declined, although the absolute number of those aged 15 to 24 years who are without a job has risen and there are some indications that the number without vocational training might also have increased. Unemployment statistics do not, however, fully reflect the number who would be available for work if jobs were available. A wider measure including ''hidden unemployment'' would cover those in early retirement on account of unemployment, those in training or on make-work programmes, those available for employment but not receiving benefits or those receiving benefits but not obliged under current circumstances to actively seek work (115 000 persons at end-1997).[46] One such measure places the overall level of unemployment at around 16 per cent in 1997, although the contribution of hidden to total unemployment has declined somewhat during the year.[47] Estimates of discouraged workers account for a further 2½ million persons. In the new Länder, official and hidden unemployment amounts to some 30 per cent of the labour force and the overall jobs deficiency remains very high even though net emigration and commuting to the old Länder have lowered the available work force (Figure 28).

Despite the large number of unemployed, flows into and out of unemployment remain quite significant. Inflows into unemployment rose by 1.8 per cent in 1997 (7.2 million persons) while outflows increased by 1.6 per cent (6.8 million persons). Although the Federal Labour Office was important in placing unemployed in jobs, self-placement remained the most used channel. Especially noticeable in 1997 was the high rate of job placement for the long term unemployed (415 000 out of a level of 1.4 million).[48] Nevertheless, a consideration of inflows and outflows in relation to the level of unemployment suggests a slowing of dynamics in recent years when compared with levels observed in the 1980s.[49]

Table 14. **The German labour market**

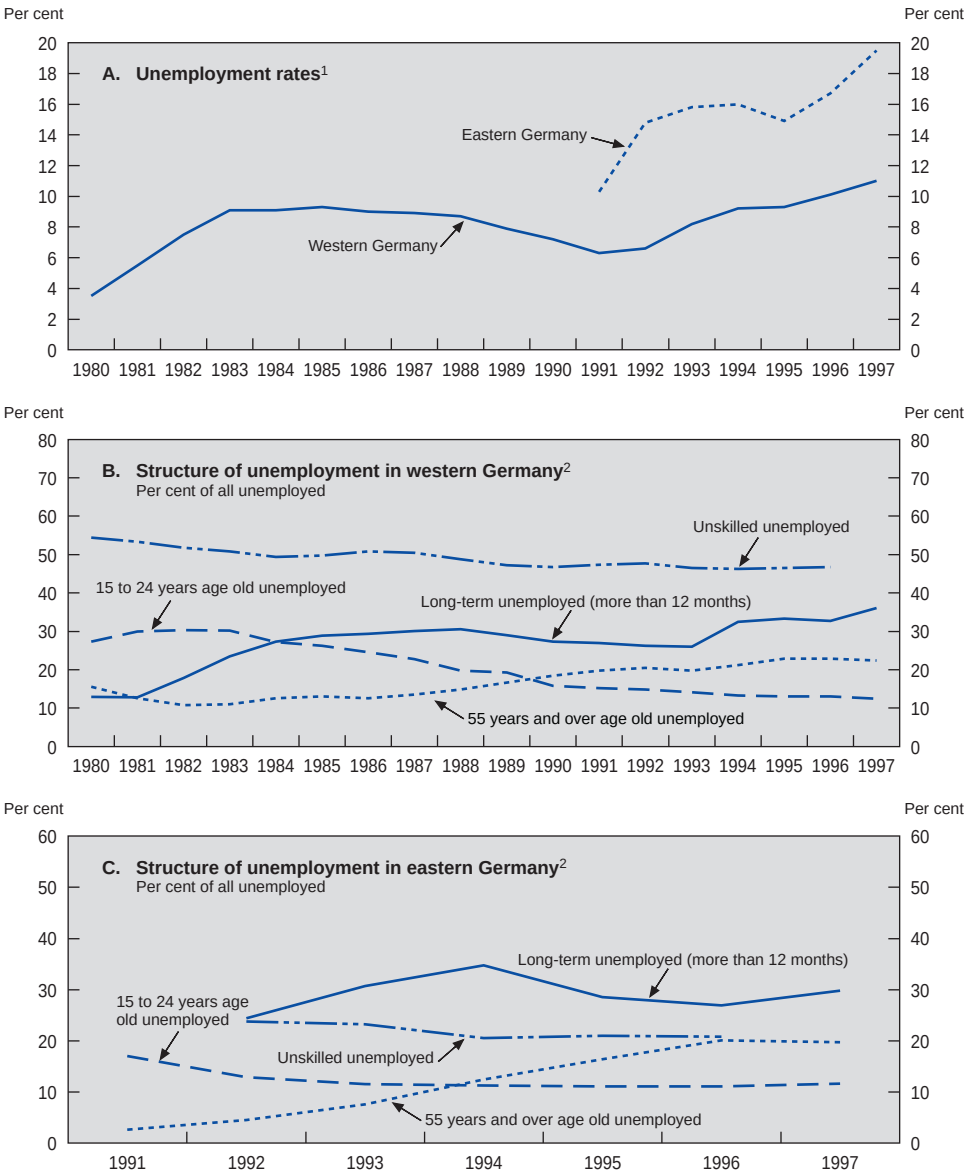
Thousands of persons

	Germany		Western Germany		New Länder (eastern Germany)	
	1996	1997	1996	1997	1996	1997
	Change from previous year[1]					
Labour force	–205	66	–97	–1	–108	67
of which:						
Unemployed[2]	348	407	279	138	69	269
Commuters[3]	–3	11	–7	8	4	3
Employed (domestic)	–556	–330	–383	–131	–173	–199
Memorandum item:						
''Hidden unemployment''	–31	–230	36	–38	–67	–192
of which:						
Short-time workers[4]	–3	–30	8	–23	–11	–7
Subsidised jobs (ABM)[5]	–9	–69	2	–12	–11	–58
Retraining courses[6]	–3	–150	–12	–55	9	–94
Early retirement programmes[7]	–42	–35	56	54	–98	–90
	Annual average					
Labour force	38 329	38 261	30 569	30 517	7 760	7 744
of which:						
Unemployed[2]	3 965	4 385	3 022	3 022	1 169	1 363
Commuters[3]	51	52	383	380	–332	–328
Employed (domestic)	34 415	33 928	28 156	27 875	6 259	6 053
Memorandum item:						
''Hidden unemployment''	2 138	1 953	1 060	1 023	1 078	930
of which:						
Short-time workers[4]	112	80	74	55	38	26
Subsidised jobs[5]	354	302	76	68	278	235
Retraining courses[6]	505	400	276	223	230	177
Early retirement programmes[7]	849	809	392	447	457	362
Estimates of discouraged workers[8]		2 400		2 000		400
	Unemployment rate (per cent)					
Official unemployment[9]	10.3	11.5	9.1	9.9	15.1	17.6
Official and hidden unemployment[10]	15.9	16.5	13.4	13.3	28.9	29.6

1. Fourth quarter to fourth quarter; according to national account definitions.
2. Source for 1996 and 1997: Federal Labour Office (Bundesanstalt für Arbeit).
3. Balance of commuters: persons commuting between East Germany and West Germany are included.
4. Programmes subsidised by the Federal Labour Office; number of persons calculated as equivalent to unemployment.
5. Public employment programmes subsidised by the Federal Labour Office.
6. Full-time participants in further education programmes and retraining courses, subsidised by the Federal Labour Office.
7. Programmes subsidised by the Federal Government, by the Federal Labour Office and by the Public Pension Systems. Excludes invalidity pensions.
8. As derived by the Institut für Arbeitsmarkt und Berufsforschung to include discouraged workers. This is derived not from survey data but by assuming a normal employment rate in a reference year and projecting forward.
9. Unemployment as a percentage of the labour force (without unpaid family workers).
10. Registered and hidden unemployment (excluding discouraged workers) as a percentage of the labour force (without unpaid family workers).

Source: Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, 1997; OECD.

Figure 27. **STRUCTURE OF UNEMPLOYMENT**

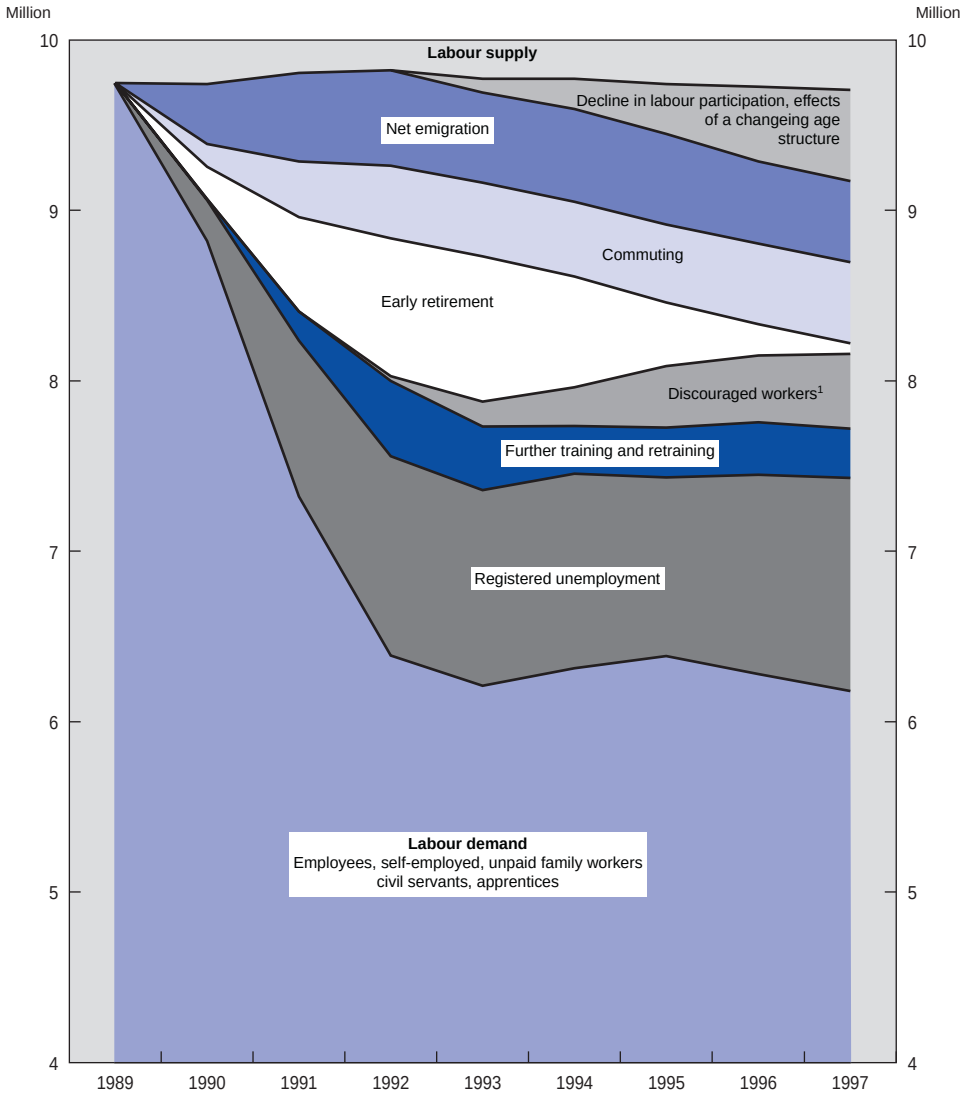


1. Annual averages.
2. September figures.

Source: Institut für Arbeitsmarkt-und Berufsforschung (IAB).

Figure 28. **LABOUR SUPPLY AND DEMAND IN EASTERN GERMANY**

Million of persons



1. Includes all persons who a) are not registered as unemployed at the labour offices, but are looking for work, or b) gave up temporarily looking for work due to discouragement. It is not derived directly from the Labour Force Survey.

Source: IAB.

Progress in labour-market reform

Increasing wage and labour cost flexibility

Although evidence is necessarily anecdotal, it appears that there has been a continued tendency for companies to conclude wide-ranging agreements, often at plant level, which guarantee employment and investment for several years in return for lower labour costs and improved flexibility *(Standortsicherungsverträge)*. In some cases such contracts have strained existing institutional and legal possibilities.[50] In part as a result, debate about the suitability of the system of branch-wide wage determination has gained force, with increasing criticism from all sides. Employers' associations and individual companies have made a number of proposals which, at a minimum, provide for greater flexibility at firm level subject to an overall branch agreement and such a system is already in force in the chemicals sector. Moreover, the largest trade union *(IG Metal)* has opened negotiations on more flexible agreements. One trade union proposal is for a ''tool kit'' approach by which the overall agreement would establish the blocks or components which can then be fashioned into different combinations at plant level. A major issue dividing the parties remains the role of the work's councils *(Betriebsrat)*: whether these should become increasingly involved directly in wage negotiations, in which case could they lose their existing character in the work place; whether a change in the law is necessary and at what stage of the reform process would such a change prove useful. In addition, there is a question of how greater decentralisation would in fact work, with companies and plants probably needing to negotiate ultimately with very large unions rather than just their workforces. A principle of German industrial relations law has been the need to preserve the equality of bargaining power – although some have argued that court decisions over the years have eroded this principle. Further, more fundamental, moves towards decentralisation would be likely to require changes in the industrial relations law and cartel regulations, making political agreement difficult to achieve.

Despite lack of agreement on the strategy of reform, the wage system continues to evolve rapidly in the direction of more flexibility. In 1997, the number of enterprises with their own contracts increased from 4 700 to 5 000 (it had been 2 500 in 1990), although the general branch agreements continue to dominate: there are currently 47 000 contracts registered. Contracts are now

much more complex, covering not only wages, bonuses and sick pay but also, *inter alia,* training, work time, part-time employment for older workers and employment security. Some contracts go so far as to allow – subject to approval by the social partners – increases in work time without compensation in return for employment security, as well as wages up to 10 per cent lower than in the agreement and starting wages for newly-employed and the unemployed up to 10 per cent less than normal. The administrative extension of wage agreements to non-contracting parties *(Allgemeinverbindlichkeitserklärung)* has declined in importance, although it has been in effect in the building industry since mid-1997 to support an agreement on minimum wages in the context of restricting competition from foreign workers.

Wage agreements in the western Länder resulted in tariff wages rising by an average of 1.5 per cent in 1997 (1.8 per cent in 1996) and once account is taken of cuts in some wage costs, such as calculating sick pay without reference to overtime and the reduction in some cases of the level of the thirteenth month's pay, actual growth of compensation has been lower than negotiated wage rates would suggest (*i.e.* more in the range of 1 per cent). In some cases, total compensation has declined in absolute terms. As before, however, actual wage payments usually exceed the negotiated norms, though by a declining margin.

In eastern Germany negotiated agreements apply to a much smaller proportion of the work force, with 30 per cent of employees currently being paid at rates lower than tariff wages (up from 23 per cent in 1996); many companies have left the employers' federations and do not take part in collective agreements, while union membership is also lower. Wage agreements lifted wages by 1.9 per cent on average in 1997 (3.6 per cent in 1996) and convergence with tariff wages in western Germany slowed, the average level rising from 89.0 per cent to 89.5 per cent in 1997. Unlike in the old Länder, actual wages are lower than those which have been negotiated so that in effective terms the average ratio of wages is much lower at 77 per cent. Moreover, over the longer term, relative wages in the new states have altered significantly. Over the period 1992 to 1997, negotiated wage increases at the branch level ranged from 51 per cent to 178 per cent. On the other hand, the administrative extension of minimum wages throughout the building industry represents a significant step away from greater flexibility, given the size of that sector in the eastern Länder.

Increasing working-time flexibility and easing employment security provisions

Work-time flexibility has been achieved by the widespread introduction of time accounts, allowing periods of overtime work to be compensated at other times by reduced hours of work.[51] The period during which hours need to be equalised varies between six months and two years. In addition, some wage agreements have introduced greater flexibility in the organisation of shifts. The system of time accounts appears to be working well, although there are still some concerns about the protection of accumulated hours in the event of bankruptcy. The level of overtime has declined markedly and it appears that machine operating times, which were low in comparison with other European countries, have probably increased.

Possibilities for increased flexibility have also arisen with respect to part-time employment of older workers prior to retirement *(Altersteilzeitgesetz).* In addition, shifting older workers from full-time to part-time work with compensating hires of the unemployed or by apprentices is supported financially by the Federal Labour Office. The law currently permits part-time work over a period of five years but the details are to be set in wage agreements. The maximum permitted period has now been extended to ten years, to take into account the steady rise in the statutory age for early retirement, which will reach 65 years in 2000. In addition, the possibilities for organising ''part-time'' employment of older workers outside of tariff agreements have been widened: paid-time can now be divided between work and leisure for a period of up to three years without the need for a wage agreement, which should serve to open the flexibility to small and medium-sized enterprises. In branches which are not normally covered by wage agreements, the division of time can be negotiated on an individual basis. Experience with the new system is limited, with negotiations still under way in many sectors. Up to now these wage agreements have usually envisaged a period of full-time work for half the time (*i.e.* full-time work for two and a half years) and no work at all for the remainder of the time to retirement. In addition, the agreements envisage a further top up of wages by the employers. For example, in the metal industry wages are topped up to 85 per cent of their previous level. In the chemical sector, which is marked by very flexible wage agreements, 15 000 workers have taken advantage of the new scheme and the number of apprenticeships has been increased significantly (24 per cent).

The ability to offer minor employment (jobs entailing a small number of hours per week *(geringfügige Beschäftigung)*) is an important source of flexibility, but in the past year the controversy surrounding such employment, which is exempt from social security contributions, has increased, with a number of proposals being put forward to restrict this potential ranging from a lower limit for social security-free jobs than the DM 620 per month at present, to the introduction of quotas by branch or firm specifying an upper limit to the number of such jobs.[52] The reason for the increased controversy was the publication of estimates by the Ministry of Labour indicating that the number of these jobs grew by a quarter between 1992 and 1997. The concern was expressed that this form of employment was being used to undermine the social security base and similar concerns have been voiced with respect to some forms of self-employment through contract work. It should be noted, however, that this area of labour statistics is exceptionally poor so that neither the level of casual employment nor its development over time is known with much confidence: depending on the source, there were between 1.6 million and 6 million such jobs in 1996.[53]

Flexibility needs to be preserved in this segment of the labour market and it should be noted that a reason for introducing a threshold for social security contributions in the first place was to avoid the administrative cost associated with maintaining individual accounts for those with only small contributions. These reasons are still valid. Moreover, charging full social security contributions would give the casually employed rights to unemployment benefits which could prove costly and difficult to administer. On the other hand, there could be a need to regularise taxation and social security arrangements. The obligation for the employer to pay a flat 20 per cent tax when an employee opts not to have personal income tax deducted may be open to abuse. With respect to social security contributions, the requirement that such jobs be added together to determine the social security liability for the individual seems inconsistent with there being no obligation to cumulate casual with full-time employment when determining contributions. This situation could be regularised. The casually employed are generally already covered by health insurance, leading to suggestions that they at least pay health contributions. However, any policy change in this area raises issues which go far beyond the immediate question of altering conditions concerning casual employment (see Chapter III of *OECD Economic Survey*, 1997 for a discussion of reforms to the health contribution base).

Job protection legislation was loosened at the end of 1996 by raising the threshold for the size of company covered from five to ten employees. The more restrictive conditions will nevertheless continue to apply to 70 per cent of all employees and 20 per cent of companies, while workers who were employed in their enterprise at the time the law was introduced will not be affected by the new provisions until September 1999. Small firms are also able to make greater use of fixed-term contracts. There does not yet appear to be any change in the size distribution of enterprises which could have been expected as firms expanded up to the level of the new threshold. However, as the recovery gets under way, the effects may become more pronounced.

Reducing the distortions arising from unemployment insurance and related benefits

Since April 1997, changes to the *Employment Promotion Law* have introduced stricter criteria for receiving unemployment-related income support, narrowing the grounds for recipients of unemployment benefits or assistance to refuse job offers *(Zumutbarkeit)* (see *Economic Survey*, 1997 for details). In particular, under conditions of structural change and an excess supply of labour, the unemployed are no longer permitted to refuse a job offer simply because it does not correspond to previous qualifications. Offers must also be accepted during the first three months of unemployment if the wage offer is within 20 per cent of the previous wage and this discount rises to 30 per cent in the following three months. From the seventh month, job offers must be accepted if the net income is greater than unemployment benefits.

Experience with the new provisions is still limited and strongly influenced by the continuing difficulties in the labour market. With respect to the new job-acceptability criteria, the Federal Labour Office has had difficulty enforcing the measures and faces extensive litigation in the labour courts. Moreover, some of the requirements for proving that benefit receivers are actively seeking work (*e.g.* visiting the labour office every three months) have been relaxed for some difficult groups such as older workers. A large number of recipients have now been unemployed for over six months and so are formally required to accept offers where the net salary is greater than the level of benefits. However, there is some anecdotal evidence that employers are reluctant to take on workers whom they regard as ‘‘over-qualified’’. The number of sanctions imposed by the labour

office has increased during the course of the year, especially for those receiving unemployment assistance, but statistics relating these to refusal to accept job offers in accordance with the new criteria are not available.[54]

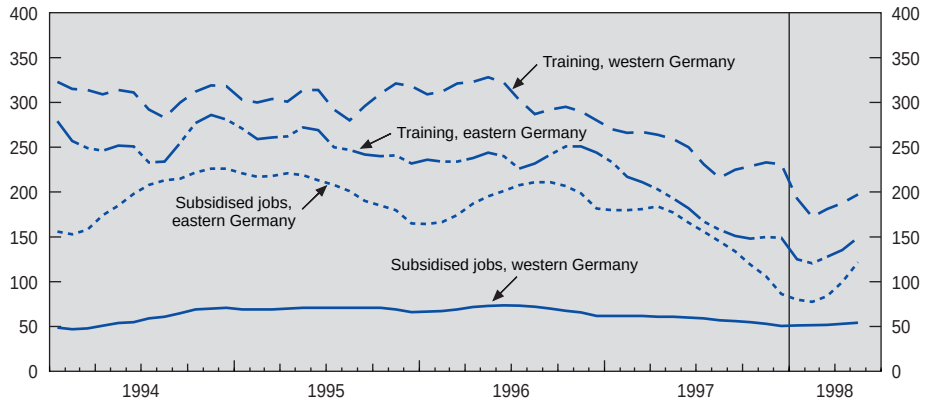
Since October 1996, the legal minimum for sick pay has been limited to 80 per cent of normal pay for the first six weeks of sickness or alternatively, the employee can opt to set aside one day of annual holidays against five sick days. As noted above, wage negotiations have mostly retained the sick pay at 100 per cent of normal pay, with other compensating cost reductions. In the process, it appears that both employers and sickness funds have sought to control the incidence of sick-leave more closely (see Chapter II for a discussion of reforms which make sickness funds more sensitive to costs). The number of days lost has declined rapidly from a peak of 11.3 per year in 1995 to 10.9 days in 1996 and possibly 9.7 days in 1997. Another reason for the decline could be increased job insecurity.

Enhancing active labour market provisions and re-entry to employment

The past year has been marked by major swings first to reduce and then to expand active labour market (ALM) measures. During 1997 the numbers on ALM programmes fell significantly (Figure 29) and expenditures declined from DM 46 billion to DM 36 billion. Two factors account for the fall. Budget pressures on the Federal Labour Office and local authorities were partly accountable.[55] But policy changes were also important in explaining the decline in numbers. Over recent years, evaluations of individual programmes have generally pointed to disappointing overall results with ALMs, especially in the new states.[56] This criticism also extends to further training *(Fortbildung)* and retraining programmes (3 million workers have participated in such measures in recent years in the new states).[57] Job-creation measures have frequently been criticised for effectively splitting the labour market (in Germany reference is made to the primary and secondary labour market, the latter covering subsidised workplaces). As a result, the reformed *Employment Promotion Law* which came into force in April 1997 had as its objective the reorientation of measures towards establishing a more explicit bridge back into the normal labour market, encouraging job search and re-entry. With respect to make-work programmes and short retraining measures, there was to be a greater link with the primary labour market, but this

Figure 29. **PARTICIPATION IN ACTIVE LABOUR MARKET MEASURES**[1]

Thousand



Note: The vertical line indicates a change in the definition of training support programmes in 1998.
1. Not seasonally adjusted.
Source: Ministry of Labour.

has proved difficult to implement. Job creation measures have always been subject to the test that they should not displace normal jobs, but with the new measures in 1997 this was strengthened: projects could only be approved if private firms were found to administer them and the local chambers of commerce had to issue certificates for all projects, certifying that no private activities would be displaced *(Unbedenklichkeitsbescheinigung).* However, in the new states especially there have been major implementation problems: private firms have not been available to administer the job-creation projects and chambers have been reticent about issuing approvals.

Despite negative assessments of such programmes in the past, active labour market schemes are being expanded again in 1998, the government aiming for participation in the programmes to increase by some 200 000 persons, while expenditures are budgeted to rise by some DM 5 billion to DM 40 billion. To achieve this target, the tighter link with the primary labour market has been relaxed and a number of exemptions introduced allowing local governments greater flexibility to administer schemes themselves without private participation. A renewed emphasis is being placed on retraining schemes, even though they

have in the past been regarded as ineffective, because they compete with private enterprise and do not facilitate re-entry into the workforce. The retraining content of such schemes can reach up to 50 per cent of costs, and when it is more than 20 per cent the usual requirement for communes to obtain clearance from the local chambers of commerce will not apply. In addition, a new lending programme with favourable terms has been created by the federal government to help finance the materials and capital costs of job creation projects which fall on local governments.

Easing active labour market policy is an important flanking measure for the government's ''new employment initiative'' *(Neue Initiativen für Beschäftigungförderung)* announced in February 1998. This initiative is designed to encourage local authorities to implement more effectively the legal provisions on social assistance, which defines a goal to support re-entry to work. Of the 700 000 social assistance beneficiaries judged capable of working in 1997, local governments (city and commune) found or created work for some 200 000, of whom about a third might have been employed on active labour market job-creation measures (ABM). The number finding employment represents a significant increase over 1993 when 66 000 found work, but can in part be explained by communes (especially cities, which bear the full cost of social assistance) attempting to offset the cost of social assistance for themselves. Many projects run for only one year, which requalifies social assistance recipients for unemployment benefits and unemployment assistance, which are paid by the Federal Labour Office and the federal government respectively. The government's new approach, which has been received critically by local governments,[58] is to encourage employment for a further 100 000 per year. Closer co-operation between the Federal Labour Office – which itself has now been granted greater flexibility to adapt programmes for local conditions – and the communes is being promoted.

As outlined in last year's *Survey*, the new Labour Promotion Law introduced a number of programmes intended to promote reintegration into the primary labour market. Although only a year has passed, experience is already accumulating:

– *A special type of work contract (Eingliederungsvertrag)* targeted at the long-term unemployed and other difficult groups involves the Federal Labour Office in assuming risks for periods of non-work such as

sickness, and the jobs can be terminated at any time by the employer. This instrument was positively rated in last year's *Survey*, but has hardly been used, other subsidy programmes offering greater incentives for employing workers with less risk. There is, in addition, a programme to subsidise employment for the long-term unemployed by 60 to 80 per cent in the first six months and by 40 to 60 per cent in the second six months. This programme covered 47 000 persons at the end of 1997. Special support programmes, however, appear to play only a minor role in dealing with this specific labour market problem, most placements being on job-creation projects.

– *A wage subsidy for employing unemployed workers by enterprises in the new Länder (Lohnkostenzuschuß Ost).* This programme provides a wage subsidy, equivalent to unemployment benefits, depending on the worker, for up to twelve months to enterprises which employ those unemployed who have been referred to them by the Federal Labour Office. To avoid abuse, employment in the enterprise must not have declined in the previous six months and no overall reduction is permitted during the period of the support. The number of workers so supported depends on the size of the enterprise's workforce. This programme has had a high take-up rate, and at the end of 1997, 70 000 contracts (1¼ per cent of employment) had been approved or were in force. The programme is generous[59] and might have squeezed out other programmes to support longer term unemployed. It remains to be seen whether workers will obtain permanent employment or simply be rotated for new employees at the end of a year.
– *Short-term training measures to help re-entry into the labour force.* These courses are from two to eight weeks during which time the labour office continues to pay benefits and covers the associated costs. From April 1997 to February 1998, 116 000 unemployed participated in such courses.

In addition, a wide range of other programmes are in operation, one of which preserves unemployment benefits for a period for those who become self-employed. There are also numerous federal and state programmes to assist new start-ups with capital.[60] Particularly important, given the concern with untrained youth, is a programme which pays firms to give unemployed school leavers work experience for a period with a small stipend (Table 15).

Table 15. **Active labour-market measures**

Million DM

Programme description	1996		1997		1998[1]	
	West	East	West	East	West	East
Facilitating integration into work	n.a.	n.a.	n.a.	n.a.	12 308	13 026
Professional training	9 957	7 981	8 517	6 628		
Integrating immigrants	1 107	258	964	260	950	270
Rehabilitation	3 619	975	3 335	1 181	2 572	948
Help to become self-employed	780	282	681	262	838	339
Job creation	2 447	6 835	1 990	5 264	n.a.	n.a.
Support for structural adjustment	177	1 902	219	1 877	477	3 816
Wage subsidy for older workers	550	246	549	233	n.a.	n.a.
Subsidy to employ long-term unemployed	461	199	401	125	530	213
Support for those on short time	1 031	435	774	270	576	210
Support for early retirement		5 385		1 890		301
Total expenditure on active measures	21 563	25 234	18 495	18 590	20 555	20 278
Total spending on labour-market (including benefits)	86 457	51 781	85 146	49 969	87 981	53 600

1. Budget allocations of the labour office.
Source: Ministry of Labour and Social Affairs.

Remaining agenda for labour-market reform

The important progress which has been accomplished in labour-market reform together with areas where further action is required are summarised in the first part of Box 6 at the end of the chapter. As noted in Chapter I, policy changes to support improved wage and employment flexibility, in combination with more flexible bargaining by employers and employees, have contributed powerfully to the improvement in unit labour costs necessary to place the economy on a new sustainable growth path. The initial impact of some policy measures could have been to lower employment, both directly and indirectly, in the short run as firms have been able to improve machine running-times and effective labour input per worker has increased (*e.g.* through less sick leave and more flexible labour hours which have led to a decline in overtime), but the labour market has now stabilised. Adverse employment effects in the short term are characteristic of partial structural reforms, but form part of the adjustment needed to underpin a recovery of employment in the medium term. To ensure that flexibility is preserved and enhanced, there is a need to underpin labour market reforms in several areas, particularly with respect to the legal framework covering collective

bargaining. The scope for policy initiatives in the area of wage determination is limited and sensitive due to the need to support rather than hinder voluntary negotiations. Nevertheless, there are opportunities to widen the room for manoeuvre by the social partners. In this respect the German Council of Economic Advisors noted four areas where policy action could be undertaken, and these deserve consideration:[61]

- Opening clauses in branch wage agreements have become increasingly important but need to be explicitly incorporated in the agreement to be valid under current law. The legal framework could be liberalised so that if agreement is obtained at plant or enterprise level it would supersede any branch agreement.
- Without essentially changing the nature of the Work's Councils, the possibilities for them to negotiate agreements in enterprises which are not parties to collective agreements could be widened. At present, employment conditions must be settled through contracts with individual workers, which is cumbersome.
- Wage agreements can only be changed at plant level in favour of employees *(Günstigkeitsprinzip)* and this has been interpreted by the courts to mean higher pay or shorter working hours. This narrow interpretation has been used in some instances to invalidate agreements designed to preserve employment through wage reductions (relative to the branch agreement) or longer working hours. The law governing tariff contracts could be made more explicit in this regard to take account of employment risks and, to limit clear dangers of abuse, deviations from wage contracts could be limited in time and scope.
- Employers who leave the employers' federation are still bound to a wage agreement until it expires. For normal wage agreements this might only be one to two years but for framework agreements *(Manteltarifvertrag),* which cover working time and wage structure, it could be up to five years. The validity period for wage contracts for employers which have left the association could be reduced.

With respect to the disincentive effects of social security and employment protection provisions, further action is required in several areas:

- Dismissal protection legislation has only been loosened for a small part of the work force who work in small enterprises. Consideration needs to be given, after experience has been gained with the current measures, to extending the more liberal criteria to larger enterprises.

- Minor jobs which are free of social security contributions *(geringfügige Beschäftigung)* contribute to labour market flexibility and are also associated with low administrative costs for both companies and the social security administration. They should therefore be preserved although some tax issues need to be regularised.
- Most importantly, although a great deal has been achieved to reduce the job search disincentive effects of the social security system through, *inter alia*, reducing benefits over time and tightening job acceptability criteria, more needs to be done, particularly for low-income workers. Some form of in-work benefits or greater earnings disregard needs to be considered, though this would only be feasible financially in the context of an increase in dispersion at the bottom of the wage structure, either as a result of wage negotiations or spontaneous labour-market developments.

Active labour market policy has undergone major swings during the past year. To some extent this was unavoidable due to the underdeveloped state of the private sector in the new Länder. But the swings in policy have also been conditioned by budgetary considerations and by concern with the immediate consequences for unemployment, particularly in the new states. Weaknesses in the previous policy approach were correctly diagnosed – the measures did not in most cases establish a bridge for the individual back into the primary labour market and risked becoming permanent – so that the partial policy reversal in 1998 needs to be corrected at the earliest possibility. Similar objections are also relevant with respect to the new policy of encouraging communes to create employment for social security recipients. This will lower the headline rate of unemployment, but could lead to a deterioration in the performance of the labour market. Displacement of workers in the primary labour market is an important side-effect often noted in Germany in the context of job creation programmes, as has been the tendency to shift costs from one institution to another by using such programmes to requalify persons for unemployment benefits. These dangers apply to the present initiative and need to be dealt with. However, once output growth gains momentum, measures such as supporting hiring of the long-term unemployed, start-up assistance, greater possibilities for fixed-term contracts, and tightened job acceptability criteria, should prove effective in overcoming resistance to raising employment since they serve to lower adjustment costs.

Improving skills, entrepreneurial dynamism and competition

The *OECD Jobs Strategy* recognises the importance in practice of a balanced approach, whereby the attainment of labour-market equilibrium is achieved by enhanced labour-market flexibility in conjunction with equity and welfare objectives. For this to be the case requires that labour-force skills be fully exploited, technology effectively utilised and entrepreneurial dynamism promoted. In this way the marginal product of labour would rise to validate existing wage rates, reducing adjustment costs. Further ingredients needed to ensure allocative efficiency and the matching of skills to jobs are effective competition in product markets and an efficient capital market, the latter being linked to an effective system of corporate governance.

Improving skills and technological know-how

Previous *Surveys* have emphasised the need to maintain the attractiveness and relevance of the dual apprenticeship system, while for the tertiary education system there was an urgent requirement to improve performance. This judgement has been broadly confirmed during the past year, with the pressure on education and vocational training policy intensifying due to both short and long term factors. The number of applicants for apprenticeships in the dual system threatened to exceed the number of offers during 1997, provoking a sense of policy urgency, and the number seeking places is expected to increase in 1998 by 10 000 to 645 000. In addition, protests about the state of the university system have underlined the need for policy action and publication of a controversial study indicating that standards in mathematics and in sciences of tertiary students are low by international standards has intensified debate about curricula more generally.[62]

In response to the objective of placing as many applicants in apprenticeships as possible, the government has sought to encourage greater co-operation between the social partners and to obtain commitments from the employer's and trades associations to increase training places. This moral suasion has continued into 1998. Some wage agreements – particularly in the chemicals sector – have also linked the acceptance of greater flexibility and part-time employment for older workers to the creation of additional training places. More concretely, the government has acted to raise the number of apprenticeships in the public administration and has extended financial support for training places in enterprises,

including funds and advice for those wishing to start a training programme. In addition, the federal government will give preference in awarding procurement contracts to firms which are training – subject to tenders being the same, no foreign firms being involved[63] and EU regulations being observed – and has urged the Länder to do likewise. In addition, calls for some form of tax or surcharge on firms which do not train have arisen in the parliament and at state level.

Despite its acknowledged strengths, the dual apprenticeship system remains under pressure to adapt to changing technological and social developments. Regulations have been modernised over the last two years and seventeen new professions have been created, seven of which are in the media and information technology sector and, since August 1997, new courses have come into force in 35 occupations. From September 1998, an additional eleven professions will be created. Discussions are under way between the social partners and the federal government to modernise the curriculum for a further 50 occupations. A point of contention remains, as before, the appropriate balance between general and specific training. In addition, in order to increase work skills and lower costs for employers, the full-time school component has been more flexibly organised in a number of states and the youth protection laws have been altered allowing an additional 20 to 30 days per year in the workplace. Going beyond the immediate question of curricula, the apprenticeship system remains under pressure in two related ways. First, greater numbers of school leavers are opting for full-time higher education, which appears to be lowering the average educational standards of those seeking apprenticeships, thereby raising costs for enterprises.[64] Surveys suggest that firms are to some extent substituting graduates of full-time training for apprentices. Second, unlike the apprenticeship system, the full-time vocational and higher educational system is financed publicly, with minimal payments from students. Thus the pressure to expand resources devoted to the public system runs the risk of reducing the attractiveness of the dual system for both employers and for youth.

Continued criticism of the performance of the university system, including long study periods, low completion rates and inadequate productivity of the teaching staff, has led to preliminary agreement on a package of reforms *(Hochschulrahmengesetz)* between the federal government and the states, where the legal competence resides. Moreover, the states and federal government agreed in

March to make a share of the salary of academics dependent on performance. The major point of dispute is the insistence by some states that the new law forbid charging fees, universities being currently free. A key feature of the agreed reform is a change in the financing of universities in favour of performance-based payments, although poorly performing institutions would continue to receive support for a long period. At the same time, greater managerial freedom will be devolved to individual universities, which will receive greater room for manoeuvre in selecting students. Students will be placed under greater pressure to take exams earlier and the possibilities of moving between universities will be widened. Although it will take some time for the reforms to take effect once the law is passed, some states have already gone some distance in implementing their own measures to improve university performance, for example, by allowing additional revenues from patents, licences and consulting to remain with the university.

Although R&D expenditure has declined relative to GDP in recent years, indicators of innovation, and of technological strength more generally,[65] present a favourable picture: Germany has one of the highest rates of patents, and surveys indicate widespread product/process innovation activities due to the need to maintain competitiveness in existing markets.[66] There is evidence that process and product innovation is clearly associated with employment gains. Indeed, it appears that the two innovation activities are complementary in raising the level of employment.[67] A major source of concern for policy makers over the years has, nevertheless, been the relatively low level of R&D activity and the perceived lack of new high-technology companies. This perceived weakness led the government to emphasise the improvement of the innovation system and the provision of risk capital in its 50-point package of policy measures to promote innovation and employment which was announced in February 1996 (see *Economic Survey of Germany, 1996).*

Pending reform of the university sector, the government has taken steps to improve the performance of its research centres. These centres are now encouraged to improve contacts with industry – including SMEs – and are able to retain revenues earned from such activities. This also provides an incentive to patent research results and to issue licences for the exploitation of inventions. The government is also seeking to encourage the institutes to spin-off enterprises, with some success. From the network of government-owned and financed institutes, 90 firms were established in 1996 and this increased to 160 in 1997.

Supporting an entrepreneurial climate

In addition to these framework-oriented policy measures, the government has implemented a number of programmes to support new high technology start-ups via seed capital. In one scheme the government's development banks *(Kreditanstalt für Wiederaufbau* and *Deutsche Ausgleichsbank),* in different programmes, guarantee 65 per cent (75 per cent in the new Länder) of the potential loss of equity participation which they have financed for a period up to ten years. Several companies which have started with this support have now been listed on the New Market board of the Frankfurt exchange.

More generally, a lack of venture capital has been seen as inhibiting corporate dynamism in Germany so that the government's 50-point programme emphasised measures to improve the operation of the capital markets. After a great deal of preparation, the Third Financial Markets Promotion Law *(Drittes Finanzmarktförderungsgesetz)* entered into force in April 1998. In addition to promoting risk capital institutions, the law also seeks to create a more liberal framework for the financial markets to respond to competitive pressures, which will strengthen with the introduction of the Euro in 1999. With respect to risk capital, experience with a special form of investment company *(Unternehmersbeteiligungsgesellschaft)* designed to invest in unlisted firms has been disappointing. The framework governing such firms has now been made more flexible: they no longer need to hold shares for six years before capital gains are tax free (but only one year) and they are no longer compelled to go public within ten years, while the minimum number of shareholders has been reduced. Majority holdings in companies are now possible for up to eight years (two years previously). Many of these restrictive measures were originally introduced to avoid tax evasion but clearly they have proved more restrictive than intended. In addition, the abolition of the business capital tax and the wealth tax will also benefit venture funds. Liability rules have also been improved so that investment companies and other advisors are now held responsible for the information given in a prospectus and for financial advice only for three years instead of the 30 years previously in force.

With respect to financial deregulation, a wider range of investment funds is now possible, including funds especially oriented toward pension needs *(Altersvorsorgesondervermögen).* The flexibility to list securities and shares which are already traded on other exchanges without additional listing controls has been

substantially widened and the task of converting from a limited to a public company has been made less formidable. In particular, costly procedures for annual meetings have been simplified. In addition, companies may now opt for international accounting standards when preparing their consolidated balance sheet. A fourth reform law is under preparation for 1999 which will address organisational flexibility of the financial markets themselves (*e.g.* the corporate structure of and the relationship between the stock exchanges), allowing greater room for restructuring following the introduction of the Euro. With this proposed law it is intended to take up the question of company pension funds. This is a complex undertaking given the taxation advantages companies currently obtain from forming internal reserves.[68]

The Frankfurt Stock exchange now has a New Market segment *(Neuer Markt)* which favours medium-sized technology-oriented firms. By mid-1998 there were around 30 firms in the segment and it is hoped that another 25 could enter in the course of the year. The market, although small, has several important innovations. First, introduction to the exchange has been not only through the banks but also, in some cases, via other financial intermediaries. It has often been argued in the past that although the large banks dominated the business of initial public offerings, they had no incentives themselves to promote listings which could reduce their traditional lending activities. A second feature has been to accompany each newly-listed firm with a ''market-maker'', thereby attempting to improve market liquidity. Improving liquidity is also important from the viewpoint of ensuring exit possibilities for initial risk investors in the company.

It is not only in the area of venture capital and new technological start-ups where there has been a need for policy action: competitive pressures and the increasingly international orientation of large German companies are forcing them to turn more and more to the capital markets for finance as an additional option to bank credit. Greater emphasis on the capital markets implies that the formal framework of corporate governance in Germany – in which banks have played a major role – has needed to be altered with more transparency towards ''outsiders'' *vis-à-vis* insiders or ''stakeholders'' which have traditionally dominated the control of large German enterprises (reviewed in *OECD Economic Survey of Germany*, 1995). To this end, the functions of the supervisory boards have been tightened, the influence of banks has been restrained, and the practice

of multiple voting rights and voting caps for shareholders have been abolished (see Box 5). Taken together, the new laws seek to address complaints which have been made over the years by international portfolio investors in particular, who have argued that relations between banks and companies lacked transparency reducing the influence of other shareholders.

Despite the difficult economic environment during the past year, with insolvencies rising further to record levels, the number of new business start-ups remains strong: after allowing for bankruptcies, de-listings and inactive start-ups, 100 000 new businesses commenced operation in 1997, up from 87 000 in 1996.[69] Due in part to public start-up finance, new businesses usually survive the first two years but in the third and fourth years more than a third of western German and nearly half of eastern German firms are forced into insolvency. Programmes to encourage start-ups have been extended during the past year. In addition to assistance provided by the Länder, especially in the new states, the federal government has introduced subsidised loan programmes through one of the public banks, first as part of the 1996 50-point programme and then in connection with additional measures to promote employment which were announced in February 1998. For small and medium-sized enterprises, DM 13 billion has been made available in both 1998 and 1999. This amount is expected to support 30 000 start ups, a little under 10 per cent of the expected number.

The 1995 *Economic Survey* noted that the bankruptcy code reflected the emphasis on credit as an outside source of finance by giving high weight to securing creditors' interests. Moreover, the bankruptcy law in Germany is often criticised as being so severe in its approach that business mistakes occurring in the normal course of entrepreneurial activity are not easily tolerated. A new code is to be introduced at the beginning of 1999 which will define the grounds for filing bankruptcy claims in broader terms than in the past. While this is necessary to rebalance interests, especially with respect to large creditors, the immediate effect might be that the numbers filing could increase further. On the other hand, there will also be an attempt to support the entrepreneurial element by allowing directors to re-enter economic life more quickly if fraud or negligence is not involved: the period of liability of a director who has been declared bankrupt will be lowered from seven years to two years.

Box 5. **Changing the framework of corporate governance**

The 1995 *Economic Survey* examined the ownership, control and decision making in German firms in some detail, concluding that the removal of disincentives facing the development of the equity market was important. This would require tightening information obligations to increase transparency and reducing the relatively high costs of underwriting. To some extent these problems have been dealt with by the Stock Exchange and as part of the new financial markets law. Greater transparency in the operation of enterprises is also necessary and to this end a new law *(Gesetz zur Kontrolle und Transparenz im Unternehmensbereich)* came into force in April 1998 governing the exercise of voting rights and the responsibilities of the supervisory board of companies:

- *Improving the operation of the supervisory boards.* In response to criticism of the effectiveness of some supervisory boards, the new legislation tightens the work of the board which is now to meet at least four times a year (two at present). The maximum number of posts that can be held by board members at any one time remains at ten although Chairman positions carry double the weight (*e.g.* only five Chairman posts can be held concurrently). In addition, shareholders are to be fully informed about the professional posts held by board members. Proposals have been put forward in Germany to reduce the size of the supervisory boards but this was not enacted due to the opposition of the unions, which feared a dilution of the importance of union members on the board. In addition, audit functions have been tightened.
- *Clarifying the voting rights of banks.* A major issue for some time in Germany has been the role of the banks in corporate governance with arguments often being put forward to limit their direct and indirect role. The new law does not limit direct holdings of banks, but where their shareholdings exceed 5 per cent the potential for them to make use of the voting rights of shares held in custody for clients will be circumscribed. Customers will be provided with more information about the possibilities for exercising their voting rights and unless an explicit proxy is obtained, the voting right cannot be exercised by the bank. Companies will have to publish with their annual reports a list of all shareholders with holdings over 5 per cent.
- *Putting voting rights on an equal basis.* With the notable exception of Volkswagen, which has been individually excepted, shares with multiple voting rights will be repealed within five years, with suitable compensation, and upper limits on voting rights will be abolished for listed companies.
- *Greater flexibility in using equity.* Stock options may now be used as compensation for directors and firms are permitted to buy back their shares.

With respect to takeovers, the government has, for the moment, decided not to implement a legally binding take-over code to protect small shareholders but to rely on voluntary enforcement of the strengthened code adopted by the commission of stock exchange experts in October 1997. However, to date a number of firms have still not accepted the code, including some of those in the DAX.

Increasing product market competition

Competition law reform

Active enforcement of competition law – along with a tradition of liberal foreign trade policy – has long been a feature of German economic policy, and this framework is to be extended and even tightened with an updated law to take effect in 1999. In the process of adapting – but not unifying – competition policy to EU practices, cartels will now be forbidden, subject to various exceptions including rationalisation and structural crisis[70] cartels, while abuse of dominant market position will be made illegal. Unlike EU law which in general forbids vertical agreements subject to some exemptions, the new law will allow these, although abuse will constitute an offence. The scope of competition policy has also been widened with the deletion of exemptions for energy and air transport and the removal of entry barriers for new railway operators from 1999. Only limited exemptions for banking, insurance and farming remain valid but a new exemption for the central marketing of professional sports has been introduced. In addition, in a move to protect small retailers, the sale of products systematically at below cost has been made an offence. The most controversial policy question has been whether competition law should have been fully integrated with EU law, especially in the area of mergers and acquisitions. The view taken by the authorities has been that where EU law is stronger it should be incorporated, but that where German practice was tighter it should be retained since corporate behaviour would have already adapted to the stricter conditions.

Competition policy difficulties have arisen with respect to policies pursued by lower levels of government – especially the procurement policies of the Länder – and in relations with the EU. One Land has already sought to link public tenders to the wage policy of companies and their suppliers, and other states have proposed to introduce social criteria (*e.g.* the provision of training places) into the new public procurement law which is currently under consideration in the parliament. The federal government opposes such a change, including the introduction of training as a criterion, although for smaller contracts it already practises such a policy. Differences with the EU arise in the context of granting state aid to companies – often within the context of approved regional development programmes – which could result in a distortion to competition in the European internal market. The most publicised example concerns payment of aid

by the state of Saxony to Volkswagen, but there are currently some 150 main investigations by the EU under way and about 500 preliminary cases, although some of these are small.

Reducing subsidies

A continuing theme in past *Surveys* has been the need to reduce subsidies, and the 1997 *Survey* dealt with the topic in some depth.[71] During the past year, federal subsidies committed to the new Länder for the period 1999 to 2004 have been placed on a new basis, with an emphasis on supporting manufacturing and related services, while support for services and housing has been cut. Moreover, the emphasis has been switched from accelerated depreciation allowances towards investment grants, which are also intended to provide greater support to small and medium-sized enterprises.[72] Direct federal support for the new Länder will, however, remain at high levels (DM 12.7 billion in 1998), although on a gradually declining trend (Table 16). Tax expenditures related to the new states are projected by the federal government to increase somewhat in 1998 and the estimate for 1999 is for a slight decline to around DM 5.8 billion. However, in view of the sharp decline in income tax receipts in recent years there must be some concern as to whether the estimate for tax expenditures is an underestimate. With respect to federal subsidies more generally, the level is expected to decline somewhat in 1998 from DM 40 to DM 39 billion with only small reductions in direct support for coal mining from DM 9.2 to DM 8.8 (Table 16).

From the perspective of overall economic efficiency, the more important subsidy concerns relate to Länder support programmes. In some states, off-budget guarantees have been extended to protect firms in financial difficulty and their development institutes – and Länder-owned banks – are often used to support favoured activities. With respect to the Länder banks, state guarantees reduce borrowing costs and hence constitute a state subsidy which could be reduced by charging a fee. Whether such fees should be introduced has become an issue, as Länder banks compete increasingly with the private sector banks. In addition, infrastructure and environmental investments are sometimes used as non-transparent indirect subsidies. As noted above, the EU has increasingly viewed Länder subsidy arrangements, which are often extended within the legal framework of EU regional aid, as detrimental to the development of the European single market, but criticism is equally valid from a domestic perspective.

Table 16. **Federal subsidies and financial support**

DM million

	1995			1996			1997			1998		
	Total	Cash transfers	Tax expenditures	Total	Cash transfers	Tax expenditures	Total	Cash transfers	Tax expenditures	Total	Cash transfers	Tax expenditures
Agriculture and forestry	5 203	4 881	322	4 890	4 548	342	4 489	4 143	346	4 204	3 855	349
Business sector (excluding transport)												
Mining	2 662	2 614	48	9 946	9 895	51	9 171	9 128	43	8 780	8 737	43
of which:												
Operating support	2 662	2 614	48	2 446	2 395	51	2 171	2 128	43	1 030	987	43
Aid for power generation	n.a.[1]	n.a.[1]	n.a.[1]	7 500	7 500	. .	7 000	7 000	. .	7 750	7 750	. .
Energy and raw materials	181	180	1	52	51	1	29	28	1	29	28	1
Technology and innovation	729	729	. .	699	699	. .	654	654	. .	617	617	. .
Aid for branches of industry	650	650	. .	443	443	. .	455	455	. .	331	331	. .
of which:												
Shipbuilding	399	399	. .	355	355	. .	335	335	. .	291	291	. .
Aerospace	252	252	. .	88	88	. .	120	120	. .	40	40	. .
Regional aid	7 991	1 790	6 201	7 828	2 132	5 696	6 272	1 930	4 342	6 182	1 865	4 317
Business sector in general	4 141	2 106	2 035	3 800	1 825	1 975	4 093	2 035	2 058	4 103	1 969	2 134
Total	**16 354**	**8 069**	**8 285**	**22 768**	**15 045**	**7 723**	**20 674**	**14 230**	**6 444**	**20 042**	**13 547**	**6 495**
Transport	1 513	122	1 391	1 484	122	1 362	1 438	61	1 377	1 410	40	1 370
Housing	9 886	4 890	4 996	9 813	4 270	5 543	10 569	4 594	5 975	10 456	4 384	6 072
Savings and wealth	585	390	195	544	305	239	434	270	164	667	460	207
General	2 612	0	2 612	2 516	0	2 516	2 560	0	2 560	2 602	0	2 602
Total aid	**36 153**	**18 352**	**17 801**	**42 014**	**24 289**	**17 725**	**40 162**	**23 296**	**16 866**	**39 382**	**22 287**	**17 095**
of which: Aid for the new Länder	14 877	8 217	6 660	13 768	7 178	6 590	13 279	7 329	5 950	12 737	6 427	6 310

1. Prior to 1996, compensation for having to use higher-priced German coal was financed by a surcharge on electricity bills for all consumers (*Kohlepfennig*). The surcharge was replaced in 1996 by a direct cash subsidy.

Source: Federal Ministry of Finance.

Removing barriers to entry

Product market competition has always been strong in the manufacturing sector but in services and in utilities (*e.g.* telecommunications and energy) this has not been the case. With respect to services, a differentiation needs to be made between industry-related and other services which are more directly linked to the consumer. With respect to industry-based services, framework conditions now appear to be generally favourable and growth is quite strong, even though developments in this branch are not captured well by the statistics: many such services are still classified as manufacturing. With respect to consumer-related services which account for some 4 to 5 million employed, growth has not been as strong. This may be due in part to relatively high wages for unskilled workers as well as the social benefits system, which has been characterised by high marginal effective tax rates as benefits are withdrawn (See *Economic Survey*, 1996). But framework conditions could also be important. For example, the skilled trades law *(Handwerksordnung)* regulates entry into a large number of occupations and specifies in a detailed way the activities which can be carried out by each trade. In response to technological and economic change, which has in some cases made the differences between individual trades either unclear or uneconomic (*e.g.* repairing different types of electrical equipment has been separated into different trades), a new law came into force in April 1998. Trades requiring a special Masters qualification have been reduced from 127 to 94 and, in some cases, it is now possible for the one trade to handle a greater range of closely-related activities. The curricula of the associated apprenticeships have also been widened. In addition, the possibilities for master craftsmen to transfer from industry to a trade have been partially relaxed and the procedures for recognising qualifications from other countries have been improved.

Shop opening hours have been liberalised for a little over a year but, coming at a time of stagnant consumption, it is still difficult to assess the full consequences. However, as to be expected some enterprises and locations have benefited more than others (*e.g.* shopping malls in the outer city areas), and some consumers have also taken greater advantage of the new opportunities. Employment (paying social security), which should not be the primary measure for assessing reforms, has finally started to grow towards the end of 1997 and there appears to have been an increase in minor employment in the retail sector. The partial liberalisation has not been associated with many of the negative effects critics expected.

An important barrier to entry and to expansion has been the long and costly approvals procedure for opening plants and workshops. In response to long standing complaints, the federal government introduced new legislation in 1997 to streamline the process *(Verfahrensbeschleunigungsgesetz)* even though primary responsibility for approvals resides with the Länder and with local government. Although results are still sketchy, it appears that there has been a substantial reduction in approval times in some states.

Deregulating the public utilities

Liberalisation of the telecommunications sector has proceeded rapidly over the last year and with voice telephony open since the beginning of 1998, competition has spread quickly. The new regulatory authority has been established and decided quite early against the old telephone monopoly *(Telekom)* by reducing substantially the inter-connection fee it was demanding from potential competitors. Transferability of numbers has also been ensured. While progress has been substantial in telecommunications, the new postal law is not as liberal, even though the monopoly on postal services has been lifted from January 1998. Although the weight limit for mail which will remain the exclusive preserve of the old post office has been reduced and will remain in force until 2002, parliament has specified that new competitors will have to operate under broadly similar employment conditions as the post office.

As in other countries, liberalisation of power generation and distribution is proving a difficult process, with lower levels of government defending their power companies and economic interests while economic objectives need to be balanced by environmental and regional considerations. A new law was approved by parliament in April 1998 and entered into force in the middle of the year. Environmental objectives have been given equal status with economic efficiency, but are to be achieved by specifying the general framework rather than through direct control. Crucial to the new approach is the lifting of the sector's exemption from general competition law, which means that the exclusive concession and demarcation agreements, which have been widely used to prevent competition from emerging, will cease to exist. Moreover, the burden of proof has been shifted to the operators of the transmission network who wish to deny access to the net. In a compromise with the states and local authorities, the law does not opt exclusively for either the third party access or single buyer model.[73] Rather,

single buyer models are possible but will need to be approved and are valid only until 2005. A review will be conducted of the single buyer arrangements in 2003 and they will only be able to continue in operation after 2005 if ''the performance is similar to the third party access system'', presumably in terms of energy prices. The terms of third party access have been negotiated by the association of industry (BDI) and the association representing energy producers and distributors. If the implementation of the agreement is not judged to be competition-promoting by the Cartel Office, charges can be set by ordinance.

With respect to environmental concerns, renewable energy (such as wind- and tide-based systems which are particularly important in the north of the country) is to be protected by requiring network operators to accept all renewable energy offered up to a limit of 5 per cent of transmissions. The additional cost of renewable energy will be treated as if it were a network operating cost and will thus be spread over all customers of the relevant region. A similar procedure will be used to protect brown coal production in the new Länder until 2003. Environmental concerns are also important with respect to city-based, integrated generation and central heating systems,[74] which also raise the issue of the treatment of stranded costs (*i.e.* who is to indemnify the loss on investments which are no longer profitable due to the change in the regulatory system). Protection of such systems will be decided on a case-by-case basis but the generator/central heating entity must prove that the system would be non-viable under a system of third party access. Some 100 cities out of 200 currently involved in generation could qualify for such exemptions.

In response to the new competitive framework, the electricity-generating industry has already started to adapt. Some generators have already formed buying groups to reduce acquisition costs of equipment and power companies are taking steps to reduce costs by, *inter alia*, rationalising employment. Commune-based generators are in some cases being corporatised and placed on a commercial basis. Such adjustments are necessary in view of the interest being expressed by large clients to aggregate the energy needs of their separate factories and put these out to tender to the cheapest generator. On the other hand, as in air transport, there is a great difference – and perhaps a substantial delay – between establishing an enabling framework and the development of effective competition. The existing electricity generators, which are in a monopoly position, can be expected to defend their positions and this is indeed occurring through the efforts

of their association to negotiate high prices for access to the transmission grid. Effective enforcement of competition policy and close attention to regulatory provisions will be necessary to create a genuinely contestable market.

Assessment and scope for further action

The above discussion provides ample evidence that the German economy is in the process of substantial structural adjustment: reforms have been initiated on a wide front which recognise that employment-creation will need to derive predominantly from the private service sector, taking advantage of new opportunities for individual initiative. The reforms undertaken with respect to education, training, technology diffusion, together with deregulation and liberalisation in public utilities and financial services, all implicitly recognise the need for a different pattern of resource allocation. Because of the scale of the task, the results of the above programme will take time to come through. More importantly, as the *OECD Jobs Strategy* has emphasised, the impact of individual reforms depends crucially on the effectiveness of action over the totality of interrelated factors which may impede the effective operation of markets. It is thus important to assess the programme in the light of what remains to be done, and factors which could continue to impede its effectiveness.

The high level of education and vocational training continues to be a key factor supporting high living standards, a strong technology base, and competitiveness in Germany. However, pressures are apparent, and measures to upgrade the dual apprentice system and make it more flexible, and to reform the universities through a more performance-related financing system all go in the right direction, even though the speed and extent of reforms could be too modest. In particular, university reform needs to be accelerated. However, other measures go in the wrong direction. This is particularly so with respect to linking the award of public procurement contracts with training by the enterprise. While such a measure might increase training places in the short run, it is not at all obvious that the occupations would be the most appropriate from the medium run perspective: firms could simply train in order to ensure contracts, and experience shows that youth, once trained, suffer income losses if forced to change occupation. Flexibility of corporate structure would also suffer. Moreover, once the possibility is established to award tenders on the basis of other objectives, the way might be open for widespread abuse. Training is an area where, since the trainee is free to move to another company at the end of the apprenticeship, externalities could be

present leading to a sub-optimal level of apprenticeships. Levying a charge on firms which do not train, which is often proposed, could theoretically correct for any externality. However, low wages for trainees and their contribution to output during the apprenticeship should already serve to compensate for most externalities. At the same time, levying a tax on those companies which do not train would be inefficient: companies might utilise a different technology/skill mix which would not be appropriate for current apprenticeships or they might utilise the graduates of full-time training, whose employment could suffer from a general levy. These negative operational features are magnified once consideration is given to reallocating the tax revenues to promoting training in other firms. In any case, it is difficult to see how a training tax would deal with the threat to the dual apprenticeship system arising from the expansion of the publicly-financed training system, which is free of charge. Policies should instead focus on increasing the returns to companies from offering training places according to their present and prospective needs, and by placing the returns to individuals from the public and private training systems on a less distorted basis through greater use of fees or of student loans.

A problem which is starting to emerge concerns the 8 per cent of school leavers who do not have any qualification. Recent measures to directly support their placement for a period with companies to gain work experience appear promising, in the light of local experience, but will need to be carefully monitored to avoid squeezing out other less well-trained employees. In the end, however, the solution is less to introduce new support programmes than to encourage more flexible wage structures, including lower wages for first jobs which are already a feature in some branch wage agreements, while preserving the benefits of training for enterprises.

With respect to extending competition to previously-protected sectors, significant progress has been made, especially in telecommunications. In the power-generation and transmission sector, the compromise which the government needed to accept appears reasonable: the single buyer system preferred by communes is in principle limited in time, and continuation will be judged against the standard of third party access. Thus the presumption that the single buyer system could be abused has been to some extent addressed. However, the important role given to the association of energy producers in negotiating the price for transmission over long distances on the high tension network does give some grounds for concern that competition will be frustrated by the incumbents setting access

prices too high. This will raise key questions for the enforcement of competition policy. More difficult is whether environmental questions have been effectively and efficiently handled: treating renewable energy as a network cost of the region concerned could prove difficult from the point of view of industrial location, and it is not obvious that protection of brown coal is compatible with environmental objectives, and with the polluter pays principle. In addition, with so many cities likely to be excluded from competition due to combined generation/central heating systems (100 out of 200 cities which produce electricity), competition and prices will remain quite differentiated regionally, perhaps leading to increased pressures on policy makers to compensate such areas in other ways.

As for the broad spectrum of factors which affect entrepreneurial activity and technology diffusion as a whole, there are still impediments arising both from the operation of the financial markets and from the regulatory environment. The new financial markets law is an important advance since it widens the possibilities for venture capital firms and eases the path of companies which wish to expand their capital base by going public gradually. In addition, reform of the voting rights law for companies will increase company transparency, facilitating deeper capital markets. The policy response has also been in part to develop further government support programmes for new start-ups and for technology diffusion. The problem with such programmes is that they often treat only the symptoms and promote greater dependence on the state rather than creating a better set of framework conditions. To help create these conditions, other areas require further policy initiatives:

- A start has been made in streamlining the regulatory framework with an acceleration of approval procedures, but in general the German economy remains highly regulated and the government estimates that firms are subject to 80 000 regulations at the federal level alone.[75] Many of these appear to lack a clear rationale so that a systematic review remains necessary.
- Although the changes to the skilled trades law generally go in the right direction of loosening barriers to entry, while preserving the qualification structure which reduces transaction costs for consumers, the balance might still be tilted too much to protecting incumbents. Indeed, in the face of new competition arising from technological change, the reaction has been often to absorb such activities into a trade thereby erecting barriers to entry.[76]

Box 6. **Implementing the *OECD Jobs Strategy***

Since the last review a number of policy measures have been implemented and experience with reforms continues to evolve. This box summarises actions taken over the entire period since the original review in the *OECD Economic Survey*, 1996.

Job strategy proposal	Action taken	OECD assessment/ recommendations
A. PROGRESS IN LABOUR MARKET REFORM		
I. Increase wage and labour cost flexibility		
• Encourage wage differentiation, greater plant-level bargaining and opening clauses.	Minimum wage in construction was declared binding in mid-1997.	Wage bargains should incorporate greater flexibility; widen the legal framework covering wage negotiations; eliminate administrative extension of wage agreements.
• Facilitate the employment of older workers and reduce incentives for early retirement.	Early retirement age to be increased; part-time work of older workers is subsidised under some conditions; eligibility conditions for invalidity pensions tightened.	Consider further means to raise the retirement age (stricter eligibility criteria, actuarial discounts for pension benefits) and keep transitional arrangements to a minimum.
II. Increase working-time flexibility and ease employment security provisions		
• Reform regulations underpinning inflexible working practices.		Review effects of regulations.
• Liberalise terms for renewing fixed-term contracts.	Renewal period and admissible frequency of renewals increased; scope for secondments extended (period extended, approval requirement reduced).	Monitor employment effects and continue with reforms if necessary.
• Facilitate part-time work.	More flexible work arrangements in the public sector; subsidies to part-time work for the elderly.	Avoid further subsidisation of part-time work for the elderly.
• Reform dismissal protection.	Exemption from general dismissal law widened; social criteria for dismissals relaxed.	Monitor the effects on hiring and the costs of dismissals. Consider extending the more liberal conditions to larger companies.

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Job strategy proposal	Action taken	OECD assessment/ recommendations
III. Reduce the distortions arising from unemployment insurance and related benefits		
• Reduce the generosity of benefits.	Minimum ages for receiving unemployment benefits increased by three years; redundancy payments credited against unemployment benefits.	Abolish the extended period of receiving unemployment benefits for the elderly; increase the gap between social assistance payments and the earnings of low-income families.
• Restrict the criteria for not accepting job offers, strengthen penalties, tighten controls.	Qualification no longer reason for rejecting job offers; definition of acceptable wage narrowed; controls tightened, internal auditing introduced of criteria.	Monitor the revised criteria for refusing job offers or taking up employment and narrow if necessary.
• Lower social security contribution rates.	Spending curbed in the pension and health funds and in the Federal Labour Office, but social security contribution rates have increased; increase in VAT to avoid further rises in rates; pension reform legislation passed.	Curb social spending; consider further measures to secure financing of the pension system (retirement age, basic pension, funding).
• Simplify the personal tax system and lower marginal taxes.	Taxation reform proposed and rejected by upper house of parliament.	Proceed with tax reform.
• Reduce the generosity of sickness benefits and strengthen incentives to avoid abuse.	Statutory sick pay lowered from 100 per cent to 80 per cent.	Maintain policy.
IV. Enhanced active labour market provisions		
• Exclude measures from the requalifying period for unemployment benefits.	Training periods will not re-establish eligibility for another entitlement period.	Continue reform.

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Job strategy proposal	Action taken	OECD assessment/ recommendations
IV. Enhanced active labour market provisions *(cont.)*		
• Targeted ALMPs at at-risk groups.	Support schemes for training of unemployed extended; more flexible work contracts for formerly unemployed. Communes given greater powers to introduce job-creation programmes and new loan facility given for associated non-wage costs.	Monitor the efficiency of work contracts for the unemployed, in particular in relation to employment provision measures. Tighten new powers of communes to organise job-creation programmes and seel to avoid simply transferring benefit recipients from one level of government to another.
• Base employment subsidies on wages below market rates.	Admissible wage base lowered. Special conditions for the new Länder will be phased out.	
B. IMPROVING SKILLS, ENTREPRENEURIAL DYNAMISM AND COMPETITION		
V. Improve labour force skills		
• Preserve and restore the attractiveness of the dual vocational training system, clarify its relationship to higher education.	Curricula for apprenticeships revised and new occupations introduced.	Continue to revise vocational training schemes and preserve benefits from training apprentices. Do not introduce a training levy.
• Shorten higher education and focus it on more occupational-oriented studies.	Agreement in principle on university reform between federal government and states but no agreement on fees.	Shorten higher education and make it more occupationally oriented. Make universities more performance-oriented. Greater use of fees or student loans.
VI. Enhance creation and diffusion of technological know-how		
• Deepen financial markets, foster venture capital and reduce regulatory barriers.	Laws implemented to improve flexibility in financial markets, ease access of companies to equity capital and change corporate governance.	Monitor development of risk capital; introduce pension funds.

(continued on next page)

(continued)

Job strategy proposal	Action taken	OECD assessment/ recommendations
VII. Support an entrepreneurial climate		
• Facilitate the establishment of new enterprises.	Accelerated depreciation allowance for newly-founded enterprises; wage subsidies for small new-founded firms.	Continue to reduce regulatory impediments.
• Increase the knowledge of the new owners of enterprises in the new Länder.	No major changes.	Support projects to diffuse knowledge.
• Reduce the overall tax burden.	Abolition of wealth tax and the business capital tax. Reduction of the inheritance tax.	Implement a tax reform with sharply reduced tax expenditures and lower marginal tax rates.
VIII. Increasing product market competition		
• Simplify planning approvals.	Approval procedures simplified and entry into trades made somewhat easier.	Monitor the effectiveness of the new regulations; continue reform and modernisation of law on trades.
• Liberalise shop-closing hours.	Statutory shop opening hours extended.	Liberalise further statutory shop opening hours.
• Encourage competition in the network sectors.	Law passed allowing competition in energy sector and entry into telecommunications promoted. Competition law widened to cover network sectors and transport.	Monitor whether regulations to enforce the opening up of networks are effective and promote contestable markets.
• Market test services by communities need to be more closely market-tested.	No major changes.	Abolish preferential VAT treatment of public utilities.
• Pursue privatisation.	Telecom partly privatised.	Step up privatisation.
• Reduce subsidies.	Subsidies benefiting construction reduced overall but reduction of coal subsidies slowed. Subsidies to east placed on new basis.	Avoid new industrial subsidies; set up schedule for reducing existing subsidies; close tax expenditures.

- The partial liberalisation of shop opening hours is resulting in a restructuring of the retail sector to better reflect consumer preferences and should be extended.
- Despite the decline of subsidies in western Germany in recent years they remain high in some areas of activity and in eastern Germany the level of support probably goes far beyond the level objectively required to compensate for cost disadvantages. Particularly worrying has been the tendency in the new Länder to extend support for firms in financial difficulties through local development institutes. Greater efforts need to be made to first increase transparency with respect to subsidies and then to reduce their level, especially those granted through tax expenditures.

A summary of policy actions taken since the *Jobs Study* review in 1996, and recommendations for further action is given in Box 6, Panel B.

Overview of progress in structural reform

Over the past few years, the government has introduced an impressive range of structural reforms in both labour and product markets, and these have been accompanied by significant changes in behaviour of the social partners, particularly with respect to wage bargaining. The benefits of enhanced flexibility are so far apparent in productivity and competitiveness gains, rather than employment, which has fallen across a wide spectrum of activities including manufacturing, mining, energy, construction, telecommunications, insurance and banking. In nearly all cases this has reflected the need for the activity in question to restore competitiveness in the face of changing conditions in world and domestic markets, while in other cases it reflects the need to adjust to declining levels of subsidies. Structural policies have both initiated and facilitated these changes. Cost savings, lower relative prices and improved productivity will gradually lead to a recovery in output and in the level of employment, if the changed behaviour persists as the recovery picks up.

However, to ensure that employment creation will be sufficient to reduce substantially the currently-high level of structural employment, there is a need to complete the process of reform. On the labour market side, assuming that the degree of wage flexibility apparent in recent years persists, the priority is to ensure the reintegration of low-income workers, by reducing social-security

disincentives, easing employment regulations and refocusing active labour market programmes. On the product-market side, the need is to complete the process of reforming regulation, which still inhibits growth in areas of potential employment expansion.

Notes

1. Deutsche Bundesbank *Monthly Report,* April 1998, p. 28.
2. This result has also been confirmed by other researchers using different specifications of labour demand. See for example ''Trotz höherer Risiken: Aufschwung setzt sich fort'', *ifo Wirtschaftskonjunktur,* 12/97, p. A20. Their equation covers only the old states whereas the OECD has estimated an equation for Germany as a whole. Using business sector employment rather than manufacturing avoids any measurement problems arising from the tendency to spin-off services into specialist activities.
3. It is also important to realise that investment, as currently defined, is now a less useful guide to increases in productive capacity, because of the growing importance of computer software, which, for the moment, is not recorded as investment.
4. As noted in the 1996 *OECD Economic Survey of Germany,* machine running times were low in comparison with other countries in Europe while 80 per cent of companies reported that they could lower unit costs by increasing running times.
5. J. Lindlbauer, ''Kapazitätsauslastung im ifo Konjunkturtest- mehr als eine Prozentzahl'', *ifo Schnelldienst,* 7/98.
6. Average factory operating hours have increased from 60.6 hours in 1984 to 71.8 hours presently, *Handelsblatt,* 11 May 1998.
7. For illustrative results see for instance C. Giorno, P. Richardson and W. Suyker, ''Technical progress, factor productivity and macroeconomic performance in the medium term'', *OECD Economic Studies,* 25, 1995/II. Complex effects were also demonstrated in S. Blöndal and D. Pilat, ''The economy-wide effects of regulatory reform'', *The OECD Report on Regulatory Reform, Volume II: Thematic Studies,* OECD, 1997.
8. The OECD assumes that production can be represented by a Cobb-Douglas production function with factor neutral technological progress. The growth in total factor productivity is, however, treated as if it were an increase in labour efficiency.
9. For the past, there is some evidence that in Germany wage growth reacts more to the change in the rate of unemployment than its level. If this were to continue to be the case, even though a lower rate of unemployment may be compatible with stable wage inflation in the long run, this lower rate could not be reached in the short term without setting off changes in inflation. It is this process which needs to have been changed by recent structural reform.
10. S. Scarpetta, ''Assessing the role of labour market policies and institutional settings on unemployment: a cross-country study'', *OECD Economic Studies,* 26, 1996/1.

11. The target is based on the assumption that potential output will expand by around 2 per cent while the medium-term inflation assumption has been set at 1½ to 2 per cent. In addition velocity is assumed to continue its trend decrease of 1 per cent. See Deutsche Bundesbank *Monthly Report*, January 1998 for details.

12. Hoffmann J., ''Probleme der Inflationsmessung in Deutschland'', *Deutsche Bundesbank Diskussionspapier*, 1/98 and ''Problems of inflation measurement'', Deutsche Bundesbank, *Monthly Report*, May, 1998.

13. See S. Jacobs *et al.*, ''Regulatory quality and public sector reform'', in *The OECD Report on Regulatory Reform, Volume II: Thematic Studies*, OECD, Paris, 1997.

14. It is too early to say whether overall payments on medicines have declined, which is the more relevant measure from the point of view of the economy as a whole.

15. In the year to January 1998, the large regional health funds (AOK) lost 1.5 per cent of their members and some other funds raised their membership by up to 6 per cent. *Pressemitteilung, No. 13*, Bundesministerium für Gesundheit.

16. Doctors are currently permitted to charge up to a multiple of fees paid in the statutory system. Many doctors already charge the maximum permitted.

17. ''The retirement decision'', in *OECD Economic Outlook*, No. 63, Chapter VI, Paris, 1998.

18. The share of the federal government in overall general government spending belongs to the lowest in the OECD (some 20 per cent, national accounts terms). Bund, Länder and communities account for 11.5, 52.5 and 36 per cent of the territorial authorities' employment respectively.

19. Income tax sharing between the Bund and the Länder was introduced in 1955. Up to that time income tax revenue was assigned to the Länder although in exceptional circumstances the Bund could also participate. The rebalancing of revenues was largely motivated by the desire to mobilise federal funds for the purpose of financing Germany's reconstruction. In 1970, sharing of income tax revenues was extended to the communities (15 per cent share), in exchange for a participation of the Bund and the Länder in the communal business trade and capital taxes. Simultaneously, the Länder obtained a 30 per cent share in the federal turnover tax. This restructuring was largely motivated by the desire to smooth the cyclical sensitivity of their tax revenues. Subsequent extensions of tax sharing were largely driven by Länder demands to compensate them for rising spending obligations.

20. Between 1970 and 1989 the Bund and the Länder received equal shares of the local business tax although the shares of each have fallen from 20.3 per cent to 7.2 per cent. After reunification the western communities contributed to transfers to the eastern Länder by increasing the business tax share of the Länder. As a result, the Länder's share in the business tax in 1996 totalled 18.1 per cent, and the Bund's share dropped to 5.0 per cent. Also, since 1996 the Bund's mineral oil tax, accounting for 9 per cent of overall tax receipts, is shared with the Länder (18.2 per cent share). The share of the mineral oil tax serves to finance Länder investment in railway infrastructure.

21. See *OECD Economic Survey of Germany*, 1996. See also Blankart, Charles, ''Politische Ökonomie der Zentralisierung der Staatstätigkeit'', Humboldt-Universität zu Berlin, *Economic Series Discussion Paper 108*, 1998.

22. About four-fifths of the Länder's total revenues comprise taxes or income from fees, business activities or the sale of assets. The remainder is made up of transfers from other levels of government.

23. About two-thirds of the communities' total revenues comprise taxes or income from fees, business activities or the sale of assets. The remainder is made up of transfers from the Länder and the Bund.

24. For features of communal equalisation systems see Parsche, R., and M. Steinherr, ''Der kommunale Finanzausgleich des Landes Nordrhein Westfalen'', *ifo Studien zur Finanzpolitik,* Vol. 59, München, 1995; Steinherr, M., ''Das Schlüsselzuweisungsproblem im kommunalen Finanzausgleich des Freistaates Sachsen'', *ifo Studien zur Finanzpolitik,* Vol. 63, München, 1997; Steinherr, M., R. Parsche and R. Koll, ''Der Ausgleich zwischen Finanzbedarf und Finanzkraft im kommunalen Finanzausgleich des Landes Rheinland-Pfalz'', *ifo Studien zur Finanzpolitik,* Vol. 66, München, 1998.

25. For example, the Saarland which has the lowest per capita tax receipts among the western Länder prior to redistribution ranks third among the western states after redistribution (Table 12).

26. See Huber, Bernd and Karl Lichtblau, ''Konfiskatorischer Finanzausgleich verlangt eine Reform'', *Wirtschaftsdienst* 3/1998. For the Länder's own taxes (*e.g.* inheritance tax) the authors compute marginal effective tax rates of between 58 per cent and 99 per cent.

27. The study was conducted by the Arbeiterkammer Bremen. Quoted in *Süddeutsche Zeitung,* 9 March 1998.

28. Redesigning the financial equalisation system along these lines has been suggested by Huber and Lichtblau, *op. cit.* The authors show, based on official tax receipt estimates, that under their reform proposal – neglecting any efficiency gains – the new system would slightly benefit the eastern states in the medium term. Similar proposals, which do not explicitly recognise the important role of lump-sum payments to reduce the implicit marginal taxation of Länder revenues, have been made by the Council of Economic Advisors, *Jahresgutachten* 1991/92, and by Hüther M., ''Reform des Finanzausgleichs: Handlungsbedarf und Lösungsvorschläge'', *Wirtschaftsdienst* 1/1993.

29. While deficit caps are sensitive with respect to ''creative accounting'', arrangements would benefit from rules which have been developed by Eurostat to control this problem.

30. Empirical research for the United States indicates that the degree of fiscal consolidation depends on the stringency of budget caps. States with balanced budget requirements which do not allow carrying-over fiscal deficits from one year to the next have higher surpluses. See Bohn and Inman, ''Balanced budget rules and public deficits: evidence from the US states'', *Carnegie-Rochester conference series on public policy 45,* 1996.

31. This has been suggested by the Institute for Economic Research Halle which has derived a distribution of admissible deficits based on this rule. See Snelting, M., ''Der nationale Stabilitätspakt in einer Europäischen Währungsunion: ein Umsetzungsvorschlag'', *Wirtschaft im Wandel,* 12/1997.

32. A more radical solution would be to auction deficit rights among the Länder. Under the assumption that deficit rights would be acquired by those states which utilise above-average deficits for financing public investment, the auction principle should again produce deficit

allocations which are similar to the population-based rule in the medium term. It is then preferable to implement the latter, allowing for a transitory period which regards past deficits, since it is an open question whether deficit restrictions which are auctionable would be perceived by the Länder to be as binding as fixed rules or whether the risk of overruns would increase. Also, auctions could effectively amount to transfers from the poor states to the rich states.

33. In the present equalisation system the need for transfers is inflated by compensating disadvantages of ''smallness'' which derive from the fact that the cities Berlin, Hamburg and Bremen constitute states on their own. Territorial reform, which would incorporate these cities into larger states and which would result in the merger of other states, could lead to a substantial increase in public sector efficiency and better utilisation of taxpayers' money but the issues are extremely complex, involving political, social and cultural questions, and are not taken up in this chapter.

34. But the fact that communal road construction is of little importance for inter-state traffic connections suggests leaving the financial responsibilities for road construction in a narrow sense fully with the communities. Interstate road construction is fully paid for by the Bund.

35. For detailed descriptions of federal subsidies provided as investment aid or within the framework of joint fiscal tasks see: Bundesministerium der Finanzen, *Subventionsbericht 1997,* Bonn, 1997.

36. Federal ''Support of the Regional Economic Structure'' has been reduced from DM 0.5 billion in 1990 to DM 0.4 billion in 1996 in the west while the east received DM 3.7 billion.

37. See *OECD Economic Survey of Germany,* 1996.

38. Formally, administrative simplicity and a certain degree of tax harmonisation has been achieved via federal-provincial tax collection agreements. However, provinces are asking for an increasing number of special tax credits and other measures, some of which could have adverse effects on efficiency. See *OECD Economic Survey of Canada,* 1997.

39. This has been suggested by the Scientific Council at the Ministry of Finance, see, Wissenschaftlichen Beirat beim Bundesministerium der Finanzen, ''Gutachten zum Länderfinanzausgleich in der Bundesrepublik Deutschland'', *BMF Schriftenreihe 47,* Bonn, 1992.

40. Higher administrative costs can arise with respect to the wage withholding tax if firms employ residents of different states. This would imply either residence-specific withholding rates on the side of the firms, or – with no such differentiation on the employers' side – additional filings of tax returns and claims for tax refunds on the side of the tax authorities.

41. With the tax surcharges varying across the Länder, the value of tax concessions could also vary by Land, depending on the type of the concession. Whether or not this would conflict with equity goals (for example in the case of child allowances) is a matter of judgement, taking into consideration that the value of tax concessions would vary inversely with the size of the tax surcharge.

42. OECD Members countries are increasingly charging fees for services, a major objective being to relieve tax payers of costs borne by the users who benefit from a service. See OECD, ''User charging for government services: best practice guidelines and case studies'', *Public Management Occasional Papers 22,* Paris, 1998.

43. ‘‘Social partners’’ is here taken to cover employers’ associations, acting often at regional level, unions and firms which have left the employers’ associations and bargain employment and wage contracts directly. It can also cover works councils, which have often been important in negotiating working conditions.

44. This is documented in the special report of the research institutes on the service sector in the new Länder. ‘‘Gesamtwirtschaftliche und unternehmerische Anpassungsfortschritte in Ostdeutschland, Siebzehnter Bericht’’, *DIW Wochenbericht*, 3/98.

45. There is a growing debate about whether there is a services ‘‘gap’’ in Germany: whether the share of services in employment is lower than in other countries. Most agree that current statistics understate the size of the service sector since some activities are still classified as manufacturing while with other measures the classification is imprecise. Utilising direct questions to individuals about the nature of their occupation and relating this to the size of the workforce, rather than those in employment, does indeed show that the ratio is lower in Germany than in a number of other countries. This does not, however, lead automatically to the conclusion that there is an unutilised potential.

46. Unemployed above the age of 58 years are at present permitted to choose to continue to receive benefits but not to actively seek work. This measure lowers pressure on labour offices to control such persons who under current conditions are difficult to place in a job.

47. Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, *Jahresgutachten 1997/98*, Wiesbaden, 1997, Table 38.

48. Two-thirds of these long-term unemployed were placed by the Federal Labour Office and of these, 162 000 were found jobs on job-creation projects rather than in the normal labour market.

49. Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, *Jahresgutachten 1997/98*, Wiesbaden, 1997, paragraph 163.

50. The Council of Economic Advisors list several such cases to illustrate the issues involved (Table 61). Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, *Jahresgutachten 1997/98*, Wiesbaden, 1997.

51. In an analysis of 60 sectors, the limit on excess hours accumulated in time accounts varied between 40 and 600 hours and for time not worked from 20 hours to 170 hours.

52. The jobs cover regular work of less than 15 hours per week and with earnings less than DM 620 (DM 520 in the new Länder). They do not involve employers demanding individual tax cards for payment of taxes; rather, a flat 20 per cent tax is paid by employers. For the individual, all such jobs need to be cumulated in order to establish whether there is an obligation to pay social security contributions but there is no cumulating with a full-time job. The jobs have the same legal rights as any other employment.

53. R. Erbe, ‘‘Brauchen wir eine Neuregelung der ‘620 DM jobs’’’, *Wirtschaftsdienst*, 1998/II.

54. In the year to December 1997 the total number of sanctions affecting beneficiaries of unemployment benefits increased from 178 000 to 180 000 and for those in receipt of unemployment assistance from 35 000 to 41 000. Sanctions more directly related to refusing a job offer rose from 13 000 to 16 600 and from 11 000 to 16 000 respectively.

55. Additional funds became available to the Federal Labour Office late in the year after the supplementary federal budget, but without clarity about the appropriation in 1998, it was not in a position to expand the programmes.

56. One study for eastern Germany concluded that re-employment probabilities for unemployed who participated on a public works programme were worse than for those who had not participated. F. Kraus, P. Puhani and V. Steiner, ''Do public works programmes work? Some unpleasant results from east German experience'', *ZEW Discussion Paper,* 98-07.

57. However, one study found that the effectiveness of training/retraining measures in the new states appear to have risen over time as institutional arrangements have improved. The study cannot be taken as justification for continuing past policy since the question of cost benefit analysis was not addressed and trainees could simply have displaced other members of the labour force. See F. Kraus, P. Puhani and V. Steiner, ''Employment effects of publicly financed training programmes – The east German experience'', *ZEW Discussion Paper,* 97-33, Mannheim.

58. In the case of smaller communes, social assistance is paid by the local regional authority *(Kreis)* and not actually by the communes which are obliged to offer employment, sometimes at an additional cost to themselves. The incentive for them to actively seek work for social assistance beneficiaries is thus weakened (they do save in so far as contributions to the regional authorities are reduced) while at the same time they are sensitive to local disturbances on the labour market which such a promotion could entail.

59. B. Schultz, ''Lohnkostenzuschüsse fur Unternehmen – Neues Instrument mit alten Fehlen? – ein Kommentar'', *Wirtschaft im Wandel,* 4/1998.

60. There are still only a small number of assessment studies available. One study in a new state came to the conclusion that more targeted support was necessary, together with more advisory services. It also advocated greater co-ordination between the numerous support programmes. ''Förderung von Existenzgründungen aus der Arbeitslosigkeit in Mecklenburg-Vorpommern: Ein beschäftigungspolitischer Beitrag'', *DIW Wochenbericht,* 11/98.

61. Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, *Jahresgutachten 1997/98,* Wiesbaden, 1997.

62. *Performance assessment in IEA's third international Mathematics and Science Study,* TIMMS, 1997.

63. The condition that no foreign firm has tendered removes a potential conflict with EU law as does an additional restriction that the provision will only apply to tenders which are under the value set by the EU for notification in the EU's official journal (*i.e.* in practice tenders with a value of less than DM 200 000).

64. For example, see C. Büchtermann and K. Vogler-Ludwig, ''Das deutsche Ausbildungsmodell unter Anpassungszwang: Thesen zur Humankapitalbildung in Deutschland'', *ifo Schnelldienst,* 17/18, 1997.

65. *Germany's Technological Performance,* Report for the Federal Ministry of Education, Science, Research and Technology, Bonn, 1997.

66. H. Schmalholz and H. Penzkofer, ''Innovationsschub für mehr Wachstum erforderlich'', *ifo Schnelldienst,* 17/18, 1997.

67. H. Rottmann and M. Ruschinski, ''Sind Innovationen beschäftigungswirksam?'', *ifo Schnelldienst*, 17/18, 1997.

68. For a description of the issues see the *OECD Economic Survey of Germany*, 1996.

69. *Insolvencies in Europe*, 1997/1998, Creditreform, 1998.

70. Such cartels have been authorised in a few cases in the past, such as coal mining and steel when it was considered necessary for inter-firm agreements to aid the restructuring of the industry.

71. One estimate of subsidies, which includes expenditures on cultural activities, reduced social security payments by farmers and support programmes of lower levels of government, places the value at around 8 to 9 per cent of GDP. Boss A., and A. Rosenschon, ''Subventionen in der Bundesrepublik Deutschland – Bestandsaufnahme und Bewertung'', *Kieler Arbeitspapier*, No. 793, 1997.

72. The investment grant for machinery and equipment in manufacturing and in related services is 10 per cent and for small and medium sized enterprises 20 per cent. For buildings the rate of support is 10 per cent.

73. EU members have the option of organising network access via either ''third party access'' (TPA) or the ''single buyer'' (SB) model. While TPA permits direct contracting between customers and alternative generators with transmission networks being obliged to accept energy delivery, in the SB model the established transmission networks act as single purchasers which buy electricity from other generators on behalf of the customers which have entered into contracts with the lower-priced sources.

74. In these systems, the power generation entity avoids heat waste by pumping warm water through underground pipes to warm apartments.

75. As quoted in *Bericht der Koalitionsarbeitsgruppe: Globale Kapitlamarktpolitik für mehr Beschäftigung.*

76. This was, for example, the case with offset printing and copying where lack of barriers to entry have been associated with high growth. In negotiations for the new law an unsuccessful attempt was made to have them incorporated into a trade.

Annex I

Simulating business sector employment and investment

This annex describes the specification and estimation of the employment and investment equations of the INTERLINK model which underlie the simulations reported in Figures 7 and 10.

Business sector employment

The growth rate of business sector employment is specified as a function of the growth of business sector output and the longer-term growth of labour productivity. It is assumed that, in the long run, employment is adjusted so that the marginal productivity of labour equals the real compensation rate for labour (including the employers' share of social security contributions and related non-wage costs). To impose this condition in the regression, it is assumed that profit maximisation is subject to a Cobb-Douglas production technology.

The out-of-sample simulation shown in Figure 7 is based on the following regression results:

$$DN = \underset{(3.1)}{-0.032} - \underset{(3.47)}{0.07} * DN(-1) + \underset{(4.54)}{0.31} * DQ - \underset{(27.3)}{1.42} * DES + \underset{(4.0)}{0.11} * ECMN(-1) + \underset{(5.0)}{0.42} * DQ(-2) - \underset{(7.6)}{0.07} * D92$$

R^2 (adjusted for degrees of freedom) = 0.99;

t-statistics are in parentheses beneath the corresponding coefficient; and:

DN = growth rate of business sector employment;

DQ = growth rate of business sector output;

DES = growth rate of labour productivity as measured by a Hodrick-Prescott filter of actual productivity;

ECMN = an error correction term to reflect dynamics but which also imposes a unit elasticity on business sector output and real labour compensation in the long run;

D92 = a dummy variable taking on the value one in the first semester of 1992 (and zero elsewhere) to account for the steep rise in unit labour costs due to re-unification.

The equation has been estimated with semi-annual data for the period 1977/S2 to 1992/S2 so that the out-of-sample simulation shown in Figure 7 begins in 1993/S1. The equation overpredicts employment, except for the second semester in 1993 and the first semester in 1994. To perform stability tests the equation has also been estimated using the full sample period from 1977/S2 to 1997/S2. The recursive residuals test and the Chow test show a stability break in the first half of 1993. The cusum test rejects the null hypothesis of stability since the first half of 1995.

Business sector investment

The growth rate of business sector capital stock (which is subsequently transformed to yield the growth rate of investment) is specified as a function of the growth of business sector output and the change in the real long-term interest rate. It is assumed that, in the long run, the capital stock is adjusted so that the marginal productivity of capital equals the real user cost of capital. To impose this condition in the regression, it is assumed that profit maximisation is subject to a Cobb-Douglas production technology.

The out-of-sample simulation shown in Figure 10 is based on the following regression results:

$$DK = \underset{(1.9)}{-0.015} + \underset{(3.6)}{0.16} * DK(-1) + \underset{(4.4)}{0.12} * DQ - \underset{(1.5)}{0.001} * DIRLR + \underset{(3.3)}{0.023} * ECMK(-1) + \underset{(14.2)}{0.043} * D91$$

R_2 (adjusted for degrees of freedom) = 0.96;

t-statistics are in parenthesis beneath the corresponding coefficient; and:

DK = growth rate of business capital stock;

DQ = growth rate of business sector output;

DIRLR = change in real long term interest rates;

ECMK = an error correction term to reflect dynamics but which also imposes a unit elasticity on business sector output and the real user cost of capital in the long run;

D91 = a dummy variable taking on the value one in 1991 (and zero elsewhere) to account for reunification.

The equation has been estimated with semi-annual data for the period 1975/S1 to 1992/S2 so that the out-of-sample simulation shown in Figure 10 begins in 1993/S1. The equation over-predicts since 1995/S1.*

* To perform stability tests, the equation has also been estimated using the full sample period from 1975/S1 to 1997/S2. While the Chow and the cusum tests do not reject the null of stability, the test of recursive residuals rejects the hypothesis of stability in 1991 while the cusum squared test rejects the hypothesis for several years after 1991.

Annex II

Productivity and structural unemployment

This annex discusses the proposition, put forward in Chapter I, that the structural rate of unemployment may have fallen as a result of recent productivity gains and real wage trends.

The OECD model makes the usual assumption that labour is a homogenous production factor and no distinction is made between skilled and unskilled labour. Technical progress is specified through a labour efficiency index such that:

$$y = a\,(e + n + h) + (1 - a)\,k \qquad (1)$$

where: y = business-sector value added; e = labour efficiency index; n = business employment; h = average hours worked in the business sector, and k = business sector capital stock, all expressed in natural logarithms. The labour efficiency index can be expressed as a total factor productivity index: $tfp = a.e.$

The long-run labour-output ratio depends on real wage costs which depend on labour efficiency.

Real wages depend on labour productivity in the long run, but the adjustment of real wages to shifts in productivity may be only gradual and incomplete. Due to nominal rigidities, prices will not feed through instantaneously to wages, but on the simplifying assumption that they do:

$$w = g_0 - g_1 U + e - g_2\,(e - e\,(-1)) + p^* \qquad (2)$$

where: w = the change in wage costs; U = the unemployment rate, and p^* = expected inflation.

If $p^* = p = w - e$, so that inflation is stable, an equilibrium level of unemployment can be defined:

$$U = NAWRU = \frac{[g_0 - g_2\,(e - e\,(-1))]}{g_1} \qquad (3)$$

where *NAWRU* represents the natural or non-accelerating wage rate of unemployment.

Based on this approach, the structural or equilibrium rate of unemployment is influenced by a permanent change in the growth rate in trend productivity, but the ultimate effect will depend on how fully real wages respond to changes in the growth rate

of productivity (*i.e.* the value of parameter g_2) and on the level of wage demands (parameter g_0). If nominal wage growth adjusts fully to productivity growth, the structural rate will be unaffected, while if real wages grow more slowly than productivity the NAWRU will fall.

A feature which has been important in Germany is the sensitivity of wages to *changes* in the rate of unemployment rather than its level. Introducing such a factor into equation (2) (*i.e.* adding a term $-g_3 \Delta U$) still results in productivity increases serving to lower the NAWRU, in a similar manner to the longer-run definition:

$$U = NAWRU^* = \frac{g_0 - g_2\, e + g_3\, U\,(t-1)}{g_1 + g_3} \qquad (4)$$

The OECD short-run indicator of structural unemployment is calculated from a smoothed series for wages and unemployment so that the direct effects of productivity gains on the estimated NAWRU would only gradually become apparent.

Annex III

Chronology of main economic events

1997

January

The 1997 tax reform law (*Jahressteuergesetz* 1997) comes into force. Its major provisions are:

- The wealth tax is abolished.
- The inheritance and gift taxes are restructured.
- The real estate purchase tax is increased.
- Employment in private households is promoted by special tax allowances.
- Extended depreciation allowances are introduced for newly founded enterprises (*Ansparabschreibung*).

As legislated earlier, child allowances are also increased.

Contribution rates for the pension insurance are increased from 19.2 to 20.3 per cent of the wage base.

The law for the promotion of growth and employment (*Wachstums- und Beschäftigungsförderungsgesetz*) comes into force (some elements were already implemented earlier). The early-retirement age for women and those reaching the maximum years of contribution will be gradually increased to the standard retirement age of 65 from 2000 onwards. Other important measures are: the accrual of pension rights which are not based on contributions are restricted; the level of pensions for immigrants is curbed; spas are reduced with co-payments introduced, and the increase of unemployment benefits in 1997 is curbed.

A law comes into force (*Beitragsentlastungsgesetz*) restricting health care benefits and raising co-payments. The public health funds are required to lower contribution rates.

Minimum reserves are no longer required for ‘‘genuine sale and repurchase transactions’’ (repos) with a maturity of up to one year in marketable securities.

April

Parts of the revised employment promotion law come into force. A more flexible type of employment contract for hiring long-term unemployed is created; active labour market measures are more focused on long-term unemployed and unemployed with unfavourable characteristics; wage subsidies are curbed, and job search controls for recipients of unemployment benefits strengthened.

June

The Federal Minister of Finance enacts global spending controls requiring the consent of the ministry for discretionary expenditures exceeding DM 1 million (DM 10 million for investment expenditures).

July

New regulations come into force concerning employment conditions for civil servants (*Beamte*). Among other provisions, the transferability between posts is increased and tenure in management positions restricted.

August

Parliament passes a law extending modified tax concessions for the new Länder until 2004.

The government's tax reform bills are rejected by the second chamber of parliament (*Bundesrat*).

October

Parliament passes the 1999 pension reform law. Major provisions are a phased decline in the average replacement rate from 1999; higher transfers from the federal budget to the pension system; a sharpening of eligibility criteria for invalidity pensions, and increased pension claims for contribution free periods of child raising.

The Bundesbank raises the repo rate by 30 basis points to 3.3 per cent.

The government's tax reform law 1998/99 is finally rejected by the second chamber of parliament.

November

The Federal Minister of Finance enacts additional spending controls.

The government tables an updated competition law, forbidding cartels – subject to some exceptions – and adapting legislation to European practices.

1998

January

New tax laws came into force:

- The ‘‘solidarity’’ tax surcharge is reduced by two percentage points to 5.5 per cent.
- The business capital tax is abolished.
- The basic income tax allowance is increased.

Continuing the liberalisation in the telecommunications sector, the monopoly of the Deutsche Telecom with respect to wire-bound voice telecommunication is abolished and a new regulatory authority is established.

March

The lower house of parliament passes a new energy law introducing competition with respect to power generation and distribution.

April

The VAT rate is increased by 1 percentage point to 16 per cent to cover increased federal government transfers to the pension system.

The Third Financial Markets Promotion Law (*Drittes Finanzmarktförderungsgesetz*) becomes effective, focusing on deregulating the provision of risk capital to unlisted firms, and extending the admissible range of investment funds.

A revised trades law (*Handwerksordnung*) comes into force, reducing the number of trades requiring a special Master’s qualification (*Meister*) and widening the range of activities for some trades.

May

The official tax revenue estimates project further shortfalls of DM 3 billion in 1998 in comparison with the estimates of November 1997. For 1999 and 2000, the shortfalls amount to DM 28.8 billion and DM 32.1 billion in comparison with the projections which were prepared in May 1997.

A new competition law is passed by the parliament.

Decision to proceed with stage three of the European Economic and Monetary Union which will include Germany taken by European leaders.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
A. Percentage change from previous year at constant prices[1]										
Private consumption	2.7	2.8	5.4	19.0	2.8	0.1	1.2	1.8	1.4	0.2
Gross fixed investment	4.4	6.3	8.5	23.2	3.5	–5.6	3.5	0.8	–1.2	0.2
Construction	3.1	4.4	4.9	19.8	9.7	1.3	6.5	0.3	–3.1	–2.2
Public	0.0	0.0	–0.5	5.1	11.2	–2.3	–0.1	–6.2	–5.7	–8.6
Residential	3.7	4.9	8.5	15.8	9.4	4.0	11.6	2.6	0.1	–0.3
Business	0.0	0.0	5.6	2.6	9.2	–0.7	2.1	–0.6	–6.8	–2.0
Machinery and equipment	6.3	8.8	13.2	27.3	–3.5	–14.4	–1.0	1.6	1.9	3.9
GDP at market prices	3.7	3.6	5.7	13.2	2.2	–1.2	2.7	1.8	1.4	2.2
GDP implicit price deflator	1.5	2.4	3.2	3.9	5.6	4.0	2.4	2.1	1.0	0.6
Industrial production	3.6	4.7	5.2	3.0	–2.6	–7.2	3.6	2.0	0.5	4.0
Employment	0.8	1.5	3.0	28.2	–1.8	–1.7	–0.7	–0.4	–1.3	–1.4
Compensation of employees (current prices)	4.0	4.5	7.8	22.4	8.0	2.1	2.6	3.3	1.0	0.2
Productivity (GDP/employment)	2.9	2.1	2.7	–11.7	4.1	0.6	3.4	2.2	2.7	3.7
Unit labour costs (compensation of employees/GDP)	0.2	0.8	2.0	8.1	5.7	3.3	–0.1	1.4	–0.3	–1.9
B. Percentage ratios[2]										
Gross fixed investment										
As a per cent of GDP at constant prices	20.1	20.6	21.1	23.0	23.3	22.3	22.4	22.2	21.6	21.2
Stockbuilding										
As a per cent of GDP at constant prices	0.5	0.8	0.6	0.4	0.0	–0.1	0.7	1.1	0.8	1.8
Foreign balance										
As a per cent of GDP at constant prices	4.4	5.1	5.5	–0.1	–0.7	–0.4	–0.4	–0.6	0.0	1.0
Compensation of employees										
As a per cent of GDP at current prices	55.8	54.9	54.3	56.5	56.6	56.2	54.8	54.4	53.7	52.4
Direct taxes										
As a per cent of household income	8.7	8.9	8.0	8.3	8.7	8.5	8.5	9.1	8.2	8.1
Household saving										
As a per cent of disposable income	12.8	12.4	13.8	12.9	12.8	12.2	11.6	11.3	11.4	10.9
Unemployment										
As a per cent of civilian labour force	7.6	6.9	6.2	6.7	7.7	8.8	9.6	9.4	10.3	11.4
C. Other indicator[2]										
Current balance (billion $)	50.2	56.9	48.9	–17.8	–19.1	–14.0	–20.3	–22.6	–13.8	–1.0

1. From 1992 all Germany.
2. From 1991 all Germany.

Source: Statistisches Bundesamt, *Volkswirtschaftliche Gesamtrechnungen, Reihe 1*; Deutsche Bundesbank, *Statistisches Beiheft zum Monatsbericht, Reihe 4.*

Table B. **Gross domestic product by origin**[1]

DM billion

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Current prices										
Agriculture, forestry, fishing	33.7	37.2	36.7	41.0	40.6	36.5	36.1	36.4	39.0	39.9
Mining and quarrying, energy	67.8	69.3	70.2	90.1	90.6	91.5	90.4	95.1	89.7	87.6
Manufacturing	652.7	686.0	741.6	825.3	835.8	787.4	809.9	826.2	834.9	865.4
Construction	106.3	114.7	127.6	161.8	191.0	200.3	216.5	222.5	216.0	208.0
Trade, transport, communications	294.7	311.3	346.8	416.9	441.2	454.1	478.4	487.8	490.7	506.6
Government[2]	231.9	238.6	253.2	317.4	349.2	366.0	372.9	384.8	389.6	390.6
Non-profit organisations, households	51.7	53.9	58.3	69.6	78.3	83.6	88.8	95.0	99.7	102.5
Other services	657.3	713.5	791.6	931.3	78.3	83.6	88.8	95.0	99.7	102.5
Gross domestic product at market prices	2 096.0	2 224.4	2 426.0	2 853.6	3 078.6	3 163.7	3 328.2	3 459.6	3 541.5	3 641.8
At 1991 prices										
Agriculture, forestry, fishing	34.5	35.0	36.4	41.0	48.1	45.6	43.8	42.8	46.7	47.4
Mining and quarrying, energy	66.7	69.5	69.4	90.1	86.8	85.6	84.7	86.8	90.3	87.9
Manufacturing	698.6	722.6	762.3	825.3	805.4	747.5	767.9	778.7	778.0	808.6
Construction	127.1	132.5	136.8	161.8	173.0	171.7	180.6	180.0	173.0	168.6
Trade, transport, communications	315.8	330.0	355.8	416.9	426.9	426.0	432.9	442.2	449.6	463.0
Government[2]	261.7	262.6	266.9	317.4	320.8	320.7	322.7	321.8	321.9	319.4
Non-profit organisations, households	57.5	59.1	61.6	69.6	73.3	75.5	78.7	81.2	83.7	85.4
Other services	739.1	773.1	831.3	931.3	982.2	1 009.3	1 049.0	1 080.2	1 111.3	1 140.9
Gross domestic product at market prices	2 301.0	2 384.4	2 520.4	2 853.6	2 916.4	2 881.9	2 960.2	3 013.8	3 054.5	3 121.2

1. From 1991 all Germany.
2. Central and local government, municipalities and social security.

Source: Statistisches Bundesamt, *Volkswirtschaftliche Gesamtrechnungen, Fachserie 18, Reihe 1.*

Table C. **Gross domestic product by demand components**[1]

DM billion

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Current prices										
Private consumption	1 153.7	1 220.9	1 320.7	1 630.3	1 755.5	1 829.3	1 906.0	1973.9	2 040.0	2 084.0
Public consumption	412.4	418.8	444.1	556.7	616.3	634.9	658.6	686.5	704.8	705.2
Gross fixed investment	409.9	448.5	507.8	656.0	709.4	691.0	726.2	742.0	730.3	731.1
Machinery and equipment	182.5	203.5	234.6	306.8	301.8	261.5	258.1	262.3	267.9	279.5
Public	7.2	8.1	8.8	13.5	14.4	13.1	11.3	11.7	10.1	9.5
Private	175.3	195.4	225.8	293.3	287.4	248.4	246.8	250.6	257.8	270.0
Construction	227.4	245.1	273.2	349.2	407.6	429.5	468.0	479.6	462.4	451.6
Public	41.7	44.3	46.8	61.5	72.8	73.7	75.1	71.4	66.6	60.0
Residential	108.6	117.9	135.9	168.2	195.4	212.8	243.7	256.8	257.4	256.9
Other private	77.1	82.8	90.6	119.5	139.4	143.1	149.2	151.5	138.5	134.7
Stockbuilding	10.3	16.0	11.5	12.8	–1.7	–9.2	16.4	30.6	23.3	60.3
Total domestic demand	1 986.3	2 104.3	2 284.1	2 855.9	3 079.5	3 145.9	3 307.2	3 433.0	3 498.4	3 580.6
Exports of goods and services	619.8	701.4	778.9	727.1	732.3	697.6	757.0	820.8	865.8	967.9
Imports of goods and services	510.1	581.3	637.0	729.4	733.2	679.8	735.9	794.2	822.8	906.7
Gross domestic product at market prices	2 096.0	2 224.4	2 426.0	2 853.6	3 078.6	3 163.7	3 328.2	3 459.6	3 541.5	3 641.8
At 1991 prices										
Private consumption	1 264.3	1 300.2	1 370.0	1 630.3	1 676.4	1 678.7	1 699.0	1 729.3	1 752.7	1 756.7
Public consumption	462.3	454.9	465.0	556.7	579.4	576.6	588.9	600.5	616.2	613.7
Gross fixed investment	461.8	490.7	532.3	656.0	679.3	641.5	664.2	669.4	661.3	662.7
Machinery and equipment	195.6	212.8	240.9	306.8	296.1	253.5	251.0	255.0	260.0	270.1
Public	7.6	8.2	8.9	13.5	14.1	12.7	10.9	11.3	9.7	9.1
Private	188.1	204.6	232.0	293.3	282.1	240.9	240.0	243.7	250.3	261.0
Construction	266.1	277.9	291.4	349.2	383.1	388.0	413.2	414.4	401.3	392.6
Public	48.8	50.4	49.9	61.5	68.4	66.7	66.8	62.8	59.0	53.7
Residential	127.6	133.9	145.2	168.2	184.0	191.4	213.5	219.1	219.2	218.4
Other private	89.7	93.6	96.3	119.5	130.7	130.0	132.9	132.5	123.2	120.5
Stockbuilding	12.1	18.1	15.0	12.8	0.6	–3.8	19.4	32.2	24.0	57.0
Total domestic demand	2 200.6	2 263.8	2 382.4	2 855.9	2 935.6	2 893.1	2 971.4	3 031.3	3 054.2	3 090.1
Exports of goods and services	645.3	710.9	789.0	727.1	724.9	688.7	742.8	791.7	832.4	921.3
Imports of goods and services	544.8	590.3	651.1	729.4	744.1	699.8	754.0	809.2	832.2	890.2
Gross domestic product at market prices	2 301.0	2 384.4	2 520.4	2 853.6	2 916.4	2 881.9	2 960.2	3 013.8	3 054.5	3 121.2

1. From 1991 all Germany.

Source: Statistisches Bundesamt, *Volkswirtschaftliche Gesamtrechnungen, Fachserie 18, Reihe 1.*

Table D. **Distribution of national income**[1]

DM billion, current prices

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Compensation of employees	1 169.4	1 221.9	1 317.1	1 611.6	1 741.2	1 777.9	1 824.1	1 883.4	1 902.7	1 906.6
less:										
Employers' social security contributions	220.5	229.1	247.5	298.1	323.5	327.0	352.3	368.0	376.1	387.8
Employees' social security contributions	135.6	142.1	152.4	193.3	212.4	220.6	235.3	245.4	254.0	265.1
Wage tax	165.5	179.2	173.7	220.6	251.8	252.4	263.3	293.8	272.9	272.4
Net wages and salaries[2]	647.8	671.6	743.6	899.6	953.5	977.9	973.2	976.3	999.7	981.3
Income from property and entrepreneurship	466.2	516.2	575.1	615.8	632.5	622.6	685.9	731.6	771.5	840.1
less: Direct taxes on business and property income	75.9	89.9	80.9	94.4	99.4	103.6	89.5	77.5	77.2	71.7
Net income from property and entrepreneurship	390.3	426.3	494.2	521.4	533.1	519.0	596.4	654.2	694.4	768.4
of which:										
Retained profits	64.8	54.9	74.4	17.3	–2.6	–22.1	0.9	49.0	40.1	71.8
Accruing to Government	–39.9	–29.0	–29.1	–38.4	–50.5	–55.8	–60.7	–89.2	–92.2	–96.5
Distributed to households	365.4	400.4	448.8	542.5	586.2	596.9	656.2	694.4	746.5	793.1
Net national income at factor costs	1 038.1	1 097.9	1 237.7	1 421.0	1 486.6	1 496.9	1 569.6	1 630.4	1 694.1	1 749.7
Memorandum items:										
Household disposable income	1 323.2	1 394.3	1 532.7	1 871.3	2 014.0	2 083.9	2 156.8	2 226.4	2 302.0	2 339.6
Household saving ratio[3]	12.8	12.4	13.8	12.9	12.8	12.2	11.6	11.3	11.4	10.9

1. From 1991 all Germany.
2. Including voluntary fringe benefits.
3. Per cent of household disposable income.

Source: Statistisches Bundesamt, *Volkswirtschaftliche Gesamtrechnungen, Fachserie 18, Reihe 1.*

Table E. **Receipts and expenditure of general government: national accounts basis**[1]

DM billion, current prices

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Current receipts										
Income from property and entrepreneurship	19.9	31.5	34.3	38.2	49.9	48.4	51.5	41.9	39.6	39.3
Indirect taxes	252.3	279.6	267.3	328.4	361.7	360.8	364.4	387.9	365.1	359.1
Direct taxes	257.1	278.3	302.2	358.5	389.8	409.6	443.8	447.4	449.5	459.9
Social security contributions	365.8	382.3	408.3	511.5	560.9	594.2	640.3	670.1	702.1	728.3
Other current transfers received	29.4	27.9	32.0	34.5	43.4	40.4	43.7	46.0	48.4	47.6
Total current receipts	924.5	999.6	1 044.0	1 271.1	1 405.7	1 453.4	1 543.6	1 593.4	1 604.7	1 634.3
Current expenditure										
Final consumption expenditure	412.4	418.8	444.1	556.7	616.3	634.9	658.6	686.5	704.8	705.2
Wages and salaries	216.9	222.8	236.3	297.0	327.0	342.1	348.0	358.9	363.0	363.6
Goods and services	195.5	196.0	195.5	259.7	289.4	292.8	310.6	327.7	341.8	341.6
Subsidies	47.7	46.8	48.8	64.9	59.8	61.9	69.1	71.5	70.0	66.9
Debt-interest payments	59.8	60.5	63.3	76.7	100.4	104.2	112.2	131.1	131.8	135.9
Current transfers paid	392.5	409.6	472.9	556.8	605.5	658.7	695.9	729.7	742.2	755.1
Total current expenditure	912.5	935.7	1 029.2	1 255.1	1 382.1	1 459.7	1 535.7	1 618.9	1 648.7	1 663.0
Savings	12.0	63.9	14.8	16.0	23.6	–6.3	7.9	–25.5	–44.0	–28.7
Depreciation	14.7	15.6	16.7	20.0	21.9	23.5	24.5	25.5	26.3	26.7
Net capital transfers received	–23.0	–24.3	–25.7	–55.7	–45.6	–42.2	–33.3	–37.8	–31.4	–30.4
Gross fixed investment	48.9	52.4	55.5	75.0	87.2	86.8	86.4	83.1	76.7	69.5
Financial balance (net lending)	–45.2	2.7	–49.7	–94.7	–87.2	–111.8	–87.2	–120.9	–125.8	–101.9
As a per cent of GDP	–2.2	0.1	–2.1	–3.3	–2.8	–3.5	–2.6	–3.5	–3.6	–2.8

1. From 1991 all Germany.

Source: Statistisches Bundesamt, *Volkswirtschaftliche Gesamtrechnungen, Fachserie 18, Reihe 1.*

Table F. **Balance of payments**[1]

DM billion

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
A. Current account										
1. Foreign trade, net	128.0	134.6	105.4	21.9	33.7	60.3	71.8	85.2	98.5	121.7
Exports (fob)	567.7	641.0	662.0	665.8	671.2	632.2	694.7	749.5	788.9	887.3
Imports (cif)	439.6	506.5	556.7	643.9	637.5	571.9	622.9	664.2	690.4	765.5
2. Supplementary trade items	–2.8	–4.1	–3.8	–2.8	–1.4	–3.1	–1.1	–4.7	–4.2	–6.6
Balance of trade	125.3	130.5	101.5	19.1	32.2	57.3	70.6	80.5	94.3	115.2
3. Services, net	–14.6	–13.3	–17.7	–22.8	–36.0	–43.8	–52.1	–52.5	–52.5	–56.3
Receipts	85.8	97.1	105.4	111.3	110.1	109.6	109.8	120.2	130.8	143.3
Expenditure	100.4	110.5	123.1	134.1	146.1	153.4	161.9	172.7	183.3	199.6
4. Factor income, net	12.5	26.8	33.2	35.5	28.1	21.8	11.4	–1.8	–7.7	–4.2
Receipts	63.2	85.3	105.4	122.5	125.4	127.4	111.4	122.0	114.6	132.9
Expenditure	50.7	58.5	72.2	87.0	97.3	105.6	100.0	123.8	122.3	137.1
5. Transfer payments, net	–34.9	–37.0	–38.1	–61.4	–54.1	–58.4	–62.9	–58.7	–54.8	–56.4
of which:										
Remittances of foreign workers	–7.5	–7.5	–7.1	–6.4	–6.8	–6.8	–7.5	–7.6	–7.4	–7.5
Transfers to the European Community, net[2]	–13.7	–14.3	–13.6	–21.9	–25.3	–27.3	–31.7	–30.0	–27.5	–28.4
Balance on current account	88.2	107.0	79.0	–29.6	–29.8	–23.2	–32.9	–32.5	–20.7	–1.7
B. Capital account										
6. Capital transfers	0.0	0.1	–2.1	–1.0	0.9	0.8	0.3	–0.9	0.0	3.6
7. Financial transactions	–125.6	–134.7	–90.5	20.2	91.5	14.0	66.4	72.3	28.1	–12.6
German investment abroad (increase: –):	–155.2	–248.6	–183.4	–106.0	–116.8	–295.9	–107.9	–175.6	–198.2	–385.7
Direct investment	–21.2	–28.5	–38.7	–39.3	–30.5	–25.3	–27.9	–55.6	–44.5	–57.5
Portfolio investment	–71.7	–50.1	–25.1	–29.9	–75.6	–52.8	–84.3	–32.6	–54.0	–161.3
Credit transactions	–60.6	–167.8	–117.5	–33.6	–8.1	–215.2	6.6	–83.3	–95.8	–163.5
Long-term	–2.5	–14.2	–43.0	–26.2	–14.1	–33.4	–26.3	–27.1	–19.8	–57.5
Short-term	–58.2	–153.6	–74.5	–7.4	5.9	–181.8	32.8	–56.2	–76.0	–106.0
Other investment	–1.7	–2.2	–2.1	–3.2	–2.6	–2.5	–2.3	–4.1	–3.9	–3.3
Foreign investment in Germany (increase: +):	29.7	113.9	92.9	126.2	208.3	310.0	174.3	247.9	226.3	373.1
Direct investment	2.0	13.3	4.0	6.8	4.2	3.2	2.9	19.3	–4.1	–0.3
Portfolio investment	7.4	45.7	19.4	71.2	122.5	235.7	45.0	84.6	140.8	150.7
Credit transactions	20.4	55.0	69.8	48.3	81.9	71.1	125.6	145.0	89.7	223.7
Long-term	1.7	14.3	20.7	–0.2	27.6	50.2	34.2	63.6	41.2	44.3
Short-term	18.7	40.7	49.2	48.4	54.3	20.9	91.4	81.5	48.5	179.3
Other investment	–0.0	–0.1	–0.4	–0.1	–0.2	–0.0	0.8	–1.0	–0.1	–1.0
8. Unclassifiable transactions	2.7	8.5	24.7	10.7	6.1	–27.4	–21.6	–21.3	–8.9	2.3
C. Change in the Bundesbank's net external assets (transaction values) (increase: +)	–34.7	–19.0	11.0	0.3	68.7	–35.8	12.2	17.7	–1.6	–8.5
9. Balancing item of the Bundesbank	2.2	–2.6	–5.1	0.5	–6.3	1.5	–3.7	–2.6	0.1	14.0
D. Change in the Bundesbank's net external assets (balance sheet values) (increase: +)	–32.5	–21.6	5.9	0.8	62.4	–34.2	8.6	15.1	–1.5	5.5

1. From July 1990 including the external transactions of the former German Democratic Republic.
2. Excluding collection expenses, EAGGF (Guidance Funds) and Regional Fund.

Source: Deutsche Bundesbank, *Statistisches Beiheft zum Monatsbericht, Zahlungsbilanzstatistik.*

Table G. **Imports and exports of goods by regions**

DM billion

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
	Imports, fob									
OECD	363.0	417.3	458.5	531.7	529.3	459.5	499.7	544.8	560.0	607.7
EU	257.8	293.9	327.5	380.0	376.2	314.8	340.9	372.1	385.3	406.4
Austria	18.5	20.6	23.9	26.8	27.7	25.9	28.6	25.7	27.0	27.6
Belgium-Luxembourg	30.9	34.7	39.7	45.7	44.7	34.0	38.0	43.9	43.9	46.4
France	52.4	59.8	65.1	78.5	75.8	65.2	68.3	72.8	73.0	78.2
Italy	40.0	45.1	52.0	59.8	58.4	48.1	51.8	56.8	58.3	58.9
Netherlands	44.4	51.1	55.2	61.9	60.4	49.3	51.0	57.4	60.5	63.3
United Kingdom	29.7	34.0	36.6	42.4	42.7	35.5	38.9	43.4	47.4	52.7
Other European countries	29.2	32.7	36.7	39.9	40.9	39.1	43.1	46.2	47.4	53.6
Switzerland	19.1	20.5	23.0	24.9	24.9	23.2	25.7	27.3	26.9	29.4
Other OECD	76.0	90.7	94.4	111.8	112.2	105.6	115.7	126.5	127.4	147.7
Japan	29.6	33.3	33.3	40.8	38.5	35.0	34.8	36.1	34.3	37.2
United States	28.5	37.1	37.0	41.2	41.7	39.6	43.3	44.3	48.6	57.4
Central and eastern European countries	20.8	25.2	32.6	39.2	39.9	39.4	48.1	56.4	60.2	72.1
Non-oil developing countries	45.3	51.5	55.1	63.2	58.6	58.2	63.4	67.1	66.3	74.0
OPEC	9.9	11.4	13.2	13.9	14.0	12.8	12.1	10.3	11.8	13.0
China	4.2	5.8	7.7	11.3	11.1	13.4	15.0	15.8	17.6	21.0
Dynamic Asian Economies	20.9	23.2	25.6	31.9	30.0	31.3	34.0	36.7	34.9	39.1
Total imports	431.3	497.9	550.2	636.5	627.9	564.5	613.9	656.7	679.2	755.2
	Exports, fob									
OECD	486.2	549.8	560.1	566.3	568.8	518.7	571.7	618.2	649.5	723.3
EU	361.8	413.9	413.4	422.6	425.6	369.6	404.0	437.3	451.1	489.2
Austria	32.1	35.6	37.3	39.9	40.0	37.2	39.7	41.1	44.9	45.3
Belgium-Luxembourg	41.3	45.4	47.5	48.2	49.0	42.5	46.7	48.8	49.4	51.1
France	70.8	84.4	84.7	89.8	89.0	80.0	86.4	90.1	88.0	94.9
Italy	51.4	59.6	60.1	61.2	62.3	47.0	52.3	56.7	59.0	65.0
Netherlands	48.5	53.9	54.3	55.7	55.4	48.1	52.5	56.9	59.3	61.1
United Kingdom	52.7	59.1	54.7	50.6	51.0	50.1	55.2	62.1	63.8	74.5
Other European countries	44.9	49.1	51.6	51.1	48.6	47.9	49.9	55.8	56.9	62.2
Switzerland	34.4	38.3	38.5	37.6	35.2	33.3	36.5	39.3	37.4	39.1
Other OECD	79.5	86.8	95.1	92.6	94.6	101.3	117.8	125.0	141.6	172.0
Japan	12.9	15.0	17.2	16.3	14.5	15.2	17.3	18.4	20.7	20.1
United States	45.2	44.9	46.3	41.1	41.9	45.7	53.2	53.6	59.0	75.4
Central and eastern European countries	26.4	32.6	48.7	46.0	43.4	47.7	53.8	61.5	72.0	92.0
Non-oil developing countries	44.4	51.2	52.6	55.9	58.8	63.7	74.4	82.3	84.9	95.7
OPEC	15.3	16.2	18.3	21.3	22.9	18.4	17.7	17.0	16.9	20.2
China	4.9	4.6	4.0	4.0	5.7	9.5	10.2	10.7	10.8	10.6
Dynamic Asian Economies	16.3	20.2	22.3	25.1	26.2	30.2	37.2	42.0	44.3	47.9
Total exports	565.4	638.8	662.0	668.1	671.4	632.7	696.4	749.9	786.5	885.2

Source: Deutsche Bundesbank, *Statistisches Beiheft zum Monatsbericht, Zahlungsbilanzstatistik.*

Table H. **Foreign trade by main commodity groups – customs basis**[1]

DM billion

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
	Imports, cif									
SITC classification										
0. Food and live animals	41.4	43.8	46.4	53.6	54.7	45.3	50.2	52.4	53.9	55.9
1. Beverages and tobacco	4.4	4.7	5.4	6.6	6.7	5.8	6.3	6.2	6.4	7.3
2. Crude materials, except fuels	28.5	33.5	29.6	28.6	28.5	22.7	26.8	28.4	26.0	30.2
3. Mineral fuels, lubricants and related materials	33.6	38.3	46.1	53.6	47.6	45.1	42.7	40.6	52.8	57.5
4. Animal and vegetable oils, etc.	1.5	1.7	1.5	1.6	1.7	1.5	1.8	1.9	2.0	2.3
5. Chemicals	42.6	47.7	49.7	54.2	54.6	45.4	53.7	59.3	59.4	66.3
6. Manufactured goods classified chiefly by material	80.5	94.6	98.4	106.7	106.2	84.7	97.9	108.3	100.0	109.8
7. Machinery and transport equipment	128.2	154.4	178.4	226.2	220.6	187.8	205.8	217.9	229.9	258.4
8. Miscellaneous manufactured articles	65.7	73.3	83.1	102.7	103.3	93.6	97.3	94.6	99.6	107.7
9. Other	13.4	14.6	13.1	11.7	13.8	12.8	17.5	24.7	39.0	60.6
0-9. Total imports	439.7	506.6	551.7	645.4	637.8	544.8	600.1	634.3	669.1	755.9
	Exports, fob									
SITC classification										
0. Food and live animals	23.6	26.0	25.3	29.2	29.7	27.3	28.7	29.3	31.5	33.7
1. Beverages and tobacco	3.2	3.6	3.7	4.2	4.7	4.2	5.2	4.9	5.3	5.4
2. Crude materials, except fuels	10.6	12.4	11.7	12.5	12.5	10.5	12.4	13.8	12.7	14.5
3. Mineral fuels, lubricants and related materials	6.9	7.9	8.3	8.2	8.2	7.4	7.5	6.7	9.7	10.3
4. Animal and vegetable oils, etc.	1.6	1.8	1.5	1.5	1.5	1.4	1.9	2.1	1.9	2.2
5. Chemicals	76.9	83.1	82.0	84.7	84.7	79.0	92.2	98.1	101.6	117.0
6. Manufactured goods classified chiefly by material	102.5	117.2	113.8	113.2	111.0	98.3	110.0	120.7	119.1	134.1
7. Machinery and transport equipment	272.8	311.9	319.2	326.4	332.8	299.0	336.3	360.8	383.1	440.0
8. Miscellaneous manufactured articles	62.4	69.7	72.3	74.5	75.5	67.9	71.8	72.6	76.8	85.1
9. Other	7.0	7.8	8.4	11.6	10.2	8.9	14.7	18.8	30.2	44.6
0-9. Total exports	567.7	641.3	646.3	666.2	670.8	604.0	680.8	727.7	771.9	886.8

1. From 1991 all Germany.

Source: Statistisches Bundesamt.

Table I. **Money and credit**[1]

End of period, DM billion

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Consolidated balance sheet of the banking system:										
I. Bank lending to domestic non-banks	2 346.8	2 483.9	2 888.6	3 160.7	3 497.6	3 839.8	4 149.0	4 446.6	4 781.7	5 067.2
Bundesbank	14.5	13.9	13.5	13.7	19.3	13.4	11.9	9.6	8.7	8.7
Credit institutions	2 332.3	2 470.0	2 875.1	3 147.0	3 478.3	3 826.4	4 137.1	4 437.0	4 773.0	5 058.5
To public sector	541.4	547.2	603.6	629.2	739.3	840.4	926.2	1 067.5	1 148.7	1 209.2
To private sector	1 790.9	1 922.8	2 271.5	2 517.8	2 739.0	2 986.0	3 210.9	3 369.5	3 624.3	3 849.3
Short-term	341.5	375.2	521.0	575.8	571.2	544.2	549.1	584.0	617.2	625.8
Medium- and long-term	1 412.3	1 506.4	1 697.4	1 876.0	2 077.2	2 307.6	2 458.7	2 592.6	2 801.3	2 989.3
Securities	37.1	41.2	53.1	65.9	90.6	134.2	203.2	192.9	205.9	234.1
II. Net foreign assets	255.9	288.4	325.4	334.4	338.1	414.0	287.5	280.6	328.5	310.5
Bundesbank	67.1	45.8	51.8	52.4	114.8	80.4	89.3	104.3	102.7	96.9
Credit institutions	188.8	242.6	273.6	282.0	223.3	333.6	198.2	176.3	225.8	213.6
III. Domestic monetary capital holdings	1 369.5	1 482.8	1 670.9	1 852.8	1 988.4	2 146.1	2 338.1	2 561.2	2 745.0	2 903.3
Time deposits (more than 4-year notification)	452.6	491.3	524.9	560.1	564.3	603.1	669.5	726.1	796.6	840.2
Public sector	165.2	169.3	173.7	185.4	173.3	178.6	187.5	192.7	197.3	197.7
Private sector	287.4	322.0	351.2	374.7	391.0	424.5	482.0	533.4	599.3	642.5
Saving deposits[2], banks saving, bonds and bank debt securities outstanding	777.5	835.5	955.7	1 080.8	1 184.6	1 280.9	1 382.3	1 526.3	1 613.5	1 693.0
Share capital and reserves	139.4	156.0	190.3	211.9	239.5	262.1	286.3	308.8	334.9	370.1
IV. Public sector claims on the Bundesbank	3.5	6.9	19.1	12.7	0.4	13.5	3.0	2.2	6.7	0.1
V. Other items, net	–40.0	–26.9	–20.9	–31.8	–128.1	–187.6	–158.5	–156.2	–176.9	–214.5
VI. Money supply M3 (M3 = I + II – III – IV + V)	1 189.6	1 255.6	1 503.0	1 597.7	1 718.7	1 906.7	1 937.0	2 007.4	2 181.8	2 259.7
VII. Domestic non-banks' savings deposits[3]	493.5	479.0	515.4	513.2	522.2	587.5	654.3	749.7	865.9	928.8
VIII. Money supply M2 (M2 = VI – VII)	696.1	776.6	987.6	1 084.5	1 196.5	1 319.2	1 282.7	1 257.7	1 315.9	1 330.9
IX. Time deposits (less than 4-year notification)	269.1	325.8	403.3	480.5	527.0	592.9	518.6	441.6	399.0	392.9
X. Money supply (M1 = VIII – IX)	427.0	450.8	584.4	604.1	669.6	726.3	764.1	816.1	916.9	938.0
Sight deposits	284.4	303.8	425.8	432.3	469.1	514.3	538.2	578.6	670.1	691.0
Currency in circulation	142.6	146.9	158.6	171.8	200.5	212.0	225.9	237.5	246.8	247.0
Memorandum item:										
Central bank money[4]	207.2	216.6	246.1	266.3	305.0	294.3	280.9	271.9	286.2	288.6

1. From July 1990 the time series cover the entire Deutschemark currency area.
2. At over three months' notice.
3. At three months' notice. Untill 1992 savings deposits at statutory notice.
4. Defined as currency in circulation plus minimum reserves on domestic bank liabilities at current reserve ratios. Data reported here are averages of daily figures for December.

Source: Deutsche Bundesbank, *Monatsbericht.*

Table J. **Population and employment**[1]

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
	Thousands									
Population	61 449	62 063	63 253	79 986	80 596	81 181	81 423	81 663	81 895	82 061
Working-age population (15-64 years)	42 960	43 258	43 947	55 302	55 149	55 244	55 549	55 714	..	..
Labour force, total	29 608	29 799	30 369	39 166	38 836	38 641	38 669	38 428	38 328	38 260
Self-employed	3 001	3 011	3 026	3 424	3 485	3 533	3 587	3 614	3 614	3 616
Dependent employment, total	24 365	24 750	25 460	33 139	32 375	31 688	31 385	31 203	30 750	30 260
Employment, total	27 366	27 761	28 486	36 563	35 860	35 221	34 972	34 817	34 364	33 876
	Per cent of civilian employment									
Agriculture, forestry, fishing	4.0	3.7	3.5	3.9	3.4	3.2	3.0	2.9	2.8	2.7
Industry	39.8	39.8	39.7	39.5	37.9	36.7	35.9	35.5	34.7	33.9
Commerce and communications	18.6	18.6	18.7	18.6	19.0	19.1	19.1	18.8	18.8	18.8
Other	37.6	37.9	38.1	38.0	39.7	41.0	42.0	42.7	43.8	44.5
	Thousands									
Unemployment	2 242	2 038	1 883	2 602	2 978	3 420	3 697	3 611	3 964	4 384
Short-time workers	2 073	1 071	555	17 588	6 526	9 454	3 705	1 982	2 767	1 819
Vacancies	190	256	322	364	354	277	287	321	328	339
	Per cent of total labour force									
Unemployment	7.6	6.8	6.2	6.6	7.7	8.9	9.6	9.4	10.3	11.5
Vacancies	0.6	0.9	1.1	0.9	0.9	0.7	0.7	0.8	0.9	0.9

1. From 1991 all Germany.

Source: Statistisches Bundesamt, *Wirtschaft und Statistik and Volkswirtschaftliche Gesamtrechnungen*, Fachserie 18, Reihe 1; and OECD, *Labour Force Statistics.*

Table K. **Wages and prices**[1]

Indices 1991 = 100

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Wages and productivity, whole economy										
Monthly contractual pay rates	87.8	90.2	94.1	100.0	111.0	118.2	121.7	127.0	129.9	131.7
Monthly gross wages and salaries per employee	87.6	90.2	94.5	100.0	110.5	115.5	118.3	122.6	125.3	126.7
Output per employee	93.1	95.0	97.6	100.0	104.1	104.7	108.3	110.6	113.6	117.7
Unit labour costs	94.2	94.9	96.8	100.0	106.2	110.2	110.4	112.2	112.1	110.1
Wages and productivity, manufacturing, western Germany										
Hourly contractual pay rates, blue collar	86.4	89.7	93.9	100.0	106.6	112.0	115.1	118.9	124.6	126.3
Hourly gross earnings, blue collar	86.2	89.6	94.3	100.0	105.4	110.7	114.2	118.7	122.7	..
Hours worked, blue collar[2]	97.1	98.7	100.2	100.0	97.5	87.9	84.1	100.0	95.2	92.8
Output per man-hour	90.1	93.4	96.8	100.0	100.8	103.1	111.2	..	..	..
Unit labour costs[2, 3]	92.4	93.3	95.9	100.0	104.9	108.7	102.1	100.0	99.2	93.6
Prices										
Agricultural producer prices	97.9	106.4	101.0	100.0	98.8	90.6	91.7	91.6	91.1	92.5
Industrial producer prices	93.0	96.0	97.6	100.0	101.4	101.6	102.2	104.0	103.5	104.7
Costs of dwelling construction	84.9	88.0	93.7	100.0	106.4	111.7	114.3	117.1	117.0	116.4
GDP deflator	91.1	93.3	96.3	100.0	105.6	109.8	112.4	114.8	115.9	116.7
Private consumption deflator	91.2	93.9	96.4	100.0	104.7	109.0	112.2	114.1	116.4	118.6
Consumer prices										
Including food	91.4	94.0	96.5	100.0	105.1	109.8	112.8	114.8	116.5	118.6
Excluding food	91.1	93.8	96.2	100.0	105.7	111.2	114.6	116.8	118.7	120.9
Foreign trade prices										
Exports	96.2	98.9	99.0	100.0	100.7	100.7	101.6	103.3	103.5	105.1
Imports	97.1	101.5	99.2	100.0	97.6	96.1	96.9	97.3	97.8	100.9

1. From 1991 Germany.
2. From 1995 Germany, index 1995 = 100.
3. Including mining and quarrying.

Source: Statistisches Bundesamt.

Table L. **Structure of output and performance indicators**

	1992	1993	1994	1995	1996	1997	1992	1993	1994	1995	1996	1997
	Share of GDP						Share of total employment					
A. Structure of output (constant prices)												
Agriculture, hunting, forestry and fishing	1.6	1.6	1.5	1.4	1.5	1.5	3.4	3.2	3.0	2.9	2.8	2.7
Energy, water supply, mining	3.0	3.0	2.9	2.9	3.0	2.8	1.7	1.7	1.6	1.5	1.4	1.4
Manufacturing	27.6	25.9	25.9	25.8	25.5	25.9	28.5	26.9	25.7	25.3	24.8	24.5
Construction	5.9	6.0	6.1	6.0	5.7	5.4	7.7	8.2	8.6	8.7	8.4	8.1
Trade, transport, communications	14.6	14.8	14.6	14.7	14.7	14.8	19.0	19.1	19.1	18.8	18.8	18.8
Services	28.7	30.6	31.0	31.8	32.8	33.7	19.1	20.3	21.2	22.0	22.9	23.6
Total traded goods and services	89.0	88.9	89.1	89.3	89.5	89.8	79.4	79.3	79.2	79.3	79.1	79.0
General government non-traded sector	11.0	11.1	10.9	10.7	10.5	10.2	16.1	16.0	15.8	15.5	15.5	15.4
	Productivity growth						Share of total investment					
B. Economic performance (constant prices)												
Agriculture, hunting, forestry and fishing	37.7	3.0	0.5	1.8	15.8	5.5	1.9	1.7	1.6	1.6	1.6	..
Energy, water supply, mining	6.4	6.1	5.1	7.4	10.0	3.4	5.4	5.7	6.1	4.9	5.2	..
Manufacturing	6.2	–0.1	8.3	3.6	3.2	6.8	18.6	15.9	14.2	15.0	15.3	..
Construction	2.3	–4.9	0.7	–1.6	0.6	2.8	2.0	2.1	1.9	1.8	1.7	..
Trade, transport, communications	2.1	0.7	2.7	3.7	3.3	4.5	15.6	15.7	14.7	13.2	..	..
Services	6.0	1.6	0.1	1.0	1.9	3.2	43.5	45.7	48.7	51.3	..	..
Total traded goods and services	6.6	1.2	3.8	2.8	3.6	5.0	87.0	86.7	87.1	87.8	23.8	..
General government non-traded sector	2.0	2.3	2.4	1.8	1.7	1.2	11.7	12.0	11.6	11.0	10.5	..
	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996		
C. Others indicators (current prices)												
Total R&D expenditure as % of total GDP	2.90	2.90	2.90	2.75	2.61	2.48	2.43	2.33	2.30	2.28		
R&D as % of GDP in business enterprise sector	2.60	2.60	2.60	2.48	2.29	2.16	2.08	1.97	1.92	..		
Government-funded R&D as % of total	34.7	34.2	34.1	33.9	35.8	36.0	36.7	37.0	37.4	37.3		

Source: Statistisches Bundesamt, *Volkswirtschaftliche Gesamtrechnungen, Fachserie 18 Reihe 1;* OECD, *Main Science and Technology Indicators.*

Table M. **Labour market indicators**

A. Trend

	Peak (before 1991)	Trough (before 1991)	1994	1995	1996	1997
Standardised unemployment rate	1983: 7.7	1980: 3.0	8.4	8.2	8.9	9.7
Unemployment rate[1]						
Total	1985: 9.3	1979: 3.8	10.6	10.4	11.5	12.7
Male	1985: 8.6	1979: 2.9	9.5	9.6	11.0	12.2
Female	1986: 10.6	1980: 5.2	12.0	11.4	12.1	13.3
Youth (15-19 year old)	1983: 9.1	1980: 3.2	7.2	7.9	9.0	9.6
Share of long-term unemployment[2]						
Western Germany	1988: 32.6	1981: 12.8	32.5	33.3	32.7	34.0
Eastern Germany			34.7			
Registered vacancies (thousands)	1991: 331	1983: 76	285	321	327	339
Monthly hours of work, hrs mln[3]	1980: 763	1988: 644	598	574	540	524

B. Structural and institutional features

	1990	1991	1992	1993	1994	1995	1996	1997
Labour force (% change)	1.9	..	–0.8	–0.5	0.1	–0.6	–0.3	–0.2
Participation rate[4]								
Total	69.1	71.6	71.7	71.4	71.3	70.7	70.3	..
Males	80.8	81.8	81.4	81.0	80.9	80.1	79.9	..
Females	57.0	61.2	61.6	61.5	61.5	61.0	60.4	..
Employment/total population	45.0	45.7	44.5	43.4	43.0	42.6	42.0	41.3
Employers, self-employed and family workers (as % of total)	10.6	9.4	9.7	10.0	10.3	10.4	10.5	10.7
Wage-earners and salaried employees (as % of total)	89.4	90.6	90.3	90.0	89.7	89.6	89.5	89.3
Civilian employment by sector (% change)								
Agriculture	–3.2	..	–14.9	–8.0	–4.3	–3.9	–6.0	–3.7
Industry	2.8	..	–5.9	–4.8	–3.0	–1.4	–3.6	–3.4
Services	3.5	..	1.9	0.6	1.0	0.4	0.3	–0.2
of which: General government	0.7	..	–1.0	–2.3	–1.8	–2.1	–1.6	–1.9
Total	3.0	..	–1.8	–1.7	–0.7	–0.4	–1.3	–1.4
Civilian employment by sector (as % of total)								
Agriculture	3.5	3.9	3.4	3.2	3.0	2.9	2.8	2.7
Industry	39.7	39.5	37.9	36.7	35.9	35.5	34.7	33.9
Services	56.8	56.6	58.7	60.1	61.1	61.6	62.5	63.3
of which: General government	15.1	15.9	16.1	16.0	15.8	15.5	15.5	15.4
Short-time workers[5]	2.3	2.6	1.0	0.6	0.8	0.5	0.6	0.8
Non-wage labour costs[6]	16.4	17.0	17.0	17.3	18.4	18.5	18.8	19.6

1. Total unemployment as a per cent of total dependent employment.
2. People looking for a job one year or more as a percentage of total registered unemployment.
3. Data refer to the total number of hours worked in the month by wage-earners in the mining and manufacturing industries.
4. Labour force as a percentage of population from 15 to 64 years.
5. Short-time workers as percentage of total employment.
6. Employers' social security contributions as a percentage of total wage, from 1991 all Germany.

Source: Statistisches Bundesamt; Bundesanstalt für Arbeit, *Amtliche Nachrichten, Jahreszahlen;* OECD, *National Accounts, Labour Force Statistics and Main Economic Indicators.*

Table N. **Public sector**[1]

	1992	1993	1994	1995	1996	1997
Budgetary indicators: general government accounts (% of potential GDP)						
Cyclically adjusted primary current receipts (excluding interest)	46.2	44.9	45.4	45.0	44.1	43.8
Cyclically adjusted primary current disbursements (excluding interest)	43.0	42.5	42.5	42.6	42.2	41.4
Cyclically adjusted primary budget balance	–1.7	–0.6	0.3	0.1	0.2	1.2
General government budget balance	–4.3	–3.1	–2.3	–3.0	–2.8	–1.9
Structure of expenditure and taxes (% of GDP)						
General government expenditure[1]	49.2	50.2	49.7	50.3	49.6	48.5
Consumption	20.0	20.1	19.8	19.8	19.9	19.4
Subsidies	1.9	2.0	2.1	2.1	2.0	1.9
Investment	2.8	2.7	2.6	2.4	2.2	1.9
General government revenue[2]	46.4	46.7	47.1	46.8	46.1	45.8
Direct taxes	11.7	11.4	10.9	11.2	10.3	9.9
Indirect taxes	12.7	12.9	13.3	12.9	12.8	12.6
Social security contributions	18.2	18.8	19.2	19.4	19.8	20.1
Other indicators[3]						
Income tax as a per cent of total tax	47.9	46.7	44.9	46.2	44.5	43.7
Income tax elasticity[4]	1.3	–0.1	0.3	1.9	–1.7	–0.7
Tax rates (%)						
Average effective personal income tax rate	25.5	24.9	24.8	25.6	24.0	23.5
Average rate of employees'social secutity contribution paid[5]	18.3	18.3	19.4	19.7	20.2	. .
Standard VAT rate	14.0	15.0	15.0	15.0	15.0	15.0

1. Defined as current disbursements plus gross investment plus net capital transfers paid.
2. Defined as tax and non-tax receipts plus consumption of fixed capital.
3. Households. Income tax elasticity is with respect to household disposable income.
4. TIn 1995 the increase in the tax elasticity is due to the introduction of the solidarity tax surcharge.
5. For an unmarried average production worker, per cent of gross earnings.

Source: OECD, *National Accounts and The Tax/Benefit position of production workers;* Deutsche Bundesbank, *Monatsbericht.*

Table O. **Financial markets**[1]

	1970	1975	1980	1985	1990	1994	1995	1996	1997
Structure of financial flows[2]									
Share of intermediated financing in total financing	88.9	73.2	88.6	59.4	57.4	58.2	85.1	76.4	78.7
Financial institutions' share of financial assets	45.2	50.3	44.1	44.3	43.4	50.9	52.3	54.3	55.8
Structure of private non-financial sector's portfolio:									
Deposits	55.4	61.2	44.9	43.6	36.4	8.4	31.6	44.2	25.4
Bonds and bills[3]	12.1	7.4	15.5	17.0	31.7	48.7	18.3	2.7	17.6
Equities	5.1	3.6	3.9	2.8	8.2	5.6	4.9	5.2	4.0
Non-financial corporate financial structure	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Own-financing	56.7	67.0	58.1	67.3	60.8	61.4	98.7	67.7	73.3
Debt and equity	43.3	33.0	41.9	32.7	39.2	38.6	1.3	32.3	26.7
Long-term debt	22.9	30.6	19.0	20.5	19.9	27.7	–17.5	15.5	15.4
Equity	2.2	3.2	2.3	2.5	4.8	4.3	4.8	5.8	1.9
Short-term debt	18.3	–0.8	20.6	9.7	14.6	6.7	14.0	11.0	9.4
Internationalisation of markets									
Foreign business of the domestic banking sector[4]									
Assets	6.6	7.1	7.4	8.7	11.7	11.2	11.4	11.9	13.3
Liabilities	4.3	4.1	6.2	5.6	6.3	8.4	9.1	9.1	11.0
International banking networks:									
Foreign banks in Germany[5]	..	..	..	127	177	197	198	195	..
German bank branches abroad (without subsidiaries)	..	44	88	118	177	241	255	264	..
Share of long-term capital transactions:									
Net purchases of domestic bonds by non-residents[6]	..	–6.8	0.5	29.8	8.1	7.7	37.8	41.7	39.0
Net purchases of foreign bonds by residents[6]	6.8	2.7	13.7	26.0	10.0	9.0	10.6	8.6	23.5
Efficiency of markets									
Divergence between Euro rates and domestic interest rates[7]	–0.98	–0.42	–0.42	–0.14	–0.02	–0.02	–0.09	–0.07	–0.04

1. From 1991 all Germany.
2. Credits from domestic and foreign banks as percentage of total funds raised on the credit market by domestic non financial sectors.
3. 1991-1994 including investment certificates.
4. As a percentage of deposit banks' balance sheets (without foreign business of German bank branches and subsidiaries abroad).
5. Number of branches and subsidiaries.
6. As a percentage of total purchases.
7. Three-month Euro-DM interest rate minus three-month interbank rate.

Source: Deutsche Bundesbank, *Monatsbericht* and *Statistisches Beiheft zum Monatsbericht, Zahlungsbilanzstatistik.*

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

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